

Death, Destruction, and Data

An Intro to Using Data to Study International Conflict

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Table of contents

Preface	4
Part I: Trends in conflict	4
Part II: Who fights whom and why?	5
My hope	5
1 What is War?	6
1.1 Politics by Other Means	6
1.2 The Peace Science Tradition	8
1.3 Defining War and Other forms of Conflict	9
1.4 How We'll Measure War In This Class	10
1.5 Conclusion	11
2 The Data	12
2.1 Nothing Good Lasts Forever	12
2.2 The Militarized Interstate Events Dataset	14
2.3 Accessing and Cleaning the Data	17
2.3.1 Reading the Data into R	18
2.3.2 Cleaning the Data	20
2.3.3 Some Simple Descriptive Summaries	23
2.3.4 Making Some Graphs of Data Summaries	26
2.4 Summary	32
3 Is War Disappearing?	33
3.1 The Decline of War Thesis & Why It Matters	33
3.2 Braumoeller's Critique	36
3.3 What You Count Matters	37
3.3.1 Measuring Incidents	38
3.3.2 Measuring Opportunities	41
3.3.3 Measuring War Propensity	44
3.3.4 Looking at the Trend	45
3.4 Accounting for Random Chance Matters	46
3.5 Summary	52
4 Is War Less Deadly?	53
4.1 Decline of War Thesis and War Severity	53

4.2	Three Problems with Studying War Severity	54
4.2.1	Measurement	54
4.2.2	Detecting Changes	55
4.2.3	Explaining Changes	56
4.3	A Low-tech Replication of Braumoeller's Work	57
4.3.1	Get the Data	57
4.3.2	Data Processing	59
4.3.3	Analysis	61
4.4	Summary	67
5	Dyadic Models of Conflict	68
5.1	Why do countries fight?	68
5.2	Studying dyads	69
5.3	Modeling dyadic data	77
5.4	Some additional recommendations	86
5.5	Summary	90
6	The Balance of Power and War	92
6.1	What is power?	92
6.2	Why measure power?	94
6.3	Testing BoP and PT	95
6.4	Summary	103
7	The Democratic Peace	105
7.1	What is the Democratic Peace?	105
7.2	Measuring Democracy	107
7.3	Testing the Democratic Peace	110
7.4	Summary	117
8	The Trading Peace	118
8.1	Introduction	118
8.2	Quantifying Trade Interdependence	120
8.3	Analysis	123
8.4	Summary	132
	References	133

Preface

My name is [Miles Williams](#), and I authored this free and open access book for students taking a 300 level undergraduate class at Denison University called “Death, Destruction, and Data.”

This provocatively named course is an introduction to using data to study international conflict. This is a specialized class that assumes some prior knowledge of the R programming language and, ideally, some experience with the `{tidyverse}` dialect. If you don’t have prior experience with either, that’s not the end of the world. I’ll go slowly, and because this class is focused on studying one issue (international conflict) that also means that much of the material will center on conceptual issues rather than an emphasis on technical know-how (though there will be some of that, so don’t rest too easy, and I will expect you to do some original data analysis and writing along the way.).

The book is divided into two parts, which I’ve summarized briefly below.

Part I: Trends in conflict

Part I is all about studying historical trends in international conflict. I begin with an introduction to conflict (including war) and how international relations scholars define armed conflict between countries.

I then talk about how conflict is measured using data, and show how to use this data to answer a pair of important questions: is war becoming more or less common, and is it becoming more or less deadly?

Both questions are foundational to the quantitative study of war, but they have faded in relevance as conflict scholars turned their attention to dyadic (country-pair) analyses of factors that explain why some countries tend to fight while others don’t. These questions have witnessed some renewed interest, however, due to recent debates about the “decline of war theory.”

On the technical side, we’ll spend a lot of time thinking about how conflict is measured, how to visualize trends in conflict onset and deadliness, and detect whether changes in each have taken place.

Part II: Who fights whom and why?

Part II moves the focus to country dyads with the goal of answering the question: why do certain countries fight each other more than others?

The literature on why countries fight is much too broad to cover in a short book about using data to study conflict, but I will draw on a sample of prominent theories of conflict and peace to motivate a variety of analysis examples and expose you to some of the ins and outs of constructing datasets for studying international conflict. I'll also introduce you to the standard set of statistical tools conflict scholars use in their research (so get ready to learn about logit models!).

My hope

I will not be able to address every single technical issue or measurement question that comes along with conflict research. I do hope, however, that these instructional materials will give you a solid enough foundation to start asking your own questions, testing novel hypotheses, and conducting empirical research. This is the start of your journey rather than the end (assuming you choose to keep going long after this course wraps up).

1 What is War?

Main ideas:

- A useful way to think of war is as a continuation of politics by other means (but what does that mean, exactly?).
- Peace science is a research tradition in the field of international relations that seeks to use data and the scientific method to study war and peace.
- War, and international conflict more broadly, is treated by peace scientists as a specific kind of political event that can be measured and, about which, systematic comparisons can be drawn.
- The Correlates of War Project, started in the 1960s, had (and continues to have) outsized influence over how international conflict is defined and empirically operationalized.
- A recent controversy surrounding one of the most popular COW datasets has called into question the reliability of the data, and the outgrowth of this controversy is a brand new dataset that we'll use in the rest of this class.

1.1 Politics by Other Means

It's a bit foolhardy of me to think I can address the question "what is war?" in a single chapter of a set of open source lecture notes. If this were my goal, I'd have to affix a big asterisk to the chapter with the ubiquitous phrase spoken in nearly every *Star Wars* movie ever produced: "I've got a bad feeling about this."

War has been part of the human experience since well before recorded history. For thousands of years, countless thinkers have opined on the nature of war.

However, one of the more recent takes (by historical standards) that you may have heard of is the idea that "war is a continuation of politics by other means." This idea comes from the famed Prussian General, Carl von Clausewitz (2003). Though this sentence is sometimes is taken out of context, and he didn't actually say this exact phrase (he didn't even write in English), this idea that war is just an extension of politics has had a profound impact on contemporary thought about war.

While not all translations or analyses of Clausewitz's writing agree on every detail, few contest the idea that he thought war was inherently political. At least, this is true of the most common

type of war. As the political scientist Bear Braumoeller (2013a) nicely summarizes, Clausewitz drew a distinction between two kinds of war: *absolute* war and *real* war.

Absolute war is rare, and considered to be a kind of duel to the death. It also isn't the type of war Clausewitz considered an extension of politics. When he said this, he meant *real* war. So what's the distinction?

As Braumoeller (2013a) further summarizes, real war is a form of political communication meant to gather information necessary to assess the hypothetical outcome of absolute war. Another, more intuitive way to put it, is that real war is about sizing up your opponent. By engaging in brief skirmishes, sides of a conflict learn information about the other side (its strengths, weaknesses, resolve, and so on). This information is necessary so that both sides can agree to the terms of a bargain.

In short, war is intimately connected with bargaining between countries. Countries have some issue, whether that's territory, trade, or resources, that they have a dispute over. As they try to bargain with each other to find a compromise that satisfies them both, there's some uncertainty about how much each side should actually ask for and deserves to get. Real war can serve as a means by which sides get this information. *Ergo*, war as a continuation of politics by other means.

The idea that war is related to bargaining would later be famously formalized by the political scientist James Fearon (1995), who developed a game theoretical model to help him show why countries would choose to go to war when negotiating a deal would be mutually better for both sides. A few years later, R. Harrison Wagner (2000) would extend Fearon's formalized logic of bargaining to include war as not just the result of bargaining failure, but also a costly means by which sides negotiate a deal.

In some ways, these game theoretic approaches nicely encapsulated the ideas Clausewitz proposed long ago. Braumoeller (2013a) puts it most succinctly: "information, not attrition, is the goal of conflict" (9).

Now, if you're at all tuned in to current events, you probably have some doubts about this way of understanding war, and rightly so. You can make a good case that Israel's recent war with Hamas went far beyond communicating information. It seemed an awful lot like attrition. It also demonstrated that war can be a means of decisively changing the balance of power by weakening an opponent, which happened, by extension, with Israeli strikes against Hezbollah in Lebanon and against the Iranian regime.

These kinds of conflicts, however, align more with Clausewitz's definition of absolute war, which he considered rare, as these kinds of conflicts in fact are. But they do happen, and most observers agree that when you initiate a conflict of the continuation-of-politics variety you risk escalation to the duel-to-the-death sort. This is why German Chancellor Theobald von Bethmann-Hollweg referred to the start of war as *rolling the iron dice* on the eve of World War I. Actually, what he is supposed to have said exactly is: "If the iron dice must roll, may God help us."

Iron dice seems an apt analogy, because much to everyone's surprise at the time, the conflict that became known as World War I was predicted by nearly all Great Power elites at the time to be a short conflict that would be over before Christmas (Tuchman 1994). The lesson of World War I is that wars can escalate to attrition, even when those involved had no hopes or intentions of making that happen.

War, in short, is more than just a costly form of bargaining; it comes with existential risks. The choice to roll the iron dice is political, but what happens next can quickly fall beyond anyone's ability to control.

1.2 The Peace Science Tradition

Because war is a form of politics, and of a sort that has fatal implications, it's natural that a group of political scientists would be motivated to carve out a niche area of research focused on trying to study war in a systematic way. This area of research became known as "peace science."

Even up to the mid-20th century, the study of war was mostly the purview of historians, whose analyses generally focused on particular wars or periods in time. The focus was on close examination of historical documents and other source materials to construct a narrative about why a war happened, why one side won or lost, or the special role that particular individuals or social forces played.

These are all important things to document, and the field of peace science would not have been able to get started without the painstaking work of historians to document past conflicts.

However, peace scientists wanted to go further. They wanted to measure wars across time and space in a consistent and rigorous way so that they could then use statistical analysis and the scientific method to systematically test arguments about war and peace. They wanted to come up with theories, generate testable hypotheses, and empirically verify them against observable regularities in conflict.

The political scientist J. David Singer and the historian Melvin Small were among those at the forefront of the peace science tradition. Singer founded the Correlates of War Project in 1963, which remains alive and well today, and with the help of Small they began work to collect data on all wars that took place since 1816 (the post-Napoleonic-Wars period).

On the Correlates of War Project [website](#) (most people just refer to it as COW, and pronounce it like "cow"), it says that "[t]he original and continuing goal of the project has been the systematic accumulation of scientific knowledge about war." This project rests on the assumption that international conflicts can be *compared* across time and space. To make this possible, Singer and Small, as the COW website continues, "needed to operationally resolve a number of difficult issues such as what is a 'state' and what precisely is a 'war.'" Based on their work,

Singer and Small published a book in 1972 called *The Wages of War*, which would establish a standard definition of war that would guide the work of countless future peace scientists.

As a result of their efforts, decades of research looking at trends in the data and assessing how various factors predict conflict onset have supported the accumulation of knowledge about war. One of the most comprehensive, yet succinct, resources you can find that summarizes all this accumulated knowledge is the edited volume *What Do We Know About War?*, which is now in its third iteration (Mitchell and Vasquez 2024). While there is still a lot that peace science has failed to figure out about war, the field knows a lot more now than they did in 1963 when Singer and Small began their work.

1.3 Defining War and Other forms of Conflict

So how do peace scientists define war? How do they define other instances of conflict between countries as well?

Singer and Small considered war an instance of “sustained combat involving substantial fatalities,” according to [documentation for the latest COW wars dataset](#). More precisely, they recorded that a war occurred if more than 1,000 battle deaths were documented, combat was sustained, and it involved organized armed forces.

They would go on to develop a typology of war based on the participants: (1) interstate war, (2) civil war, (3) extra-systemic war.

Interstate war involves wars between and among sovereign countries considered part of the international state system (I’ll focus on interstate conflict in this class). Civil war involves conflicts between the government of a state and an armed group within its borders. Extra-systemic war involves conflict between a state and a separate actor not currently recognized as a state, such as a colony.

COW maintains datasets for each of these kinds of wars on [their website](#).

There are other prominent datasets on international conflict as well. Probably the most popular is the [Militarized Interstate Dispute](#) (MID, pronounced “mid”) dataset, which is also maintained by COW. MIDs are distinct from wars but are often used to study the potential for war, or warlikeness, of countries.

MIDs are a comparatively more recent innovation, and the intended goal of measuring MIDs is to capture more variation in international conflict than is possible using the more stringent criteria used to measure war. The concept and original dataset were proposed by Gochman and Maoz (1984), and the data would go through several rounds of updates and revisions. The most recent version of the dataset (version 5) just dropped in 2020 (Palmer et al. 2022).

COW notes on their webpage that MIDs, citing Jones, Bremer, and Singer (1996):

‘...are united historical cases of conflict in which the threat, display or use of military force short of war by one member state is explicitly directed towards the government, official representatives, official forces, property, or territory of another state. Disputes are composed of incidents that range in intensity from threats to use force to actual combat short of war.’

MIDs, which are not wars, are the most common measure of international conflict used by peace scientists today, and this is for two key reasons.

The first is pragmatic: MIDs are more common than wars. That makes them more amenable to statistical analysis.

The second is conceptual. I think Braumoeller (2013a) puts it best:

[T]he best way to gauge the warlikeness of nations [is] to measure the frequency with which they run the risk of war. Initiating a war, or issuing a threat that creates the risk of war, is indicative of the willingness, in the worst case, to spill quite a bit of blood; that willingness is the best indicator of the propensity of the state to engage in violent conflict; and the individual conflict incident, or *militarized dispute*, is the most reliable objective indicator of that willingness (10, emphasis mine).

Linking back to the idea of *real* war (war as a costly means of bargaining), MIDs seem to capture the essence of this definition much better than the one proposed by Singer and Small, who require that a war involve actual use of armed military forces and substantial fatalities (at least 1,000). Real war, by Clausewitz’s standards, might only need a minimum of a few shots across the bow at sea, or targeted strikes on military bases, for both sides to learn what they need to learn about the other side’s ability, resolve, and the possible outcome of absolute war. Threats may even be enough.

Going further, Braumoeller (2019) later concluded in a book he wrote about trends in war that escalation to absolute war (like one of the World Wars) is hard to predict and can happen even when most or all the actors involved wish to avoid it. The time when purposeful political action is the most significant on the path the war is the initial stages when countries issue threats or make the decision to start shooting. MIDs capture this idea by definition. To relate this back to another idea in the last section, MIDs show the willingness of countries to roll the “iron dice.”

1.4 How We’ll Measure War In This Class

Going forward in this class, we’ll primarily use the concept of the MID to study international conflict. However, as I’ll discuss more in the next chapter, we won’t use the COW MIDs dataset.

Without giving too much away, a recent controversy in peace science emerged in the last decade when a group of scholars undertook painstaking work to validate and recreate the COW MID dataset from scratch. What these scholars found surprised them.

They discovered thousands of data entry errors, including phantom conflicts (conflicts that had no supporting evidence to verify they happened) and double counting. So bad were these issues that this team of researchers claimed all MID data for years prior to 2002 were unreliable for analysis.

This was a bold claim, and a damning one if true. The maintainers of the MID dataset obviously responded, and the result was a series of publications that would slowly come out over several years documenting the argument that ensued. Eventually, the group of scholars who documented these data issues would get a National Science Foundation grant to reconstruct all the data from the ground up, and along the way they made some conceptual and technical improvements.

Perhaps the two most relevant updates are their rigorous reporting of battle fatalities and their choice to redefine wars as a subset of all conflicts rather than a distinct phenomenon. (In the MID dataset, an observation leaves the dataset the moment it escalates to a war.) Their new concept is called the militarized interstate confrontation (or MIC), which can include threats, shows, or uses of force short of war, but also all-out wars.

Even better, these scholars documented specific militarized interstate events (MIEs), their specific timing, and deadliness. These events are the individual threats, uses of force, and battles that comprise MICs.

In short, the data that I'll introduce in the next chapter is incredibly rich and represents an important new advancement in the study of international conflict. I therefore want you to become familiar and comfortable with this dataset as we work with it throughout this class.

1.5 Conclusion

In sum, many in peace science conceptually treat war as political, and in particular as a means of costly bargaining. To measure this concept well, it isn't necessary to only look at massive and deadly conflicts—you can examine threats or even small scale uses of force, too.

Importantly, the field of peace science, which is responsible for providing us with data on international conflicts, bases this exercise on the idea that conflicts are comparable across space and time. By assuming this is true, it's possible to approach the study of war using statistical analysis and the scientific method. And while many questions remain, the field of peace science has been able to use this approach to build up quite a lot of knowledge about war over the decades.

In the next chapter I'll introduce the data that we'll use the rest of the semester, and discuss a recent controversy about this data.

2 The Data

Main ideas:

- The Militarized Interstate Disputes (MID) dataset has supported decades of peace science research on international conflict.
- But a dispute over its quality in the last decade spurred the creation of a an alternative set of data: jointly, Militarized Interstate Events (MIEs) and Militarized Interstate Confrontations (MICs).
- We'll use the latter (MIEs in particular) in this course given its purported quality and high level of granular detail.
- Once you download the data, you can see it has lots of useful information, which we'll spend some time exploring.

2.1 Nothing Good Lasts Forever

In the last chapter I talked about war, how it's defined, and how it's measured. I also ended that chapter by noting a controversy that arose about the most commonly used conflict dataset available for studying international conflict: The Militarized Interstate Dispute (MID) dataset maintained by the Correlates of War Project (COW).

MIDs are ubiquitous in conflict research, so when a group of scholars come along and say the data is unreliable for vast stretches of time (every year from 1816 to 2002 to be exact), that creates a stir. It also led to the publication of a series of articles where the competing research teams behind the MID data and the group of discontents duked it out. [Many of the articles](#) in the exchange were published in the journal *International Studies Quarterly*, one of the top journals in the field of international relations.

The most significant development to come out of this back-and-forth was the creation of novel datasets by the group of scholars that raised concerns about MIDs. In a pair of articles, the political scientists Douglas Gibler and Steve Miller (2024a, 2024b) introduced the Militarized Interstate Confrontations (MIC) dataset and the Militarized Interstate Events (MIE) dataset. The former is actually an aggregated version of the latter: MIEs are embedded in larger MICs.

The rationale for these new datasets is summarized pretty well [in the codebook](#) (a manual about what's in the dataset) for MICs:

We found an overwhelming number of errors in the original [MID] data and corrected those data, providing all of our suggested changes to the CoWMID Project dataset hosts. However, after more than nine years of communications with CoWMID and several exchanges in *International Studies Quarterly*, it is apparent that CoWMID admits their pre-2002 data is incredibly error-prone but that they have neither the resources nor inclination to fix those thousands of admitted errors. Therefore, we are introducing the Militarized Interstate Confrontation (MICs) data, which are based on proper use of MID coding rules and include numerous advancements in both information and presentation.

While the MIE and MIC datasets include some innovations in the way events are coded, and what information is either included or excluded from the data, both datasets are premised on the foundational ideas behind the concept of MIDs crystallized by Jones, Bremer, and Singer (1996), along with updates made by subsequent iterations of the COW MID project.

In short, Gibler and Miller didn't try to measure a new concept; they tried to measure an old concept better.

That doesn't mean they didn't innovate. One of the most important updates Gibler and Miller made was to keep wars in the data. Recall from the last chapter that MIDs are considered threats, shows, or uses of force between countries *short of war*. Once a conflict escalates to a war, this is noted in the data and then everything from there on out is dropped. Gibler and Miller decided to keep conflicts in the data, even after they escalate to a war. This is why their main aggregate dataset is called the Militarized Interstate *Confrontations* dataset instead of Militarized Interstate *Disputes*. They wanted to signal that they aren't sticking to the original criteria for disputes (events short of war). Instead, they're capturing everything short of war, and all-out-war as well.

I think this is a really useful innovation, because if someone wants to study something closer to the original definition of MIDs, they can do that with this data. And if someone wants to just study all-out-wars based on their own fatality threshold, they can use this data to do that, too.

The way that disputes and wars seamlessly line up in the MIC/MIE datasets is, itself, another helpful innovation. One of the oddest features of the COW MID dataset and the COW Wars dataset is that they can't be easily merged together (ReidSarkees and Wayman 2010). This isn't good. Remember from the last chapter that the COW Wars dataset uses a 1,000 battle death threshold to mark if a dispute escalates to a war. Unfortunately, the COW Wars and COW MID datasets don't totally overlap in their death statistics or even cases they consider conflicts. The MIC and MIE datasets, to the contrary, are unified and completely overlap, and they don't base inclusion on fatality statistics. Instead, they leave it to the researcher to

make a judgement call about how to analyze the data, using their own death thresholds or other criteria to study conflicts if that's what they want to do.

I admit that I don't really have a dog in this fight, but for practical reasons I favor the MIE/MIC datasets over the COW MID dataset.

First, I like the consistency of all events, ranging from threats to all-out war, being included in one dataset. That makes my life easier because I don't have to download multiple datasets or toggle between them if I want to study only low-level disputes versus wars.

Second, while I haven't personally validated either the MID or MIE/MIC datasets, I find the case made by Gibler and Miller in support of their data compelling—for two reasons.

For one, as they noted in the quote I shared above, the folks at COW MID admitted that the errors Gibler and Miller found exist, and they said they don't plan to fix them. When people with a vested interest in the validity of a dataset own up to the fact that errors others found were real, that's telling, and then when they say they don't plan to do anything about it, that's frustrating.

Also, while this isn't the most scientifically valid justification, I find it hard to believe that Gibler and Miller would have bothered to get a National Science Foundation grant, put together a research team, and engage in the tedious and lengthy process of reconstructing a dataset that purports to capture *every single* conflict that has taken place between countries from 1816 to 2014 just to win an argument when a good-enough dataset already exists. This seems like the work of sincere researchers who want to make sure fellow peace scientists have the best data possible for doing their work, and who feel like the community of conflict researchers is being let down by current resources.

For these reasons, we will use the Gibler and Miller data in this class. In particular, we'll use the MIE dataset since the MIC dataset is just an aggregated version of its MIE counterpart.

Speaking of which, let me introduce you to the MIE dataset and give you a tour of what it has to offer.

2.2 The Militarized Interstate Events Dataset

First, some background. The Militarized Interstate Events (MIE) dataset and its Militarized Interstate Confrontations (MIC) counterpart were created as a part of the [International Conflict Data Project](#) hosted at the University of Alabama. (COW MID also is hosted by a university, previously by the University of Michigan, and now by Penn State.)

When you go to the download page for the data, you'll see that multiple datasets are hosted on the site ([you can check that out here](#)). In addition to the MIE and MIC datasets, there are: Militarized Interstate Participants, Militarized Interstate Confrontation Endings, Militarized

Interstate Confrontation Names (useful if you want to know the historical names of many conflicts that escalated to wars), Protest-Dependent Militarized Interstate Disputes, and Formal War Declarations (which is treated as a separate action from fighting, since declaring war is technically a distinct political behavior from actually going to war).

There's a lot of rich material to draw on here for studying international conflict. Some datasets are just useful summaries of MIE, while others, like the endings dataset or confrontation names dataset provide additional information. For someone like me, it's very easy to spend hours poking around all the files.

But, I have some things to teach you, and you probably want to get a move on, so let me show you the relevant stuff you need to know about the MIE dataset.

First, here's a screenshot of what the download page looks like focusing just on the MIE dataset:

Militarized Interstate Events (MIEs), 1816-2014



The MIE data includes daily events data that correspond to the Jones, Bremer, and Singer (1997) coding rules, with some additional coding changes as described in the codebook.

CODEBOOK

LINK TO ARTICLE

SUMMARY STATS

DATA DOWNLOAD

You can see there are four tabs associated with the dataset:

- The [codebook](#) (basically, the manual for the dataset)
- A [link to the article](#) Gibler and Miller published about the data
- Some [bare-bones summary statistics](#) about what's in the data
- A link that will download the data in a zip file to your computer

Don't worry about the last one because I did you a solid and downloaded the data myself and then saved it on my GitHub. This will make it possible for you to read the data into your software (R) directly from online without having to deal with downloading and storing it permanently on your computer.

Before we get to that step though, I think it's a good idea to just look at the information on offer in the dataset. I've copied below the relevant details from the [MIE codebook](#). Each bullet point tells you the name of a column in the dataset and what it represents.

- **micnum**: The Militarized Interstate Confrontation (MIC) number for each case. If the event is part of a Militarized Interstate Dispute (MID) originally coded by CoW, we use that number. Note that we have also added numerous confrontation cases that are not in the CoWMID data as events or disputes. These cases begin with micnum values in the 9000's to demonstrate completely new cases.
- **eventnum**: This is the individual event number within the confrontation. Note that these numbers are meaningful. An event number value of "1" denotes the first threat, display, or use of force in that particular confrontation. The highest event number is the last onset of a threat, display, or use of force that confrontation—previous events may still last longer and cause the duration of the confrontation to extend. A large number of events have missing start and/or end dates. We researched the cases in which the missing-day event could be the first or last participation of the state in the confrontation and confirmed the temporal placement of the missing-day event using a meaningful event number. We have not yet sequentially ordered the missing-day events that could not have been first or last events, but we hope to add that feature soon.
- **ccode1**: This is the CoW country code for the state. **ccode2**: This is the CoW country code for the state that was targeted by a threat, display, or use of force. In clashes, the higher number ccode is ccode2.
- **stmon**: The start month of the event, with values ranging from 1 to 12. There are no missing month values; if the month of a militarized action could not be determined, the action was not coded.
- **stday**: The start day of the event, with values ranging from 1 to 31 and missing days reported as values of -9. We have done our best to appeal to the historical record to eliminate missing days in the data, but there are numerous cases where only the month of the militarized action is known.
- **styear**: The start year of the event, with values ranging from 1946 to 2014. There are no missing year values; if the year of a militarized action could not be determined, the action was not coded.
- **endmon**: The end month of the event, with values ranging from 1 to 12. There are no missing end month values.
- **endday**: The end day of the event, with values ranging from 1 to 31 and missing days reported as -9. Again, we have done our best to appeal to the historical record to eliminate missing days in the data, but there are numerous cases where only the end month of the militarized action is known.
- **endyear**: The end year of the event, with values ranging from 1946 to 2014.
- **sidea**: Side A in the confrontation is defined by whichever state initiated the first militarized action in the confrontation. All states coordinating with the original Side A state are also considered members of Side A. The variable is dichotomous, with a "1" indicating Side A and "0" indicating Side B. Reciprocated confrontations will have one or more actions by states with a Side A value of "0".
- **action**: The type of action ccode1 targeted ccode2 with in the event. Possible values [with **hostlev** in brackets] include the following: 0 No militarized action [1] 1 Threat to use force [2], 2 Threat to blockade [2], 3 Threat to occupy territory [2], 4 Threat to

declare war [2], 5 Threat to use CBR weapons [2], 6 Threat to join war, 7 Show of force [3], 8 Alert [3], 9 Nuclear alert [3], 10 Mobilization [3], 11 Fortify border [3], 12 Border violation [3], 13 Blockade [4], 14 Occupation of territory [4], 15 Seizure [4], 16 Attack [4], 17 Clash [4], 19 Use of CBR weapons [4], and 22 War Battle [5]. Note that this scale is actually not ordinal. See above for a discussion of how we changed how wars are coded, and see the last section for additional notes on wars and battles.

- **hostlev**: The hostility level of the event. Possible values include the following: 1 No militarized action, 2 Threat to use force, 3 Display of force, 4 Use of force, and 5 War.
- **fatalmin1**: This is an estimate of the minimum number of military battle-deaths for ccode1 in the event.
- **fatalmax1**: This is an estimate of the maximum number of military battle-deaths for ccode1 in the event.
- **fatalmin2**: This is an estimate of the minimum number of military battle-deaths for ccode2 in the event.
- **fatalmax2**: This is an estimate of the maximum number of military battle-deaths for ccode2 in the event.
- **version**: The MIE version number. We are constantly reviewing and updating our datasets, and we ask users of our data to always report which version number of the data they are using in their research.

This is a lot of detail to take in, but once you have a moment to absorb the information, it boils down to a few key things: identifiers for confrontations and events, identifiers for the country on either side of the event (where the country on side 1 is responsible for engaging in an action against side 2), information about the timing of the event, a classification for what kind of action took place, a classification for how severe the action was, and information about how many people died on each side.

There's enough here that a person can write a lengthy report based solely on what's in this dataset (which you'll do at the end of Part I of this class).

To do that, however, you first need to get the data into your software, and clean it up a bit to make it useful for analysis, which I'll show you how to do in the next section.

2.3 Accessing and Cleaning the Data

Data analysis is impossible without specialized tools, and for the time being the most ubiquitous tool in the political scientist's (and peace scientist's) toolkit is the R programming language and the RStudio IDE (integrated development environment). I'm a political scientist, and I don't drift too far from the herd. I've been using R for more than a decade now.

I can tell you from personal experience that R has its shortcomings, but I'm [with the computer scientist Clause Wilke](#) in saying R is among the best programming languages for data science.

More importantly, it's probably one of the best languages for doing statistical analysis and data visualization, which is what most social scientists need to use it for.

Going one step further, as you'll see in Part II of the book, Steve Miller (one of the people responsible for the MIE dataset) created an R package specifically designed to make creating datasets for doing peace science in R easier. This package is going to be a real time-saver when we start incorporating a bunch of different variables into our conflict data to assess which factors predict patterns in international conflict. This package will also help out immensely in the next two chapters when we start visualizing trends in conflict occurrence and deadliness.

But all that in due time. Let me start by showing you how to get the data.

First, you need to make sure you're in RStudio. If you're a Denison student, you can access RStudio remotely by going to r.denison.edu, where you can sign in using your Denison credentials. You can also install the software directly to your computer. Just follow [these instructions here](#).

Second, I recommend opening a Quarto file (.qmd) which will give you a single document where you can write in plain text and write and run code. To do this, just go to the "File" tab, then "New File," and then select "Quarto Document."

Because my goal in this class is to focus on doing peace science research rather than introducing you to all the basics of using R and navigating RStudio, I'm going to proceed with the naive assumption that you have some background in using these tools.

However, if you don't have this background, all is not lost. Because I have a habit of making textbooks (glorified lecture notes) for all the technical classes I teach, I have written a bunch of guides already about R and RStudio, like this one called "[R Basics](#)." If you need some help getting up to speed, I recommend checking this out, and I strongly recommend visiting me during office hours. I'll do my best to coach you up.

One more thing, for those not familiar with R. If you follow along with the code I'll show you for accessing the data and cleaning it up, I think you'll quickly get a sense for how the language works.

2.3.1 Reading the Data into R

The first step in any data analysis task is to access the data you need. In this case, this step is pretty easy, both because a clean dataset already exists, and because I have saved that dataset in a format that makes it possible to read it directly into your computer using some simple R code. I've done so below.

The MIE dataset exists as a CSV file on my GitHub, so I need to give my computer (or if you're using the cloud-based version of RStudio, the server you have remote access to) the location of the file online and instructions to download its contents into its working memory. The below R code allows me to talk to my computer (or the server) to tell it what to do. (This,

by the way, is the proper way to think about what R is doing: it's a language that lets you interface directly with your computer, or a remote server, to give it explicit instructions about what to do.)

```
## open {tidyverse}  
library(tidyverse)  
  
## read in data and save as 'dt'  
read_csv(  
  "https://raw.githubusercontent.com/milesdwilliams15/foreign-figures/refs/heads/main/_data/  
) -> dt
```

The first thing I did in the above code is write `library(tidyverse)`. The R language is supported by a vast ecosystem of specialized software packages that allow you to run different kinds of routines (*functions*) in your R session. The `{tidyverse}` package is one of the most popular, and it offers more than just a useful set of functions. It supports an entire dialect in the R language. To install it (if you don't have it already), you can run the following code in the R console: `install.packages("tidyverse")`.

This is the R dialect that I personally speak, and it's become increasingly popular over time. The reason is that speaking R the “tidyverse” way is meant to improve the human readability of code. In short: to help you be nice to the people who need to read your code at a later point in time. I'll use this approach to coding throughout most of the examples in this book.

Once I opened the `{tidyverse}`, I then ran the following code:

```
read_csv(  
  "https://raw.githubusercontent.com/milesdwilliams15/foreign-figures/refs/heads/main/_data/  
) -> dt
```

`read_csv()` is a function (a thing that has instructions for a particular routine for your computer to run) that reads a CSV file into your computer's working memory saved at whatever location you tell it to go look. In this case, I gave it the url to the raw CSV file saved on my GitHub. Note that I made sure this url was in quotation marks (“”). This is important for locating files in R.

Note, too, that after I closed the parentheses of the `read_csv` function around the url (and this is also important: make sure you close parentheses), I wrote `-> dt`. In R-speak, this means I assigned the content of the CSV file to be saved as an *object* called `dt`. The arrow is what lets me assign names, and its direction controls what output gets saved with what name. Because the arrow is pointing away from `read_csv()` to `dt`, my computer gets the message that I want it to read a CSV file into its working memory and use “dt” as shorthand for anytime I want to tell it to do something else with this data.

This step is crucial. If I don't save the data as an object, my computer will just read the contents of the CSV file and report them back to me, which isn't helpful. Below, you can see what happens when I don't assign the output to an object and just run the function. Underneath the code you can see a snippet of the output. I told my computer to read in the content of the MIE dataset, and then it just read its contents back to me. Because it isn't saved as an object, I now have no way to refer back to it (except by just re-running the `read_csv` function every time I need the data, which is just painful).

```
read_csv(
  "https://raw.githubusercontent.com/milesdwilliams15/foreign-figures/refs/heads/main/_data/mie.csv"
)
```

```
# A tibble: 28,011 x 18
  micnum eventnum ccode1 ccode2 stmon stday styear endmon endday endyear sideal
  <dbl>   <dbl>   <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl>
1     2     1     2    200     5    -9   1902     5    -9   1902     1
2     3     1    300   345    10     7   1913    10     7   1913     1
3     4     2    339   200     5    15   1946     5    15   1946     1
4     4     3    200   339    10    22   1946    10    22   1946     0
5     4     4    200   339    10    22   1946    10    22   1946     0
6     4     5    200   339    10    22   1946    10    22   1946     0
7     4     6    339   200    10    22   1946    10    22   1946     1
8     4     7    200   339    10    23   1946    10    23   1946     0
9     4     8    200   339    11     1   1946    11     1   1946     0
10    4     9    339   200    11     1   1946    11     1   1946     1
# i 28,001 more rows
# i 7 more variables: action <dbl>, hostlev <dbl>, fatalmin1 <dbl>,
#   fatalmax1 <dbl>, fatalmin2 <dbl>, fatalmax2 <dbl>, version <chr>
```

If you're new to the R language, I think the exercise of reading some data into your computer is instructive. There are things called R packages that have useful functions you can use. Functions serve as short-hand for telling your computer to run specialized routines (like accessing data). If you want to save the output of functions, you need to assign their content a name (make it an object, in R-speak), otherwise you'll have to run the function again anytime you need to do anything with its output.

If you can wrap your head around this simple framework, then I think you'll be surprised about how well you pick up coding in the R language.

2.3.2 Cleaning the Data

Compared to many real-life datasets, the MIE dataset is in great shape. First of all, it's **tidy**, which isn't synonymous with "clean." Tidy is a technical term used to describe a dataset

where:

1. Each row is one observation.
2. Each column is one variable.
3. Each data entry per row and column is one value.

The MIE dataset ticks these boxes. But what, exactly, is an observation in the dataset?

According to the MIE codebook, each row is a unique militarized event that took place between a pair of countries. The below code will let me look at just the first few rows of data. Take a look at the output.

```
slice_head(dt, n = 5)
```

```
# A tibble: 5 x 18
  micnum eventnum ccode1 ccode2 stmon stday styear endmon endday endyear sideal
  <dbl>   <dbl>   <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl>
1     2     1     2    200     5    -9   1902     5    -9   1902     1
2     3     1    300    345    10     7   1913    10     7   1913     1
3     4     2    339    200     5    15   1946     5    15   1946     1
4     4     3    200    339    10    22   1946    10    22   1946     0
5     4     4    200    339    10    22   1946    10    22   1946     0
# i 7 more variables: action <dbl>, hostlev <dbl>, fatalmin1 <dbl>,
#   fatalmax1 <dbl>, fatalmin2 <dbl>, fatalmax2 <dbl>, version <chr>
```

For each event in the data, there are variables (columns) to help identify the event, indicate which countries were involved, the date it happened, and other details about the kind of military action that took place along with fatality statistics.

Again, in essence, it's tidy: each row is an observation (event), each column is a variable (a different bit of info about the event), and each row-column entry is a single value.

However, even though the data is tidy, it isn't fully ready for analysis. Notice that all the variables are numerical (except the **version** column). For some variables, this is sensible, but not so much for others. For example, the **action** column is meant to be a set of categories for the kind of militarized action that took place, but it's all numbers. The same is true for **hostlev**. These numbers are stand-in numerical codes for categories, and the codebook provides information for what each number represents.

This is a common database management strategy. It's annoying from the perspective of the analyst, but sensible from the perspective of people who store and manage data. Numerical codes eat up less memory than writing out values in plain English. The MIE dataset is 28,011 rows long. This isn't exactly "big data," but it's big enough that some measures needed to be taken to make the overall data file smaller.

The downside for the analyst is that you have to do some cleaning (or “recoding”) of variables to make them more useful. Fortunately, in a dataset with only 18 columns, this process isn’t too involved. And a nice bonus for you is that I’m about to show you code for cleaning up this data so you don’t have to figure it out from scratch. (And down the road I’m going to give you tools that automate this cleaning process even more, so you can focus on the details of your analysis.)

Alright, here’s the code. It’s a bit more involved than simply reading in data, and I introduced a new kind of operator into the mix called a pipe. This is the `|>` symbol. This is R-speak for “give this thing to that thing.” Instead of putting an object inside a function to perform a routine on it, the pipe lets you put the object first. Piping is a way to improve the human readability of code. It also is a way to conveniently chain together a succession of operations you want to perform on your data. The below code tells my computer to convert the numerical codes for `action` and `hostlev` into their qualitative category names. Then it creates a `date` column that is a specialized “date class” variable at the year-month level for when each event started (there are missing values for days, so I’m going to just get as precise as months). Finally, it creates a column called `year`, which is just the event start year (this will help with merging this data with other data down the road). On the tail end of the code I write `-> dt` which saves the output by writing over the old content saved in `dt`.

```
library(socsci)
dt |>
  mutate(
    action = frcode(
      action == 1 ~ "Threat to use force",
      action == 2 ~ "Threat to blockade",
      action == 3 ~ "Threat to occupy territory",
      action == 4 ~ "Threat to declare war",
      action == 5 ~ "Threat to use CBR weapons",
      action == 6 ~ "Threat to join war",
      action == 7 ~ "Show of force",
      action == 8 ~ "Alert",
      action == 9 ~ "Nuclear alert",
      action == 10 ~ "Mobilization",
      action == 11 ~ "Fortify border",
      action == 12 ~ "Border violation",
      action == 13 ~ "Blockade",
      action == 14 ~ "Occupation of territory",
      action == 15 ~ "Seizure",
      action == 16 ~ "Attack",
      action == 17 ~ "Clash",
      action == 19 ~ "Use of CBR weapons",
      action == 22 ~ "War Battle"
```

```

),
hostlev = frcode(
  hostlev == 2 ~ "Threat to use force",
  hostlev == 3 ~ "Display of force",
  hostlev == 4 ~ "Use of force",
  hostlev == 5 ~ "War"
),
date = ym(paste0(styear, "-", stmon)),
year = styear
) -> dt

```

One more thing: I started the code by telling my computer to pull out some tools from a package called `{socsci}`. Remember, packages contain groups of functions (specialized routines). In this case, I want to use a function called `frcode()` to help me turn numerical codes into qualitative categories, and this function isn't already in `{tidyverse}`. To install `{socsci}`, you'll need to run the following code:

```

install.packages("devtools")
devtools::install_github("ryanburge/socsci")

```

2.3.3 Some Simple Descriptive Summaries

Now that I have some cleaned up data, I can use R to tell my computer to do various summaries with the data. For example, I can ask it how many rows are in the data, which will also be a summary of how many unique militarized interstate events (MIEs) took place since each row is a unique event. I can do this by running the function `nrow()` on `dt`. According to the output, the answer is 28,011.

```
nrow(dt)
```

```
[1] 28011
```

I can also ask what time span the data covers. I can do this by running the `range()` function specifically on the `date` variable, which I can pull out of `dt` using a dollar sign (`$`). The first documented MIE took place in July of 1816, and the last happened in December of 2014.

```
range(dt$date)
```

```
[1] "1816-07-01" "2014-12-01"
```

I now can say something really important to know about MIEs. According to the dataset, 28,011 is the number of times that one country threatened to use force, showed force, used force, or did battle in a war with another country between 1816 and 2014 (a nearly 2 century-long period). In absolute terms, that seems like an awful lot of conflict (or the potential for it).

I can also ask my computer (through R) to perform even more detailed summaries. For example, I can ask how common different kinds of military actions took place, and order them by their frequency from highest to lowest. The below code will do just that, and as an added bonus, the `ct()` function that I use below (which comes from `{socsci}`) will also return proportions.

```
dt |>
  ct(action) |>
  arrange(desc(n))
```

```
# A tibble: 17 x 3
  action                n    pct
  <fct>                <int> <dbl>
1 Attack                11562 0.413
2 War Battle            4471 0.16
3 Show of force         4376 0.156
4 Clash                 2007 0.072
5 Border violation      1454 0.052
6 Threat to use force   1021 0.036
7 Seizure                694 0.025
8 Fortify border        649 0.023
9 Occupation of territory 643 0.023
10 Alert                 504 0.018
11 Mobilization          327 0.012
12 Blockade              148 0.005
13 Threat to declare war    65 0.002
14 Threat to occupy territory 53 0.002
15 Threat to blockade       31 0.001
16 Use of CBR weapons        5 0
17 Threat to use CBR weapons 1 0
```

Now I can say some other interesting things about international conflict. First of all, the most common type of MIE is an attack. From 1816 to 2014 these happened 11,562 times, accounting for more than 41% of all militarized actions by one country against another. The next most common kind of action is battle in an all-out war. From 1816 to 2014, a total of 4,471 war battles took place, accounting for 16% of all militarized actions during this period. Shows of force aren't far behind.

From there the frequency of events declines rapidly. The three least common kinds of actions are threats to blockade, use of CBR (chemical, biological, and radiological) weapons, and the threat to use CBRs. Only 31 blockade threats took place from 1816 to 2014, accounting for 0.1% of all militarized events. CBR use and threats are vanishingly rare. They were only used by one country against another 5 times (short of war) and only one country ever threatened to use them against another. With rounding error, these actions accounted for 0% of all events.

I can also ask my computer questions about the intensity of conflict. Each of the types of military actions is recorded in the MIE dataset as meeting different thresholds for intensity. These intensities are documented in the `hostlev` column in the data. I can run the following code to ask my computer tell me how common these hostility levels are:

```
dt |>
  ct(hostlev) |>
  arrange(desc(n))
```

```
# A tibble: 4 x 3
  hostlev          n    pct
  <fct>          <int> <dbl>
1 Use of force    15059 0.538
2 Display of force  7310 0.261
3 War             4471 0.16
4 Threat to use force 1171 0.042
```

Use of force (short of war) is the most common kind of event in the data, accounting for nearly 54% of all military actions by one country against another from 1816 to 2014. Displays of force were the second most common, but occur at only half the rate of uses of force. 7,310 displays of force are recorded in the data, and make up 26% of all events. War battles are even less common, with only 4,471 recorded, accounting for 16% of all events. Threats to use force are the least common. 1,171 in total took place from 1816 to 2014, accounting for only 4% of all events.

I can notch up the complexity of the summaries I ask for yet again. The data also has information about battle deaths. I can write R code that will tell my computer to summarize the minimum and maximum estimated fatalities from different kinds of events based on hostility level:

```
dt |>
  group_by(hostlev) |>
  summarize(
    fatalmin = sum(fatalmin1 + fatalmin2),
    fatalmax = sum(fatalmax1 + fatalmax2)
```

```

) |>
ungroup() |>
mutate(
  minpct = fatalmin / sum(fatalmin),
  maxpct = fatalmax / sum(fatalmax)
) |>
arrange(desc(fatalmin))

```

```

# A tibble: 4 x 5
  hostlev      fatalmin fatalmax  minpct  maxpct
<fct>      <dbl>      <dbl>   <dbl>   <dbl>
1 War      25737608  31477327  0.994   0.993
2 Use of force  163900    235873  0.00633 0.00744
3 Threat to use force    0         0  0        0
4 Display of force    0         0  0        0

```

This summary offers some really eye-catching headline figures. First of all, wars account for the greatest share of battlefield deaths from MIEs. This fact alone isn't surprising, but the disparity is. A low-end estimate of how many people died in MIEs that were battles in a war from 1816 to 2014 is 25.7 million. The high-end estimate is 31.5 million. Both of those figures in absolute terms are staggering, but they are in relative terms, too. Both the low- and high-end estimates account for more than 99% of all battle-related deaths from international conflicts over a two-century period.

Uses of force short of war can still be deadly though. While making up less than a percentage point of all battle-related deaths from MIEs, in absolute terms the totals are still large. On the low-end, 163,900 deaths from 1816 to 2014 can be attributed to uses of force short of war between countries. On the high-end, the figure is 235,873.

Threats and displays of force aren't deadly at all, which isn't a big shock. Neither involves using actual military force. Therefore, they aren't fatal.

2.3.4 Making Some Graphs of Data Summaries

Another thing you can do with your data is make graphs of your summaries. This process is called data visualization, and it involves taking tabular data or data summaries and representing them using geometry, colors, and other graphics.

The R programming language is a great way to interface directly with your computer to get it to produce nice looking graphs. The `{tidyverse}` way to do this is with `{ggplot2}`, which is in the universe of packages supported by `{tidyverse}`. That means you can start using its functions once you've written and run `library(tidyverse)`.

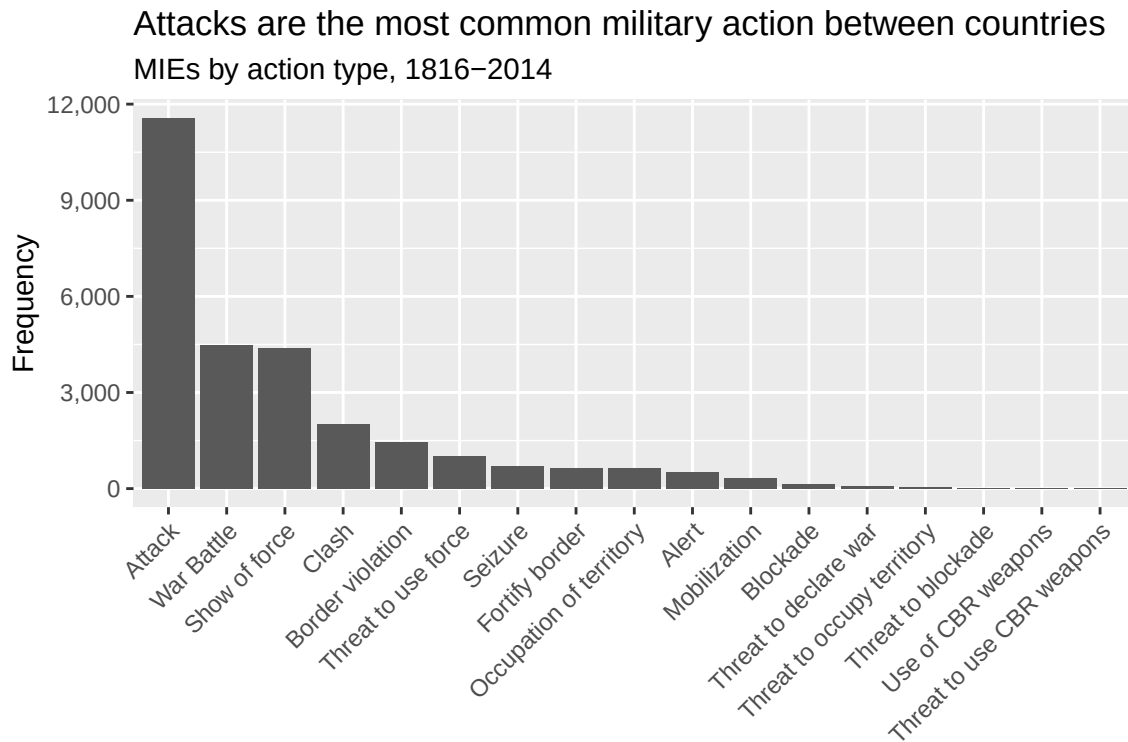
I'm not going to get too bogged down in the details of data visualization here. But if you want to know the whole range of things you can do in terms of data visualization, you can check out [some brief lecture notes](#) I've written for my 200-level class on research design, or [more detailed notes](#) I've written for a 100-level class I teach on data visualization.

I won't have you do anything too fancy with data visualization in this class, so don't think you need to master everything there is to know. I'll provide you with plenty of examples that you can borrow from and build on throughout this book.

Speaking of examples, it might be nice to have some decent looking graphs of the summaries I showed you in the previous section. Since these examples just involved counting things up, a nice data visualization choice is a simple bar/column plot. These let you summarize how some quantity of interest differs across different qualitative categories.

Here's how I might make such a plot using the summary for militarized actions:

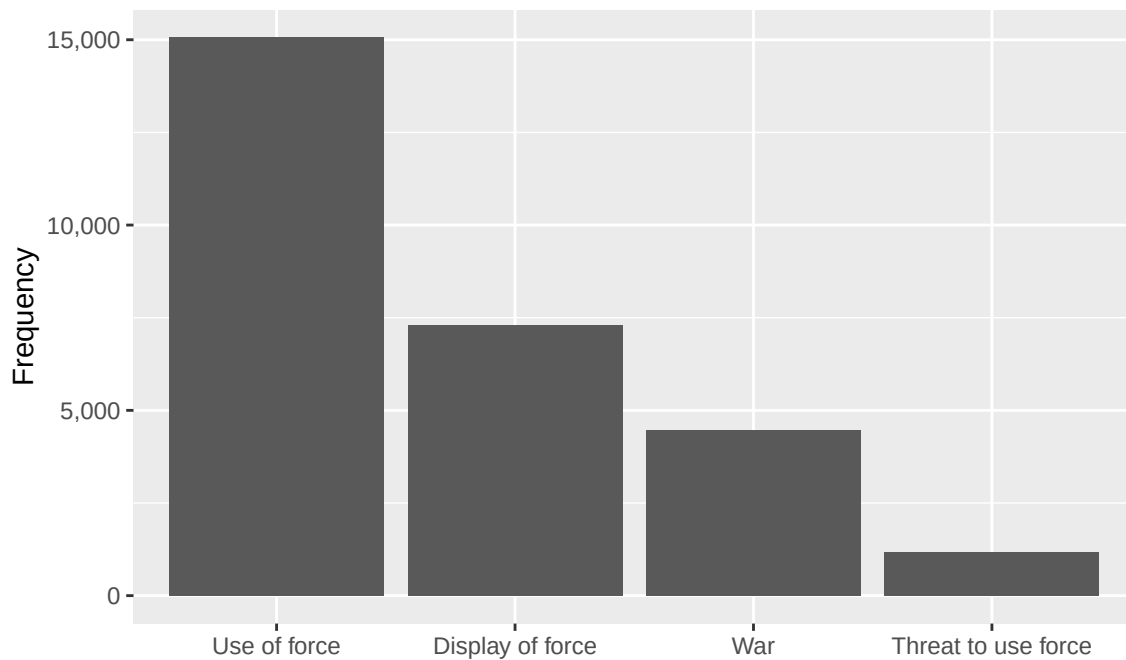
```
dt |>
  ct(action) |>
  ggplot() +
  aes(x = reorder(action, -n), y = n) +
  geom_col() +
  labs(
    title = "Attacks are the most common military action between countries",
    subtitle = "MIEs by action type, 1816-2014",
    x = NULL,
    y = "Frequency"
  ) +
  scale_y_continuous(
    labels = scales::comma
  ) +
  theme(
    axis.text.x = element_text(
      angle = 45, hjust = 1
    )
  )
)
```



Here's another one for hostility level:

```
dt |>
  ct(hostlev) |>
  ggplot() +
  aes(x = reorder(hostlev, -n), y = n) +
  geom_col() +
  labs(
    title = "Use of force is the most common hostility level between countries",
    subtitle = "MIEs by hostility level, 1816-2014",
    x = NULL,
    y = "Frequency"
  ) +
  scale_y_continuous(
    labels = scales::comma
  )
```

Use of force is the most common hostility level between countries
MIEs by hostility level, 1816–2014



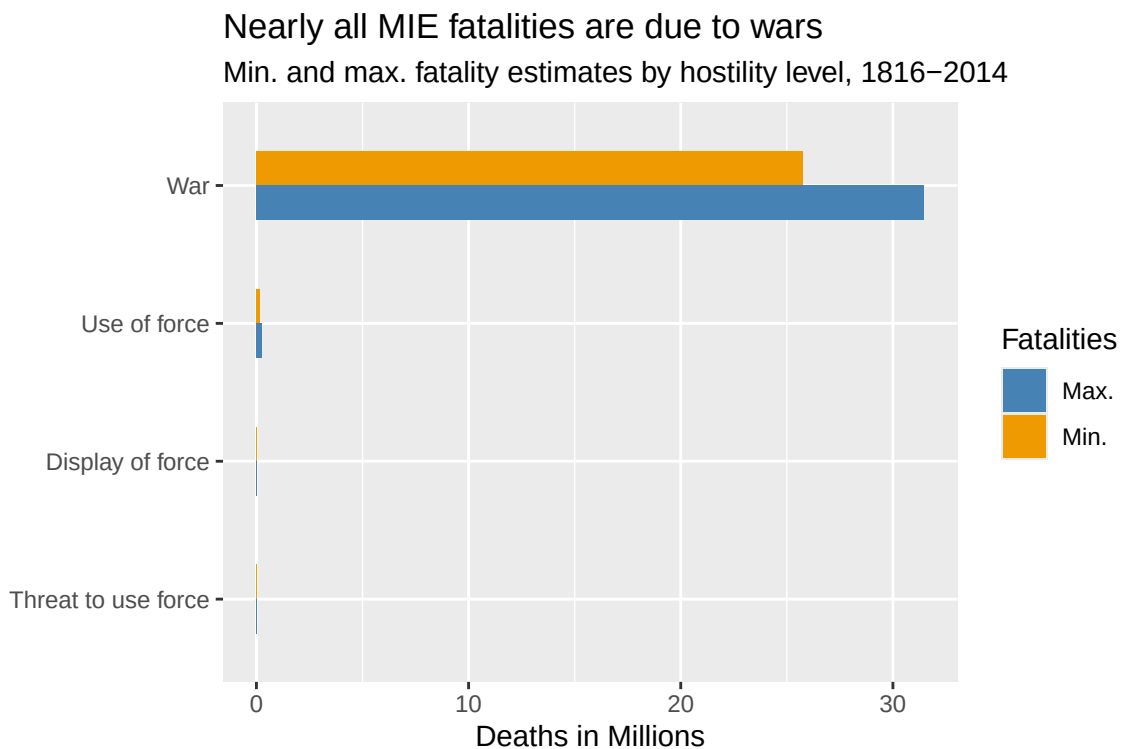
Finally, I can use a bar chart to show fatalities by hostility level:

```
dt |>
  group_by(hostlev) |>
  summarize(
    fatalmin = sum(fatalmin1 + fatalmin2),
    fatalmax = sum(fatalmax1 + fatalmax2)
  ) |>
  ungroup() |>
  mutate(
    minpct = fatalmin / sum(fatalmin),
    maxpct = fatalmax / sum(fatalmax)
  ) |>
  pivot_longer(
    fatalmin:fatalmax,
    names_to = "minmax",
    values_to = "fatal"
  ) |>
  ggplot() +
  aes(x = fatal, y = hostlev, fill = minmax) +
  geom_col()
```

```

width = .5,
position = "dodge"
) +
labs(
  title = "Nearly all MIE fatalities are due to wars",
  subtitle = "Min. and max. fatality estimates by hostility level, 1816–2014",
  x = "Deaths in Millions",
  y = NULL,
  fill = "Fatalities"
) +
scale_x_continuous(
  labels = ~ scales::comma(.x / 1e06)
) +
scale_fill_manual(
  values = c("steelblue", "orange2"),
  labels = c("Max.", "Min.")
)

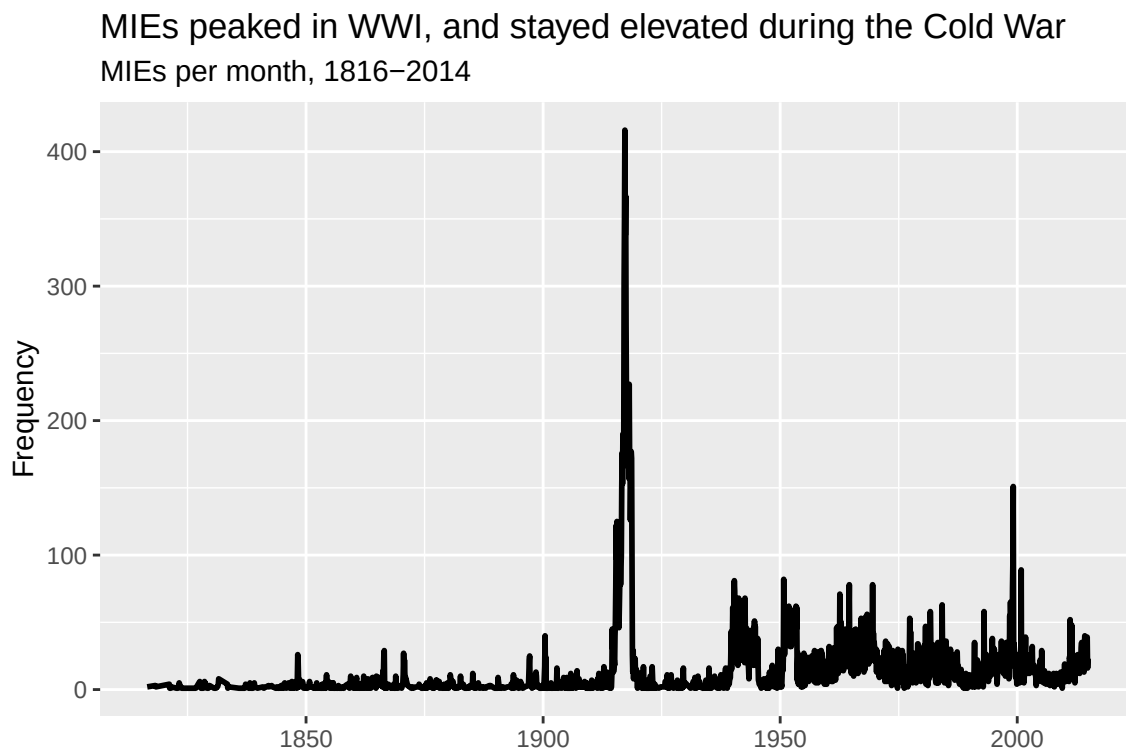
```



One more graph you can put together, which provides a nice segue to the next chapter, is to summarize the number of events by the `date` column in the dataset and then show the results

using a line plot. Line plots are a good way to show how some quantity varies over time. The below code shows you one way to do this. The results offer two interesting bits of trivia. First, World War I witnessed the highest number of unique militarized events per month than any other period of time from 1816 to 2014 (far more than even World War II). Second, the monthly frequency of events during the Cold War was elevated compared to previous periods, suggesting a higher baseline level of conflict relative to the previous century.

```
dt |>
  ct(date) |>
  ggplot() +
  aes(x = date, y = n) +
  geom_line(linewidth = 1) +
  labs(
    title = "MIEs peaked in WWI, and stayed elevated during the Cold War",
    subtitle = "MIEs per month, 1816-2014",
    x = NULL,
    y = "Frequency"
  )
```



In the coming chapters I'll talk more about measuring conflict occurrence and deadliness over time, and I'll use ggplot to show these trends. As I said before, I'll also provide you with

plenty of examples that you can build on to suit your goals.

2.4 Summary

The Militarized Interstate Events (MIE) dataset is going to serve as the foundation for all subsequent analysis we do in this class. This is a high quality dataset that ostensibly corrects thousands of documented errors in other commonly used datasets by peace scientists (such as the COW MID dataset) and offers some helpful innovations (like keeping events in the data even after escalation to war).

I provided some helpful code for accessing the data, cleaning it, and doing some interesting data summaries. In the chapters that come, I'll provide you with additional resources that help to automate the process of accessing and cleaning the data so you can spend more time thinking about analysis choices and what the data actually says.

3 Is War Disappearing?

Main ideas:

- The decline of war thesis is the argument that conflict is becoming less common and deadly, and that this trend is decades (if not centuries) old.
- A well-known criticism of this position was put forth by the political scientist Bear Braumoeller, first in a well-circulated conference paper (Braumoeller 2013a), and then in a well-received book that fleshed out his criticisms even more (Braumoeller 2019).
- Braumoeller raises many concerns with the decline of war thesis, but two key issues for him were: (1) paying attention to what quantities are counted over time and (2) accounting for the role of chance in war and peace.
- What you count over time concerns not just the numerator (conflict) but the denominator (how you adjust for changes in opportunities for countries to fight).
- Doing statistical inference matters because changes can occur over time that *look* like real changes, when in fact they could just as easily be the result of random chance.

3.1 The Decline of War Thesis & Why It Matters

In this section, I plan to introduce you to the *decline of war thesis*. But before I do, I have a meandering anecdote for you...

I first learned about the decline of war thesis when I read Bear Braumoeller's (2019) book, *Only the Dead: The Persistence of War in the Modern Age*. I was starting my third year as a Ph.D. student, and I learned that Prof. Braumoeller would be visiting my school, the University of Illinois at Urbana-Champaign, to give a talk at the political science department about his book.

I also learned that I'd been volunteered to pick him up from his hotel to take him to campus. I wanted to be prepared when I met Braumoeller for the first time, so I bought and read his book before he flew in from Columbus, OH where he was a professor at *The Ohio State University*. I think I read the whole thing, cover to cover, in just three days (including the R code in the book's appendix for some of the analyses he did).

I devoured the book this quickly for two reasons, neither of which were a pathological need to appear like a smart graduate student to a well-known and respected scholar (or so I tell myself).

First, Braumoeller's previous book, *The Great Powers and the International System: Systemic Theory in Empirical Perspective* (2013b), solidified my desire to study international relations for a living. He published it in 2013, and I read it in 2016 while working on my master's. I had been wondering what would come next once I graduated. Reading Braumoeller's book made my path very clear: I wanted to get my Ph.D. and do what Braumoeller was doing. I've been a fan of his work ever since. So, when I learned that the scholar whose work inspired me to go down the path of getting a Ph.D. to study international relations had just published another book, I couldn't get my hands on it fast enough.

Second, with this new book (much like his first) Braumoeller sought to answer a *big* question in international relations: *is war going away?* Braumoeller had a reputation for doing incredibly smart and cutting edge quantitative work in international relations. If he had applied this same aptitude to one of the most important issues in the field, I had to know what he found.

Needless to say, I had really high expectations going into my first meeting with Braumoeller. That meant I also had a lot of nerves. Because of those nerves, I never directly asked him questions about his book while I drove him from his hotel to UIUC's campus. Also, I thought I'd appear far too creepy expressing how inspiring his work was to me, so I never mentioned it. He asked me some questions about my own research interests, and I think my brain fell out of the back of my head when he did, because I neither sounded eloquent nor coherent as I explained the volatile brew of interconnected research questions percolating in my head.

I kick myself every time I think back on this experience. I met my hero, and I was too star-struck to make the most of it.

So, when I learned that I got a job at Denison, and that the suburb I would live in was the same that Braumoeller lived in (I promise this wasn't on purpose), I thought I could try again. Instead, I got a tough lesson in not letting fear or nerves get in the way of the opportunities life gives you. When I first moved to the area, I learned Braumoeller was overseas on an extended leave doing further research, building on the work he did for *Only the Dead*. And then I learned some tragic news. In the summer of 2023 he unexpectedly passed away while abroad.

I was shocked, as was seemingly the whole field of international relations and even those in political science more generally. I think nearly the whole discipline new him in some way. I also felt incredibly sad, because he had a wife and young daughter that he'd be leaving behind.

I didn't intend to stumble into some "real-talk" when I sat down to write this chapter, but you never truly know the thoughts working themselves out in your mind until you start writing. Consider this detour a nudge to meditate on the value of courage and of taking advantage of the chances life gives you. This nudge comes from a (*still*) young man who's unfortunately already old enough to have accumulated some regrets in life. The experience that fueled this

detour (and many others) serve as my own reminders that a little courage will take you a long way.

As off topic as that story may seem, I think it helps me make a key point about studying war.

It's this: studying war with data has an ethical valence to it. Yes, the goal of a peace scientist is to be objective and quantitative about studying war, but because the object of this study is *war*, a life-and-death matter, it should be taken seriously. Above all, I think it requires some *courage* (I knew I could link this back to my anecdote in some way). To make claims about war, using data, is to make claims about the "iron dice" that, when rolled, may lead to no bloodshed and a quick resolution, or to thousands or (in rare cases) millions of deaths. You should make sure that you're serious about the claims you make about war, but you also need to have the courage to make said claims if you think you're right.

In the next chapter I'll talk about how war size can be incredibly volatile and hard to predict, which makes the content of this chapter that deals with the onset of conflict all the more important. If countries are more willing, or less, to roll the iron dice today than in the past, this has existential implications for human lives.

The moral culpability of peace scientists enters the equation when considering how their findings can affect foreign policy and international politics. By making a claim that conflict is becoming rarer, more frequent, or as common as ever, they are making a policy recommendation whether they intend to or not. If the world is getting more peaceful, why should countries continue to invest in international institutions that help prevent conflict? If the world is getting more violent, shouldn't major powers double-down on building up their military might, potentially diverting funds from domestic priorities?

I don't mean to imply that the arc of international relations pivots on the conclusions drawn from a single data analysis. But the goal of data analysis for many social scientists (including those in the peace science tradition) is to make convincing arguments using data. If enough people in a field are convinced of an idea, it can start to catch the attention of practitioners operating in the policy world. Even very niche ideas can attract the attention of a decision-maker. So, while minuscule, a single analysis is part of the body of knowledge and claims accumulated about a subject, and for that reason, it matters.

This, in sum, is why the *decline of war thesis* deserves your attention. It is the argument, founded on a variety of data sources and theoretical propositions, that international conflict is becoming less likely *and* less deadly. Put another way, it's the claim that countries are less apt to roll the iron dice today than decades or even a century ago, and that when the iron dice must roll, they are far less likely to land on another world war. You can imagine that an idea like this catching on can have tangle implications if enough people in the policy world think it's true.

If true, this is great news. But if it's false, it could be a "self-defeating prophesy" (the opposite of its equally devious twin of the self-fulfilling sort). Therefore, getting the decline of war issue right matters.

Braumoeller (2019) disagreed with the decline of war thesis for a variety of reasons, including its self-defeating implications, but in the next section I want to introduce just two of his data-driven critiques, because these will be the basis for the applied examples that follow. Importantly, his data-driven critiques do not on their own disprove the decline of war thesis; they deal with methodological rigor rather than substantive findings. But, Braumoeller claimed that when he applied appropriate measures and methods, the argument in favor of the decline of war thesis looked much less convincing.

3.2 Braumoeller's Critique

Many researchers fall into the decline of war camp, but Braumoeller directs the brunt of his critique at the Harvard Psychologist Steven Pinker who wrote two important books that made their way into the public discourse on war. The first was titled *Better Angels of Our Nature* (Pinker 2012), and the second was titled *Enlightenment Now* (Pinker 2018). In the first, Pinker makes a data-driven case that international war is on the decline and has been for centuries. In the second, Pinker builds on this argument and claims that the Enlightenment movement from the 17th and 18th centuries helps explain why war is going away. He makes the additional argument that the development of modern nation-states also played a role in fostering peace due to their "civilizing" effect on societies.

Braumoeller takes issue with both the data Pinker used, and his theoretical rationales for why war is declining. With respect to the role of the nation-state in promoting peace, Braumoeller notes that the historical record is a mixed bag. On the one hand, when smaller polities organize themselves into larger states, there's more internal peace. On the other hand, states can mobilize for war on a scale that small polities never could. States also have the capacity to commit mass atrocities (think of the Holocaust, or Stalinist USSR).

Braumoeller similarly argues that the Enlightenment is a mixed bag. Some ideas, like valuing reason and respect for rights, are good, but some Enlightenment ideas have borne some rotten fruit. Karl Marx was just as much a product of the Enlightenment as Immanuel Kant, and many atrocities were committed in the name of Marx's ideas, especially in the 20th century. But even seemingly less controversial ideas, like respect for rights, can become a basis for war. The Responsibility to Protect (R2P) was a policy supported by the United Nations (driven in part in response to tragedies like the Rwandan Genocide). Endorsed in 2005, it upholds the idea that if the government of a country is committing human rights violations or mass murder, other countries have a responsibility to intervene, *which* (counterintuitively) *can include using military force*.

But these theoretical complaints about the decline of war thesis aren't the focus of this chapter. Independent of these issues, Braumoeller raises two important methodological concerns about the analysis used to support the decline of war thesis.

The first is that many prior analyses (including Pinker's) measured conflict *frequency* over time, when in fact they should have measured the conflict *rate*. Frequencies are just raw counts, while rates are counts *per* some other quantity of interest. For example, GDP *per* capita is GDP divided by a country's population. For Braumoeller, only counting conflicts over time misses half of the equation. As I'll discuss more in the next section, Braumoeller argues that you need to divide conflict counts by the number of possible chances countries have to fight each other, which doesn't remain constant over time.

The second is that many prior analyses (again, including Pinker's) don't rely heavily, or at all, on statistical inference. Researchers use statistics to quantify the role of random chance in explaining patterns in data. In any variable that goes up and down over time, some of the hills and valleys might indicate a truly new trend in said variable. They could also just be the product of perfectly natural random fluctuations. If you don't account for the role of chance, you might draw overly optimistic (or pessimistic) conclusions from data. I'll discuss this more in the section after the one below.

3.3 What You Count Matters

In the last data analysis example I showed you in the previous chapter, I produced a monthly count of the number of militarized interstate events (MIEs) over time. While this plot provides some interesting fodder for conversation, it doesn't pass Braumoeller's test for a good measure of conflict occurrence. It only shows the frequency of conflict events; not the rate. Further, Braumoeller would probably take issue with counting all the events in the data.

To measure conflict over time to test the decline of war thesis, you need a measure that accurately encodes two important quantities: (1) how often countries are willing to "roll the iron dice," and (2) how often countries have a realistic chance of reaching each other to fight. If you have both quantities, you can then calculate the conflict rate.

It's probably a good idea to express this more formally, which (sorry) means I need to throw a little math your way (it's sums and fractions, so don't sweat it too much). Let me use P to stand for the conflict rate, or what I could also call the *propensity* for war. Further, I'll use I to stand for conflict *initiations*, and O to stand for *opportunities*.

To calculate P for a given point in time t , I need to measure I and O at time t . Theoretically, both are simple to get. I_t is the sum of all conflict initiations at t , and O_t is the sum of all conflict opportunities at t :

$$I_t = \sum_{i=1}^N I_{it}, \quad O_t = \sum_{i=1}^N O_{it}$$

In the above, I_{it} is an indicator that is 1 if a country decides to roll the iron dice against another; 0 otherwise. O_{it} is between 0 and 1, and its value represents whether a country can reach another to initiate a fight. The i subscript represents a unique *pair* of countries. In the field of international relations, we call this a *dyad*.

Once you have I_t and O_t , propensity for war is as simple as dividing the former by the latter:

$$P_t = \frac{I_t}{O_t}$$

That math looks temptingly easy, making it dangerous in the hands of someone who doesn't yet know what they're doing. The real challenge is that you, the researcher, need to make some judgment calls about how to get I_t and O_t . Should you count every instance of conflict or only the first instance between a pair of countries? Should threats or shows of force count, or only at minimum when a country decides to use actual military force? How should you measure the opportunity to fight? Maybe by whether countries border each other? But don't some powerful countries have the ability to send their armed forces anywhere in the world? How should you handle that?

Let me give you some possible answers these questions below.

3.3.1 Measuring Incidents

First of all, in his own analysis, which includes his conference paper (2013a) and book (2019), Braumoeller used the Militarized Interstate Disputes (MID) dataset, which at the time his book was published was in its fourth major iteration. Now, I just spent part of the last chapter talking about issues with the MID dataset, which the MIE (Militarized Interstate Events) dataset that we'll use in this book claims to fix. At the time he did his analysis, though, the MID data was the best available. Pinker (2012) used a variety of other datasets in his own analysis to give him longer historical coverage, but many of these have not seen widespread use in the peace science community due to concerns about data quality. Compared to everything available, the MID dataset was best-in-class when Braumoeller used it.

Braumoeller didn't just use the MID data because of its superior quality over other options; he also believed that the *concept* of the MID did the best job of capturing the willingness of countries to roll the iron dice. More specifically, he thought the onset or *initiation* of a MID should be what counts. Anything beyond the initiation of a MID he treated as difficult to predict *ex ante*. Recall that a MID reflects any threat to use force, display of force, or use of force short of war by one country against another. To him this seemed as good a way as any

to empirically verify the willingness to risk war. If you recall the discussion in Chapter 1, the definition of a MID comes close to approximating Carl von Clausewitz’s definition of “real” war (that is, a continuation of politics by other means).

However, there still is some debate over which MID-like cases should be counted. On the one hand, it makes sense to count everything, including threats and displays of force short of actual violence, because acting threatening is a form of aggression meant to *communicate* resolve and power. And, remember, that use of military force to communicate information about strength and the will to fight is an important part of the Clausewitzian way of understanding war. Such threats can also backfire.

On the other hand, it makes sense to ignore any event short of use of force because use of force is the only action that empirically demonstrates a willingness to fight. Threats and displays of force might just be “cheap talk,” but using force reveals that a country will accept bloodshed to achieve its goals. Uses of force also risk reciprocation from the target far more than a threat. If you want to understand the risk of deadly war, you should focus on uses of force where the chance for escalation seems more likely.

Both of these views have merit, which is why Braumoeller (consistent with the practice of many other peace scientists) used the more liberal approach of counting everything, and the more conservative one of ignoring threats and shows of force. The idea is to use both, since each has its merits, and see if it makes a big difference in the results. In many cases, it turns out the this choice actually doesn’t matter.

So how can you measure the yearly frequency of conflict incidence using the MIE dataset that I introduced in the last chapter? In short, it’s going to require some work to whittle out irrelevant events.

Let me show you a sensible way to proceed. First, I need to get the data, which I do in the below code. Note that the code is a bit different than before. I promised you that I’d provide some tools that make getting and cleaning the data easier, and now I’m delivering on that promise. The below code, like what I showed you before, opens up the packages I need in my R session (`{tidyverse}` and `{socsci}`), but then I use a function called `source()` to which I give a link for a .R file on my GitHub. This is an R “script” file (essentially one giant code block saved as a separate file). This script contains code that creates a function called `get_mie_data()` that performs a straightforward routine when I use it: it reads in the MIE .csv file from my GitHub and then implements the recodes I showed you in the last chapter before returning the data. This is a major time-saver and makes my code easier to read. After running the `source()` function on my R script I now can use the function called `get_mie_data()` and then save the output as an object (in this case, one called `dt`).

```
## set up
library(tidyverse)
library(socsci)
source("https://raw.githubusercontent.com/milesdwilliams15/death-destruction-data/refs/heads,
```

```
## get the cleaned up data
get_mie_data() -> dt
```

Now that I have the data, I can start working on a measure of conflict incidents over time. While the data has a `date` variable that documents the unique month and year a conflict starts, most conflict research aggregates data to the year, so I'll do the same. This approach has some practical advantages, like the fact that most variables people want to use to study conflict are only at the year level, and the fact that with nearly 200 years worth of data, monthly variation isn't quite as important as yearly variation.

But what, exactly, should I count per year? To capture what Braumoeller was after, I need to count up two quantities: (1) the total number of unique conflicts documented in the data—not *events*—and (2) the total number of unique conflicts that at minimum rise to the level of a use of force. Remember that the MIE dataset documents each an every event associated with a conflict (or *confrontation*, to use their terminology). This makes for a really rich dataset, but for the purpose of simply measuring war propensity, I need to cut some fat.

The below code lets me extract the relevant quantities from the data. I start by grouping the data by `micnum` (the unique confrontation, or MIC, identifier). I then get the first year for which the MIC appeared, and the maximum hostility level observed. Next, I have to group the data by year to get a count of the unique number of MICs that occurred per year, and the sum of MICs that at least rose to the level of use of force. Finally, I use the `complete()` function to fill in gaps in the data (years that didn't witness any MICs) with zero values. I then save the output as a new object called `incident_dt`. I annotated the code so you can see where each step takes place.

```
dt |>
  # for each MIC get the start year and max hostlev
  group_by(micnum) |>
  summarize(
    year = min(year),
    hostlev = max(hostlev)
  ) |>
  # for each year get total MICs and only MICs with >= use of force
  group_by(year) |>
  summarize(
    incidents_any = n(),
    incidents_force = sum(hostlev >= "Use of force")
  ) |>
  # complete the series by filling in years without MICs with zeros
  complete(
    year = full_seq(year, 1),
```



```
fill = list(incidents_any = 0, incidents_force = 0)
) -> incident_dt # save as incident_dt
```

If you print out the first 10 rows of the data, you can see that I now have a simple dataset. Each row is a year from 1816 to 2014, and for each year I have two counts of conflict incidents—one that counts every conflict regardless of its hostility level, and another that counts only conflicts that at minimum involved use of force. In short, I have everything I need for the numerator in the equation for war *propensity*, but that still leaves the other half of the equation.

```
slice_head(incident_dt, n = 10)
```

```
# A tibble: 10 x 3
  year incidents_any incidents_force
  <dbl>         <int>         <int>
1  1816             1             1
2  1817             0             0
3  1818             1             1
4  1819             0             0
5  1820             0             0
6  1821             4             2
7  1822             1             1
8  1823             0             0
9  1824             0             0
10 1825             0             0
```

3.3.2 Measuring Opportunities

To accurately measure war propensity, a simple count of yearly conflict won't do. I need to know how common new conflicts are relative to the opportunities that exist for new conflicts to arise.

The problem (you knew there'd have to be a problem, right?) is that there isn't one universally accepted definition of what counts as an opportunity. However, the basket of options is, thankfully, limited.

The first option is to naively assume that all countries in the world can reach each other with equal ease, no matter geography or ability. If you want to make this assumption, your measure of opportunity is straightforward: just count up the total number of unique country pairs (*dyads*) that exist in the world at a given point in time. Put another way, assume that for a single country, like the United States, any other country in the world is fair game. This means that the U.S. has as many opportunities to go to war as there are countries in the

world. Multiply that total by a count of all countries in the world and there's your measure of opportunity.

You can probably see that this approach has some conceptual problems, however. For argument's sake, assume that the U.S. really can fight any other country it wants. Can you say the same of Zimbabwe? Or Jamaica? What about Switzerland? It seems silly to assume that any other country in the world is fair game for every country in the world.

This takes me to the second option, which is on the opposite end of the spectrum. One idea, which Braumoeller (2013a) explored in his conference paper, is to only count country pairs or dyads that are *contiguous*. This term just means that countries border each other (though some define it more broadly to include countries that are separated by no more than 400 miles of water). This definition of an opportunity to fight rules out the vast majority of other countries in the world for each individual country.

While this approach seems sensible for some cases, it definitely doesn't work in every case. For example, this approach would say that the U.S. has only three opportunities for war: Mexico, Canada, and Russia (Alaska is only separated from it by 50 miles of water). It should go without saying the U.S. has many more opportunities for war, as recent events attest.

There is, thankfully, a middle ground between counting everything and only counting contiguous country pairs. The solution that most peace scientists use today involves a compromise: for some countries, only contiguity applies, while for others, any other country is on the table. The countries for whom every other country in the world is a possible target are called "major powers." These are countries that, in theory, have sufficiently powerful armed forces and resources to project their military might anywhere in the world. At the moment, this list includes countries the United States, China, and Russia. This measure is known as *politically relevant dyads* (PRDs).

Admittedly, this solution also has some issues because most sensible people would agree that the U.S. has very different capabilities than Russia. Nonetheless, it's probably better than the alternatives. However, there is another option, which, funnily enough, also comes from Braumoeller along with his colleague Austin Carson (2011).

Their approach bases opportunities on quantities generated from a sophisticated statistical model that uses major power status, contiguity, and the distance between countries to quantify the probability that one country could fight another (assuming it wanted to). The difference with this measure compared with the alternatives is that opportunities are probabilistic rather than deterministic. It also means that even geography can be a limiting factor for major powers. While the U.S. can certainly send its forces anywhere it wants, it is logistically harder to do so in Iran than in Venezuela. To keep things concise, I'll just call this measure of opportunities *Braumoeller-Carson dyads* (or BCDs).

There are issues with this measure, too. For instance, it still treats major powers equally, and it assumes that distance means the same thing to the U.S. that it does to Bolivia. However, it's probably a stronger option than the other three, but given the prevalent use of the the

previous one (PRDs), it's wise to use both it and the Braumoeller-Carson measure to see if using one or the other makes a big difference.

This hard to do without data, of course, so with some measures of opportunities identified (PRDs and BCDs) the next step is to actually get some data. Thankfully, because of the hard work of the peace science community, the relevant data and tools to access it only require writing a few lines of R code. I also did some leg work by creating another R script file that gives you a pre-canned routine for generating the data based on these other tools. All you need to do is run the `source()` function on the url address for the relevant script saved on my GitHub, and then you can use the function `get_opportunity_data()` to access yearly counts of conflict opportunities.

```
source("https://raw.githubusercontent.com/milesdwilliams15/death-destruction-data/refs/heads/main/get_opportunity_data.R")
get_opportunity_data() -> opp_dt
```

```
Joining with `by` = join_by(ccode1, ccode2, year)`
```

If you take a peek at the object `opp_dt` created by the above code, you can see what's in the data. For each year, it counts up opportunities using all four of the methods outlined above (I wanted to be comprehensive, even though I don't recommend using all four). By looking across the rows you can see that these methods give very different values. As you'd expect, `dyads` (which counts every possible pairing per country) is the most liberal count, while `contdyads` (which only counts contiguous pairings per country) is the most conservative. Both the PRD and BCD measures are in the middle, with the latter a bit more conservative, which isn't surprising since BCD is probabilistic rather than deterministic, and it also accounts for the distance between countries. For this reason, the BCD sum per year isn't a whole number. Unlike the other measures, a pair of countries isn't a 0 or 1, but something in between.

```
slice_head(opp_dt, n = 10)
```

```
# A tibble: 10 x 5
  year dyads contdyads prd bcd
  <dbl> <int>    <dbl> <dbl> <dbl>
1  1816   506      120   248  199.
2  1817   506      120   248  199.
3  1818   506      120   248  199.
4  1819   506      120   248  199.
5  1820   506      120   248  199.
6  1821   506      120   248  199.
7  1822   552      120   258  205.
```

8	1823	552	120	258	205.
9	1824	552	120	258	205.
10	1825	600	126	272	215.

3.3.3 Measuring War Propensity

Now that I've made a dataset of new conflict incidents per year, and another of conflict opportunities per year, I need to combine them together so that I can construct a measure of war propensity. This step is easy, because propensity is just the sum of incidents divided by the sum of opportunities.

I do, however, have alternative measures for these different concepts. I have two ways of quantifying incidents, and another two sensible ways of quantifying opportunities. That gives me four alternative measures of war propensity.

To create all four, I first need to combine the incidents and opportunity datasets together, which I can do easily with tools from the `{tidyverse}`. The below code uses a function called `full_join()` that joins the datasets together based on matches in the `year` column in each:

```
full_join(
  incident_dt,
  opp_dt,
  by = "year"
) -> final_dt
```

I saved the combined data as an object called `final_dt`. If I print out the first 10 rows, I can confirm that the datasets merged together as intended:

```
slice_head(final_dt, n = 10)
```

```
# A tibble: 10 x 7
   year incidents_any incidents_force dyads contdyads prd bcd
  <dbl>         <int>         <int> <int>      <dbl> <dbl> <dbl>
1  1816             1             1   506      120   248  199.
2  1817             0             0   506      120   248  199.
3  1818             1             1   506      120   248  199.
4  1819             0             0   506      120   248  199.
5  1820             0             0   506      120   248  199.
6  1821             4             2   506      120   248  199.
7  1822             1             1   552      120   258  205.
8  1823             0             0   552      120   258  205.
9  1824             0             0   552      120   258  205.
10 1825             0             0   600      126   272  215.
```

The next step is to modify the data by including columns for the four possible measures of war propensity. You can do this with the `mutate()` function, as I do below. This code will add four new columns to the dataset called `prop1` up to `prop4` that are made by taking one of the incidents measures and dividing by one of the opportunity measures. Now, you have all the data you need.

```
final_dt |>
  mutate(
    prop1 = incidents_any / prd,
    prop2 = incidents_force / prd,
    prop3 = incidents_any / bcd,
    prop4 = incidents_force / bcd
  ) -> final_dt
```

3.3.4 Looking at the Trend

At long last, we have some data to measure variation in war propensity over time. So let's take a look at the results.

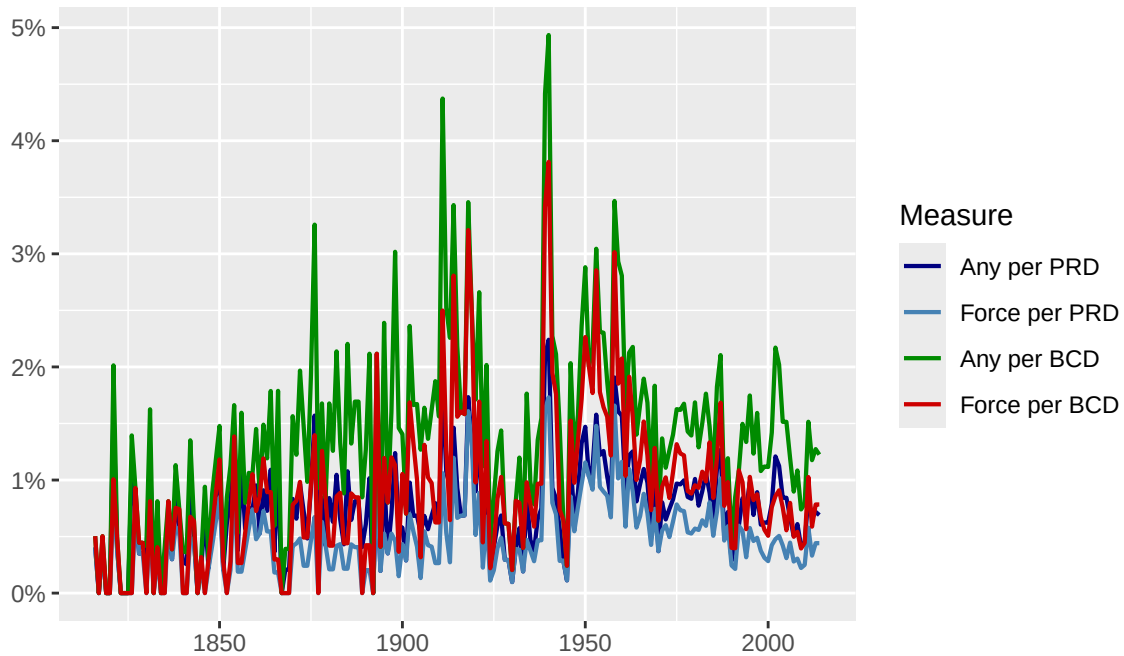
The below code will create a line plot showing, per year from 1816 to 2014, war propensity based on each of the four measures created in the previous section. As you can see, the measures aren't identical, but they do move in parallel fairly well, which tells me the choice of measure may not have too much of an impact on the conclusions you draw about conflict over time.

```
final_dt |>
  pivot_longer(prop1:prop4) |>
  ggplot() +
  aes(x = year, y = value, color = name) +
  geom_line(linewidth = .75) +
  scale_color_manual(
    values = c("navy", "steelblue", "green4", "red3"),
    labels = c("Any per PRD", "Force per PRD", "Any per BCD", "Force per BCD")
  ) +
  scale_y_continuous(
    labels = scales::percent
  ) +
  labs(
    title = "All four measures of war propensity move in similar directions",
    subtitle = "% war propensity, 1816-2014",
    x = NULL,
    y = NULL,
```

```
color = "Measure"
)
```

All four measures of war propensity move in similar directions

% war propensity, 1816–2014



This just leaves us with one final question: does the data support the decline of war thesis? Stated differently: are conflicts becoming less common over time?

One way to answer this question is to simply look at the results in the above figure. This is what folks like Steven Pinker did, so why not do the same? However, as I'll discuss in the next section, looking at the data is too subjective an approach. It's much better to rely on statistical inference to help separate random noise from true signals in the data.

3.4 Accounting for Random Chance Matters

Getting the right data and the right way to measure conflict over time is essential for testing the decline of war thesis, but for Braumoeller, this doesn't go far enough. His other methodological critique of past work on the decline of war is the failure to account for the role of chance in war and peace. Statistical analysis can help solve this problem.

There are many ways of defining statistics, but I think the most helpful is to think of it as a set of tools for reasoning about, and quantifying, the role of random chance in generating data.

Imagine that when you measure something in the world, the final number that you record is the product of two things: (1) a signal and (2) some background noise. Statistics provides a way to quantify on average how much the recorded estimate might be expected to vary from one case to the next due to background noise.

Beyond quantifying noise, when doing statistical analysis, researchers will apply certain thresholds of random variation to judge whether background noise can be rejected as the best explanation for a change or difference observed in data. These thresholds are called the *level of the test*, and they are pivotal to the practice of *hypothesis testing*. The idea is simple:

1. Start from the assumption that differences or changes you observe in data are the result of background noise.
2. Estimate a quantity, or quantities, of interest (a difference, a change, maybe multiple estimates).
3. Calculate how much you expect the quantity of interest to vary due to random noise.
4. Reject the hypothesis that background noise explains the estimates you observe if a difference is bigger than a pre-specified threshold of random variation.

If you've never taken a statistics course before, don't worry. R will do all the math for you. If you can at least wrap your head around the point of doing statistical analysis, that will take you a long way.

Let me help you out by showing you a quick example with the conflict dataset I put together in the last section (`final_dt`). Say I wanted to test whether the average conflict rate from 1820 to 1849 is statistically the same as the average conflict rate from 1850 to 1879. I would need to first calculate the observed average conflict rate in each of those periods, then I would need to use statistical tools to calculate intervals around those averages that capture how much variation I would see in the estimated average due to background noise. I can set the level of these intervals based on how high I want the threshold to be for ruling out background noise as the best explanation for any differences I observe between periods.

The below code shows how I might go about doing this using the first war propensity measure (all new confrontations per PRDs). As I told you, I'll provide a lot of tools to streamline the process of doing analysis, and this example is no exception. It starts by reading in a script file that creates a function called `mean_ci_boot()`, which does two things. First it calculates a mean for some numerical variable, then it computes intervals for that mean that reflect how much it might vary as a result of background noise. The method it uses to do this is called [bootstrapping](#), which is a computational method for simulating the distribution of some estimate as a result of random noise. It then uses this distribution to calculate upper and lower bounds for *confidence intervals*. By using the `ci =` option I can control how wide I want these intervals to be. In this case, I set them to .84 (on a scale from 0 to 1) which means it

will return 84% confidence intervals. Intervals of this size are the most common level chosen for checking if two quantities are different from each other based on whether the intervals for the quantities overlap. (There's another interval that you might be more familiar with, the 95% confidence interval, that will be relevant in the second part of this book).

```
source("https://raw.githubusercontent.com/milesdwilliams15/death-destruction-data/refs/heads/main/data/death-destruction-data.csv")

final_dt |>
  filter(year %in% 1820:1879) |>
  group_by(year < 1850) |>
  summarize(
    mean_ci_boot(prop1, ci = .84)
  )
```

```
# A tibble: 2 x 4
  `year < 1850`    mean    lower    upper
  <lg1>          <dbl>    <dbl>    <dbl>
1 FALSE          0.00715 0.00627 0.00804
2 TRUE           0.00466 0.00365 0.00570
```

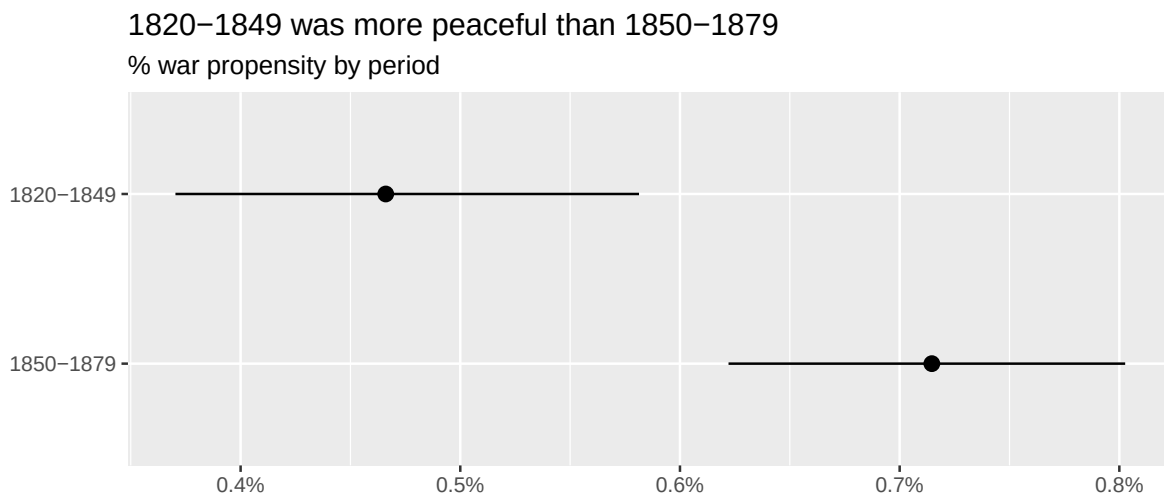
If you look at the results, it seems that the average conflict rate from 1820 to 1849 was *lower* than the rate from 1850 to 1879, and that this difference is *statistically significant*. You can tell because the upper bound for the confidence interval for the pre-1850 average is less than the lower bound of the post-1850 average.

This is much easier to see by getting R to produce a figure based on the results, which the below code does:

```
final_dt |>
  filter(year %in% 1820:1879) |>
  group_by(year < 1850) |>
  summarize(
    mean_ci_boot(prop1, ci = .84)
  ) |>
  ggplot() +
  aes(x = mean, xmin = lower, xmax = upper, y = `year < 1850`) +
  geom_pointrange() +
  scale_x_continuous(
    labels = scales::percent
  ) +
  scale_y_discrete(
    labels = c("1850-1879", "1820-1849")
  )
```



```
) +
labs(
  title = "1820-1849 was more peaceful than 1850-1879",
  subtitle = "% war propensity by period",
  x = NULL,
  y = NULL
)
```



To test the idea more generally that conflict over time has changed in an upward or downward direction, all you have to do is take the concept applied above and extend it to different periods in time, which is essentially what Braumoeller did in his own analysis; though, with a slightly different method. However, to keep things simple, let's just extend the above analysis to include the whole range of years in the dataset.

The idea with the above example is to bin the data into discrete periods in time and then see if the average of war propensity in one period is statistically different than what it is in another. You can use bigger or smaller bins, which will change the results somewhat. Obviously, smaller bins will show more variation from one point in time to the next (but they also will be subject to more background noise). Conversely, bigger bins will give you more stable results (and be less subject to background noise), but this will come at the cost of glossing over potentially important and meaningful variation across shorter time intervals.

Let me show you an example that breaks it down by 20 year intervals. I left some annotations in the code so you can see what is going on in the different steps. Essentially, it lumps the data into 20-year intervals and creates some sensible looking labels for each period. Then it calculates the average war propensity per period along with 84% confidence intervals using the first measure of war propensity (any new confrontation divided by PRDs). Finally, it gives the results to `ggplot()` for plotting.

```

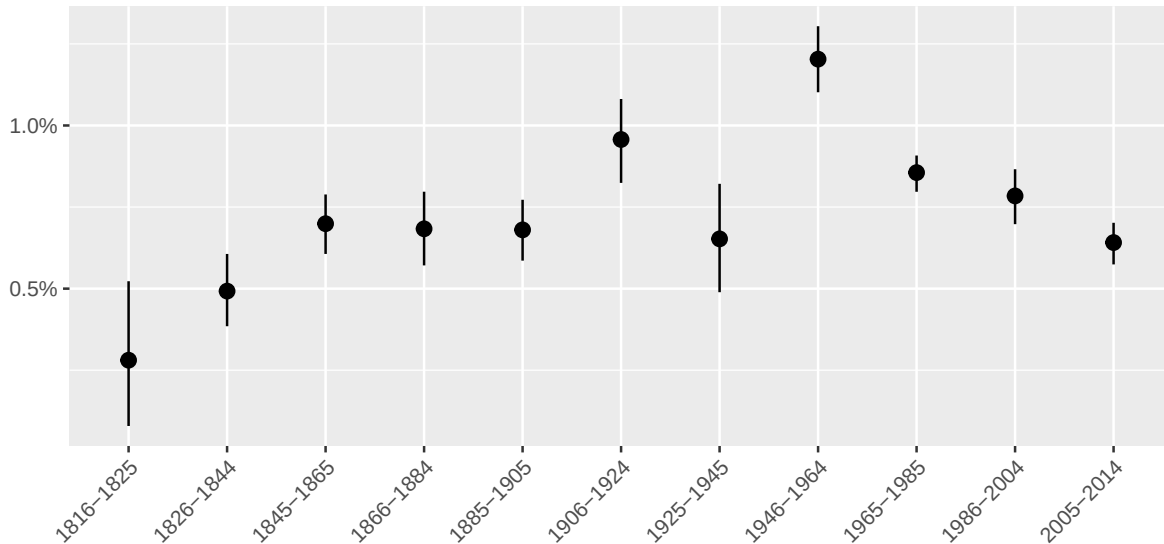
final_dt |>
  # add numerical labels for each 20 year period
  mutate(
    period = round(1:n() / 20) * 20
  ) |>
  # add labels that specify the year range per period
  group_by(period) |>
  mutate(
    period_label = paste0(min(year), "-", max(year))
  ) |>
  # make sure the labels are treated as an ordered category
  ungroup() |>
  mutate(
    period_label = factor(
      period,
      levels = period,
      labels = period_label
    )
  ) |>
  # get the mean and CIs by period label
  group_by(period_label) |>
  summarize(
    mean_ci_boot(prop1, ci = .84)
  ) |>
  # visualize the results
  ggplot() +
  aes(
    x = period_label,
    y = mean,
    ymin = lower,
    ymax = upper
  ) +
  geom_pointrange() +
  scale_y_continuous(
    labels = scales::percent
  ) +
  labs(
    title = "War propensity peaked in the mid-20th century, but propensity today is about\nas",
    subtitle = "% war propensity by 20 year increments",
    x = NULL,
    y = NULL
  ) +

```

```
theme(  
  axis.text.x = element_text(angle = 45, hjust = 1)  
)
```

War propensity peaked in the mid-20th century, but propensity today is about as high or higher than during the 1800s

% war propensity by 20 year increments



The results are consistent with a trend that is a bit more complicated than the one proposed by the decline of war thesis. It looks like the propensity for war increased between the early and mid to late 19th century. It then increased again in the early 20th century (likely due to WW1) before briefly returning to mid to late 19th century levels. There's a spike again in the mid 20th century due to WW2 and then a decline, yet again, to mid to late 19th century levels. One sensible conclusion you could draw from the data is that the most peaceful time in international politics was the early 19th century, but since then (aside from upswings during the world wars), war propensity tends to return to a stable rate of a little less than 0.75%, which was the norm for most of the 19th century and beginning of the 20th, the inter-war period between WW1 and WW2, and the early 21st century. It was slightly elevated during the Cold War, but the trend during this period just looks like a gradual decline back to the norm.

In short, the typical conflict rate over the past two centuries has remained mostly stable, with the exception of the world wars, and the unusually peaceful period in the decades after the Napoleonic Wars. Braumoeller drew about the same conclusion using an older version of the MID dataset.

This of course leaves open the question of how much the results change if you use one of the other measures of war propensity I included in the data. But I need to give you something to

investigate on your own, so I'll leave it to you to figure out how fragile the findings are to the choice of measure.

3.5 Summary

In sum, Braumoeller leveled two methodological critiques on past efforts to test the decline of war thesis: (1) ill-conceived measures and (2) the failure to account for chance.

In this chapter, I covered some of the conceptual hurdles that attend measurement, and showed you how, once you have some sensible measures, you can account for the role of random chance in explaining patterns in the data.

With respect to an appropriate measure of conflict over time, the best approach is to calculate the frequency of new conflicts *per* the number of opportunities for new conflicts to arise. The overall conflict count can go up and down for a variety of reasons, but perhaps most strongly by variation in conflict opportunities. Only once this is accounted for can you make good comparisons in war propensity over time.

With respect to the role of chance, statistical analysis provides the tools you need. The idea is to proceed as if the data you observe (war propensity over time) is the product of both a signal and random background noise. Statistical inference can help quantify how much variation you should expect due to background noise so you can more rigorously judge if a change over time can be easily explained by noise instead of a true signal (or meaningful change).

Adopting a sensible measure of war propensity and applying statistical inference yields results that don't neatly align with the decline of war thesis (at least for one possible measure of war propensity). In short, there's little evidence of a systematic decline in the likelihood of war, at least over the last two centuries.

But the chance of war is only half of the decline of war thesis. Its other major claim is that wars are now less deadly than in the past. Testing this proposition requires wrestling with a host of new conceptual and technical issues, all of which I'll discuss in the next chapter.

4 Is War Less Deadly?

Main ideas:

- The decline of war thesis holds that wars are becoming less deadly in addition to less common.
- Testing this claim is tricky for conceptual, statistical, and theoretical reasons.
- An analysis with the MIE dataset offers some mixed results that don't neatly support, but also don't clearly refute, the idea that war is becoming less deadly.

4.1 Decline of War Thesis and War Severity

Recall from the last chapter that the decline of war thesis has two main pillars:

1. War is becoming less common.
2. War is becoming less deadly.

I covered the first pillar in the previous chapter, discussing a mix of conceptual and theoretical issues, in addition to measurement and statistical inference. A reasonable conclusion you can draw from the data is that war *propensity* (the measure I adopted for quantifying the potential for war per opportunities for countries to fight) has been mostly stable over the past 200 years. Notable exceptions were the early 19th century which was unusually peaceful, and the early to middle 20th century, which witnessed two world wars. Contrary to the decline of war thesis, the late 20th and earlier 21st centuries appear about as warlike as the historical average.

In this chapter, I want to turn to the second pillar of the decline of war thesis and ask: *is war becoming less deadly?*

As you'll see, answering this question requires addressing a range of thorny conceptual issues. I'll specifically address three: (1) issues with measurement, (2) issues with statistical inference, and (3) issues with theoretical explanations.

After I discuss these issues, I'll walk through an applied example using the militarized interstate event (MIE) dataset. As you'll see by the section header, I'm calling my example a "low-tech" replication of the analysis Braumoeller (2019) did in his book, *Only the Dead*. It's low-tech because Braumoeller used a sophisticated statistical model that I think is best ignored in an undergraduate class. As it so happens, I also think that his sophisticated method was actually

unnecessary, and the new MIE dataset I'll use (which Braumoeller didn't have access to) will help me make this point.

Without further ado, let's get to it.

4.2 Three Problems with Studying War Severity

Before you can test the claim that war is becoming less deadly, you need to contend with a few issues. The first deals with measurement, which requires thinking carefully about what is meant by the “deadliness” of war. Do you mean absolute deadliness, or deadliness relative to something else? If you mean the latter, what's that something else?

The second deals with statistical inference. Braumoeller (2019) makes the argument that standard statistical tools are ill-suited to studying war fatalities because war size follows a *power-law* distribution. Power-laws apply to events that are prone to exceptionally large upswings in intensity—things that go beyond just a skewed distribution that you can handle with a simple data transformation, such as using the natural log. Because of these massive (and hard to predict) upswings, power-law data also don't obey the *central limit theorem* (CLT), which is foundational to the whole enterprise of statistical analysis. If the power-law really does apply to war fatalities, you'll have to look beyond the standard toolkit for statistical inference.

The final issue deals with theoretical explanations. The decline of war thesis holds that wars are becoming less deadly for ideological/normative reasons. But a decline in war fatalities can also be explained by other factors, like improvements in battlefield medicine, changes in the style of warfare, or a failure of war deaths to keep up with population growth (if you want to measure fatalities in relative terms). Any of these alternative explanations need not be mutually exclusive, which makes it difficult to conclude how much each mechanism contributes to an overall decline in war deaths (if such a pattern even appears in the data).

I tackle each of these issues in more detail below.

4.2.1 Measurement

The deadliness of war can be conceptualized and, therefore, quantified many different ways. Further, deadliness is just one among many dimensions of what you could call war *severity*. Wars can be severe not just in terms of battle deaths but also duration and the number of battles. I'll focus just on battle deaths here, but these alternative metrics are worth keeping in mind.

A further issue centers on whether severity should be treated in absolute or relative terms. With respect to war deaths in particular, it is important to be precise about whether you think deadliness should be defined as the *total* body count or as a *rate*. If a rate, you then

need to further specify what the reference point is. Some, like Pinker (2012), believe you should divide battle deaths by world population. Braumoeller (2019) argues that a better idea is to divide by the populations of the countries involved in a specific war. A third option could be to divide by the total military personnel of the countries involved.

These measurement choices aren't just a minor technical issue. How you measure war deadliness says a lot about how you conceptualize the risk of dying in war. Dividing battle deaths by world population treats death in war like a public health issue, in the same category as the global risk of death by heart disease or stroke. Dividing battle deaths by the combined populations of the countries involved in a war, or what you can more succinctly call *belligerent* populations, treats the risk of dying in war as a political issue where the risk only comes into play once countries make the choice to fight. Dividing battle deaths by belligerent military personnel further constrains the population of individuals you want to treat as at risk of dying in war.

Any analysis of war deadliness, therefore, needs to clearly explain how deadliness is quantified and provide a justification for why one approach was adopted over the alternatives.

4.2.2 Detecting Changes

Beyond measurement, another issue that needs to be addressed is statistical inference. In the previous chapter, I showed you how you can use statistical inference to quantify the role of random chance (background noise) in the trend in war propensity over time. Statistics adds a layer of rigor to your analysis that “eyeballing” the data fails to provide.

Statistical inference is no less important when testing whether war is becoming less deadly, but depending on who you ask, you may need to think twice before using traditional statistical tools as opposed to more specialized ones. According to Braumoeller (2019), traditional tools won't do. The reason, he argues, is that war fatalities follow a *power-law* distribution.

The idea that the power-law describes the distribution of war deaths isn't new. In fact, the argument is quite old, dating back to the birth of quantitative peace science (and even a bit before). Braumoeller, therefore, is following a well-established tradition within the study of war.

However, this tradition has an important implication for doing statistical inference, namely, that conventional statistical tests won't work. Under certain circumstances, power-law data do not have a finite mean or variance. Another term used to describe this scenario is “scale-free.” That means a variable that follows the power-law doesn't have an identifiable central tendency or quantifiable variance, which makes it impossible to draw comparisons between war death deaths in different periods in time using statistics. Scale-free data also don't abide by the central limit theorem (CLT), which is foundational to conventional statistics. The CLT holds that as you draw ever larger and larger samples, your estimates will converge toward the truth. Power-law data don't always abide by this rule.

Braumoeller therefore proposes an alternative statistical test for whether wars are becoming less deadly that is “robust” in the face of these problems. Using this test, he fails to find good evidence that wars have become less deadly over time, and he bases this conclusion both on total war deaths and deaths per world and belligerent population.

These findings are important, but they’re premised on the idea that the power-law is *the* way to study war fatalities. This idea is increasingly coming under scrutiny. In a recent working paper, Spagat, Fagan, and Weezel (2024) argue that a more conventional log-normal distribution (among other alternatives) makes more sense. Further, Cunen, Hjort, and Nygård (2020) argue that an obscure model from actuarial science known as the inverse Burr is most appropriate. Other skeptical studies exist as well, and I personally have done some research in this area and have come to a similarly skeptical conclusion. Needless to say, the idea that the power-law is *the* law of the land when studying war just isn’t true.

As I’ll show in the analysis in a later section, war fatalities in the MIE dataset can be handled quite easily by simply taking their natural log and then performing standard tests. While I admit that this is still a live question in conflict research, I think this is a sound way to proceed, and the ranks of conflict scholars who agree seems to be growing.

4.2.3 Explaining Changes

One final issue with testing if war is becoming less deadly is identifying the best explanation for why this would be happening (if, indeed, it is). Proponents of the decline of war thesis, such as Pinker (2012), argue that ideas born of the Enlightenment such as reason, liberal democracy, human rights, and pro-peace norms can explain why wars would become less deadly over time. Because societies simply cannot stomach bloody conflicts, tolerance for exceptionally deadly wars is much lower today than decades or even centuries ago.

However, this is just one explanation among many. Some political scientists, such as Tanisha Fazal (2014), have been critical of this argument. Fazal in particular argues that improvements in battlefield medicine can explain a decline in war deadliness just as well, if not better than, humanistic norms. At one point in time, the wound-to-death ratio in war was 3-to-1. Today, for countries with modern militaries (such as the U.S.), the ratio is closer to 10-to-1. Over time, it has simply become much easier to treat wounded soldiers and keep them alive. All else equal, this means wars today would be less deadly, even though no real change in the willingness to spill blood in war has occurred.

Different styles of warfare might also influence the body count. Drones, for example, are quite deadly tools that simultaneously negate the need to put your own troops in harms way. The increasing reliance by countries like the U.S. on special forces units and targeted operations likewise can limit how deadly a conflict is. Tactics and technology, in short, matter as well.

One final explanation, which Braumoeller (2019) himself raises, deals with conflict death rate measures specifically. He notes that finding a decline in war fatality *rates* could mean that

conflict really is less deadly for normative and ideological reasons, but it could just as easily mean that population growth has exceeded war's deadliness. Population growth is well known to be exponential, which, by Braumoeller's lights, makes a decline in war fatality rates much less compelling a data point than proponents of the decline of war thesis make it out to be.

In sum, to use some jargon, an observed decline in war deaths is *overdetermined*. This term refers to an event, trend, or pattern in the world has many different, equally valid theoretical explanations (far more than are necessary to actually account for it).

Keep this point in mind, not just when I show you some results from the data in the next section, but also for any analysis you encounter or personally conduct on war or some other subject. An ideal analysis simultaneously will provide a good empirical test of a theoretical argument and rule out alternative explanations. The first objective is hard enough. The second is nearly impossible.

4.3 A Low-tech Replication of Braumoeller's Work

The previous sections covered some issues with measurement, inference, and explanations for declining war deaths. In this section I want to show you some practical solutions.

I call this section a low-tech Replication of Braumoeller's work because I will not ask you to implement the specialized methods he used in his book, *Only the Dead*. As I noted in the previous discussion, I have some qualms with his approach. As you'll see, with the MIE dataset I think it's quite clear that a simple log-transformation of battle deaths is sufficient to address the problems about which Braumoeller (and many others) expressed concern.

This solves the issue with inference, but what about measurement? I'll leave the question of which measure of war severity is best to others. Here, I'll simply show you how to implement four alternatives, and offer a quick summary of their interpretations.

Finally, I sadly have not solved the overdetermination problem. So once I show you some results, you and I will have to settle for an empirical reality with a cause that remains a puzzle.

4.3.1 Get the Data

First of all, I need to get the MIE dataset. As I showed you in the previous chapter, you can run the following code in R, which will get and automatically clean it.

```
## set up
library(tidyverse)
library(socsci)
source("https://raw.githubusercontent.com/milesdwilliams15/death-destruction-data/refs/heads/main/data/mie_clean.R")
```

```
## get the cleaned up data
get_mie_data() -> dt
```

For this analysis, I need more than just the MIE dataset, however. I need to create four different measures of how severe wars are over time in terms of battle deaths: (1) total deaths, (2) the global death rate, (3) the belligerent death rate, and (4) the military death rate. I can get the first from the MIE dataset alone, but to get the last three I need more data. For the global death rate, I need the total population of countries in the world; for the belligerent death rate, I need the total population of the countries involved in a given war; and for the military death rate, I need the total military personnel of the countries involved in a war.

Thankfully, I don't need to look too far to get this data. Steve Miller, one of the researchers responsible for the MIE dataset, created an R package called `{peacesciencer}` meant to make it easier to access common variables and datasets for doing peace science research (Miller 2022). To install it, all you need to do is run `install.packages("peacesciencer")`. Once you do, you now have access to a wide variety of datasets using the `{peacesciencer}` API, which stands for *augmented programming interface*.

With this package, I can access one dataset in particular that has all the information I need to complete my analysis. The Correlates of War (COW) Project has a dataset on the military capabilities of countries across the world. It's called the National Material Capabilities (NMC) dataset. It's been around for a long time (see J. D. Singer, Bremer, and Stuckey 1972; D. J. Singer 1987), and it's used to construct an index of country military power known as the Composite Index of National Capability, or CINC (I'll talk more about this measure in the next part of this book). Most useful for the moment is the fact that both population data and military personnel counts are necessary to construct CINC scores for countries. The NMC dataset therefore has all the data I need for this analysis. Even better, `{peacesciencer}` gives me access to the most recent version of the data (version 6) which covers the whole time period from 1816 to 2014 covered by the MIE dataset.

Here's how to access the data. Once you have `{peacesciencer}` installed, you can run the following code. It opens the package, and then it creates a dataset of all the country pairs (dyads) in the world by year according to the COW state system coding rules. It then adds information from the NMC dataset and saves the output as an object called `ext_dt`. Next, it merges this dataset with the MIE dataset by each country in a dyad and the year a conflict took place.

```
## get NMC data (has population and military totals)
library(peacesciencer)
create_dyadyears(subset_years = 1816:2014) |>
  add_nmc() -> ext_dt

## merge with MIE data
```

```
dt |>
  left_join(
    ext_dt,
    by = c("ccode1", "ccode2", "year")
  ) -> dt
```

Now, I have all the data I need to test if war deaths are declining. I just need to process it a bit more to get it ready, which I do in the next section.

4.3.2 Data Processing

To get the data ready for analysis I need to transform it. First of all, each row in the dataset is a unique *event*, but the decline of war thesis claim that wars are becoming less deadly doesn't explicitly apply to the deadliness of particular battles. Rather, it applies to the body count by the end of a whole conflict. Therefore, I need to aggregate the data to the conflict level.

Remember that in the MIE dataset, individual militarized interstate events (MIEs) are embedded in militarized interstate confrontations (MICs). I need to aggregate the data to the MIC level. The below code will do this. For each MIC, it will give me the sum of all battle deaths (both the min and max estimates), the start year, the maximum hostility level, the pooled population and military personnel of belligerents, and some other summary stats that might be interesting to look at later.

```
dt |>
  group_by(micnum) |>
  summarize(
    year = min(year),
    hostlev = max(hostlev),
    fatalmin = sum(fatalmin1 + fatalmin2),
    fatalmax = sum(fatalmax1 + fatalmax2),
    events = n(),
    duration = 1/12 + max(endyear + endmon / 12) - min(styear + stmon / 12),
    bellig_pop = sum(tpop1[year == min(year)] +
                     tpop2[year == min(year)], na.rm = T) * 1000,
    mili_pop = sum(milper1[year == min(year)] +
                   milper2[year == min(year)], na.rm = T) * 1000
  ) -> war_dt
```

Next, I need to bring in the total world population per year. This next bit of code will take the dyad-year dataset with the NMC variables and extract from the data a measure of the total population of all countries in the COW system by year. It then merges this total with the MIC level dataset by conflict start year.

```

ext_dt |>
  distinct(ccode1, year, tpop1) |>
  group_by(year) |>
  summarize(
    world_pop = sum(tpop1) * 1000
  ) -> pop_dt

war_dt |>
  left_join(
    pop_dt,
    by = "year"
  ) -> war_dt

```

Finally, in keeping with tradition for this kind of analysis, I'm going to only keep MICs that actually escalate to the level of a war. The criteria for this categorization in the MIE dataset is a bit different than for the COW wars dataset I talked about a couple of chapters ago. Rather than using a strict battle death threshold, it categorizes MICs as wars that most historical records recognize as all-out wars.

```

war_dt |>
  filter(hostlev == "War") -> war_dt

```

Now, I just need to do one more thing. So that I can do some analyses more conveniently across all four different possible ways of quantifying how severe wars were, I'm going to pivot the data and generate a column that has all the war severity measures in one, and the appropriate measure label in another. I'll just take a look at maximum fatality estimates. You can think of this as a worst-case scenario for conflict severity.

```

war_dt |>
  # add a constant of 1 trillion
  mutate(const = 1e12) |>
  # pivot by pop data and the constant
  pivot_longer(bellig_pop:const) |>
  # divide by the appropriate pop/constant value (in millions)
  mutate(
    value = 1e06 * fatalmax / value,
    name = frcode(
      name == "const" ~ "Total (millions)",
      name == "world_pop" ~ "Global Rate\n(per million)",
      name == "bellig_pop" ~ "Belligerent Rate\n(per million)",
      name == "mili_pop" ~ "Military Rate\n(per million)"
    )
  )

```

```

)
) |>
# filter out a couple of cases with missing personnel
filter(value < Inf) -> piv_war_dt

```

4.3.3 Analysis

Okay, now the data is ready to go, so let's start analyzing it. First, I want to just take a look at the raw data.

The below code will produce a plot with four panels, each showing the distribution of one of measures of conflict fatalities. As you can see, war deaths follow an extreme distribution, with lots of small wars and a few exceptionally deadly ones. This is true whether looking at total deaths, or calculating the rate per world or belligerent population, or military personnel.

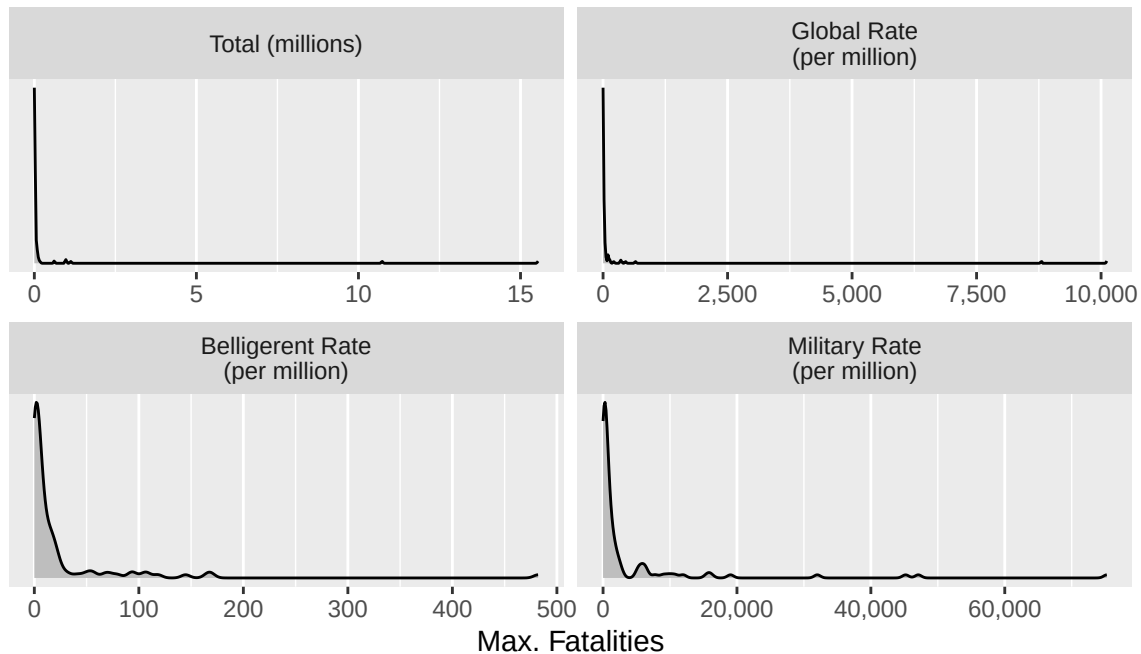
```

ggplot(piv_war_dt) +
  aes(value) +
  geom_density(fill = "gray") +
  facet_wrap(~ name, scales = "free") +
  labs(
    title = "War deaths follow an extreme distribution",
    subtitle = "Four different distributions of war severity",
    x = "Max. Fatalities",
    y = NULL
  ) +
  scale_y_continuous(
    breaks = NULL
  ) +
  scale_x_continuous(
    labels = scales::comma
  )

```

War deaths follow an extreme distribution

Four different distributions of war severity



This is consistent with Braumoeller's (2019) point that war deaths are prone to extreme upswings in deadliness, but it does not further imply that specialized statistical tools are necessary to make inferences with the data.

Here's what happens when you take the log of battle deaths. The below code produces a figure like the last one, but it puts the battle death values on the x-axis on the log-10 scale (which is proportionally equivalent to the natural log, which is more common in political science research). Once the data is re-scaled in this way, its distribution looks much closer to normal, particularly for the belligerent and military personnel fatality rates. This seems to line up with the findings of Spagat, Fagan, and Weezel (2024), who argue that the data looks log-normal—which is to say, it has an identifiable mean and variance, unlike many variables that follow the power-law. To me, this serves as a nice gut-check that using more conventional statistical methods is a viable option.

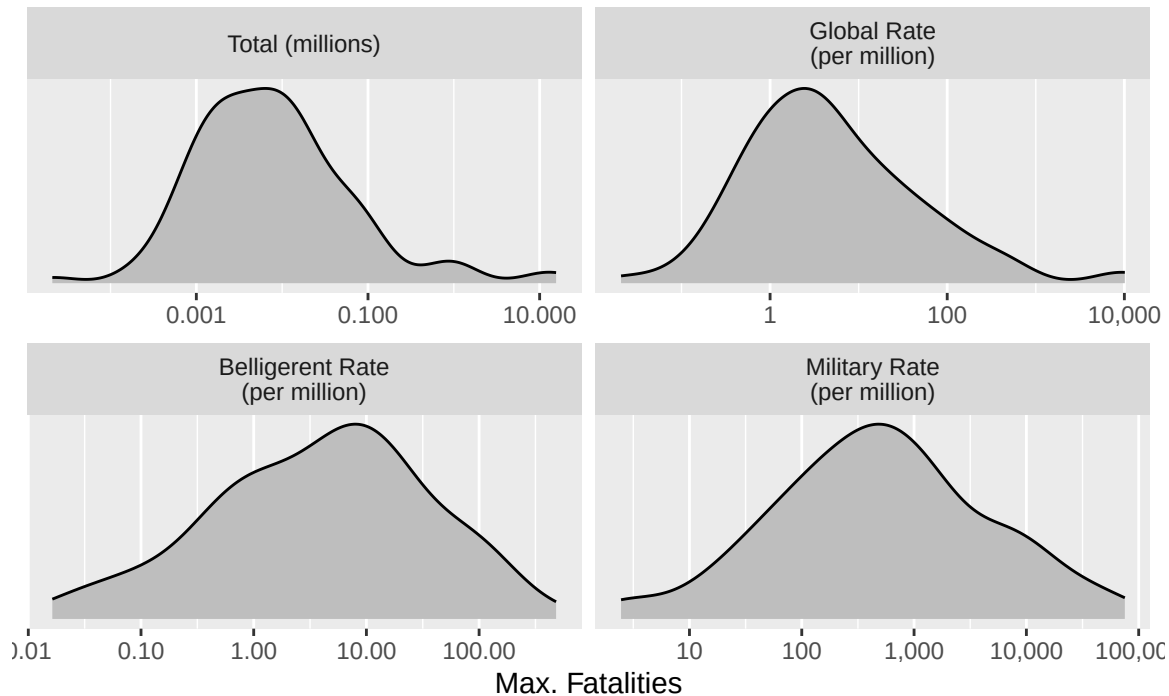
```
ggplot(piv_war_dt) +  
  aes(value) +  
  geom_density(fill = "gray") +  
  facet_wrap(~ name, scales = "free") +  
  labs(  
    title = "Four different distributions of war severity look normal-ish on the log scale",  
    x = "Max. Fatalities",
```

```

    y = NULL
  ) +
  scale_y_continuous(
    breaks = NULL
  ) +
  scale_x_log10(
    labels = scales::comma
  )

```

Four different distributions of war severity look normal-ish on the log scale



So what happens when I apply more conventional methods? I can proceed a couple different ways. First, I can just plot the data over time and run a smoothed regression line through the data to see if a meaningful trend (signal) shows up.

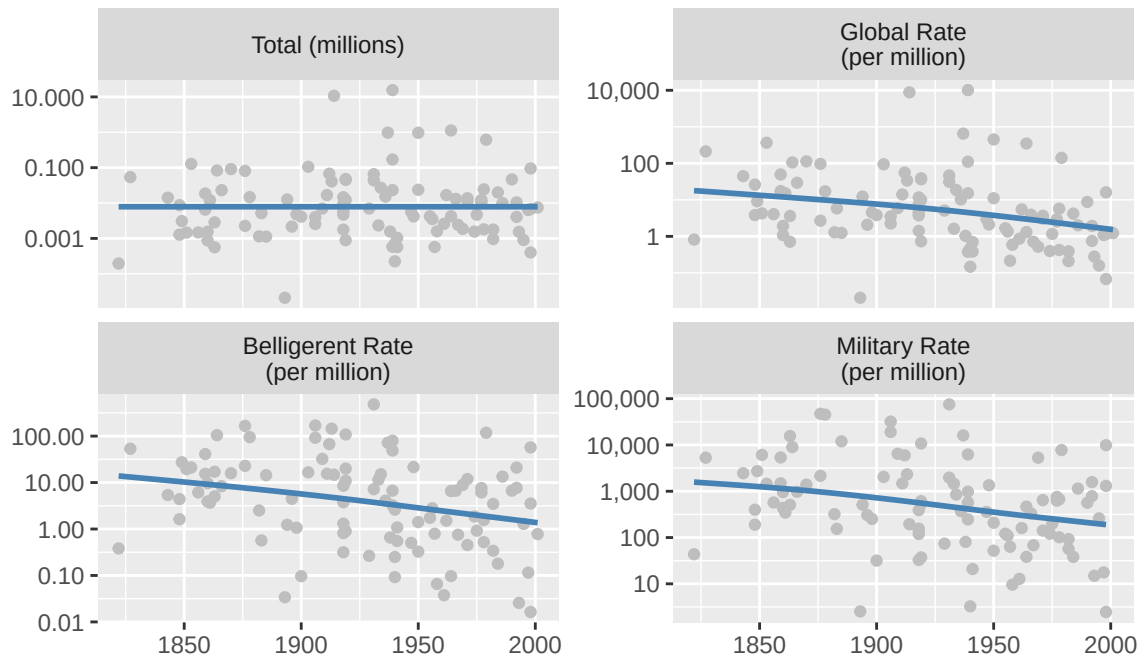
The below code produces a scatter plot and adds a smoothed line from a generalized additive model (GAM). This is my go-to method for plotting a line through data because GAMs have a really nice property of only fitting a trend if the bumps and curves in a dataset are big enough to rule them out as the result of random background noise. You can see this property on display in the figure below, which only shows a flat line for total battle deaths, but downward curved slopes for the remaining three measures.

```
ggplot(piv_war_dt) +
  aes(year, value) +
  geom_point(color = "gray") +
  geom_smooth(method = "gam", se = F, color = "steelblue") +
  facet_wrap(~ name, scales = "free_y") +
  scale_y_log10(
    labels = scales::comma
  ) +
  labs(
    title = "War is as deadly as ever in absolute terms, but less in relative terms",
    subtitle = "Max. war fatalities by start year",
    x = NULL,
    y = NULL
  )
```

`geom\_smooth()` using formula = 'y ~ s(x, bs = "cs")'

War is as deadly as ever in absolute terms, but less in relative terms

Max. war fatalities by start year



This method is different from the one applied in the last chapter looking at conflict occurrences, but you could just as easily apply it there. You could also use the binning method used in the

last chapter here. However, because there are fewer overall data points in this analysis, you might run into some issues in some bins if they're too small.

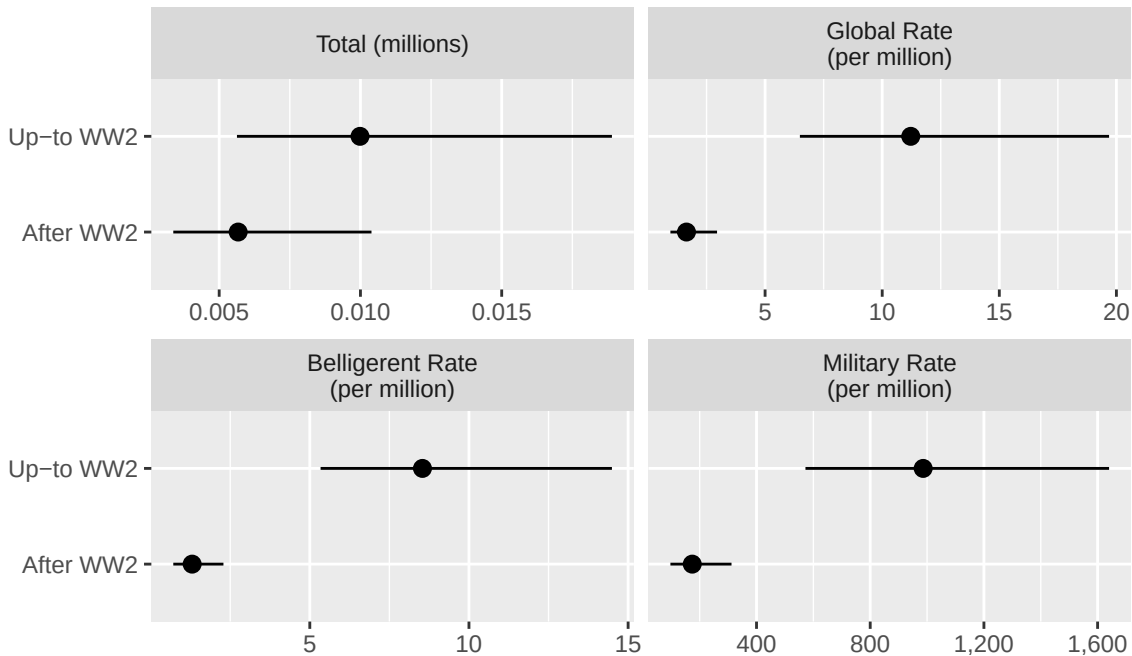
But, if you wanted to bin the data, the best place to start is just to compare conflicts after World War 2 to those before and up to it. This is the approach Braumoeller took in his book.

Here's how to implement this approach. First, read in the source file that gives you access to the function `mean_ci_boot()`, and then group the data by period and measure of war severity. Next, calculate the average of the logged fatalities with 84% confidence intervals. For plotting, it might be a good idea to convert the estimates back to their original scale so you can get a clearer sense of proportion (you can just take the exponent of the results). Consistent with the findings above, only in the case of total war deaths do I fail to rule out background noise as the cause of post- versus pre-WW2 differences in war severity. But, when expressed as a rate, the evidence looks quite clear that wars over time have become less deadly.

```
source("https://raw.githubusercontent.com/milesdwilliams15/death-destruction-data/refs/heads/main/data/processed/piv_war_dt")
piv_war_dt |>
  mutate(
    postww2 = ifelse(year > 1939, "After WW2", "Up-to WW2")
  ) |>
  group_by(name, postww2) |>
  summarize(
    mean_ci_boot(log(value)) |> exp()
  ) |>
  ggplot() +
  aes(x = mean, xmin = lower, xmax = upper, y = postww2) +
  geom_pointrange() +
  facet_wrap(~ name, scales = "free_x") +
  scale_x_continuous(labels = scales::comma) +
  labs(
    title = "Contrary to Braumoeller, wars are relatively less deadly post-WW2",
    subtitle = "Mean logged battle deaths with 84% CIs by period",
    x = NULL,
    y = NULL
  )
```

Contrary to Braumoeller, wars are relatively less deadly post-WW2

Mean logged battle deaths with 84% CIs by period



The contrast between these results and those shown in the last chapter is interesting. The contrast with Braumoeller's findings is as well. On the one hand, the analysis in the last chapter fails to show compelling evidence that war is less likely today than in the past. To the contrary, the early 19th century was the most peaceful period on record covered by the MIE dataset. On the other hand, the fatality rate from wars has steadily declined over the same period; albeit, with no apparent decline in war deaths in absolute terms.

In short, the data doesn't seem to neatly align with the decline of war thesis, but it also doesn't perfectly fit with the findings in Braumoeller's comprehensive analysis in *Only the Dead* (Braumoeller 2019). While wars seem to have a fairly stable chance of occurring over time, the risk of dying in a war has gone down for the average person anywhere in the world, for a person that's a citizen of one of the countries involved in a war, and for military personnel.

Does this mean the decline of war thesis is half right? Maybe, but remember the theoretical issues I discussed earlier. A decline in war deaths could show up in the data for reasons that have nothing to do with the mechanisms decline of war thesis proponents argue are at work. Deaths could decline because of improvements in battlefield medicine, changes in warfighting, and exponential population growth.

It's impossible to completely rule out any one of these explanations, but I'll give you my opinion. I think battlefield medicine is the most compelling argument, followed by changes in how wars are fought. Fazal's (2014) analysis provides hard to refute evidence that the typical

soldier today has a much lower chance of dying on the battlefield than in the past. Battlefield medicine (and medicine in general) has advanced so much over the past century, and even more so the past few decades, which, by the way, would explain why the trend in the data is a downward sloping *curve* instead of a simple straight line when measuring the fatality rate.

Changes in tactics also seem like a contributing factor. With few exceptions, most conflicts in recent history aren't fought on conventional battlefields with armed troops engaged in face-to-face combat. This clearly means a smaller share of a country's troops are in harms way.

I find exponential population growth less compelling. The reason is that the conflict rate also declines over time per military personnel (the sub-population most at risk of dying in war). As Braumoeller (2019) notes, militaries grow at a much slower rate than country populations, mainly because once your fighting force reaches a critical mass, adding more troops to the ranks doesn't make a massive difference in capabilities. Warfighting technologies tend to give you a bigger bang for your buck, so to speak. So the fact that war deaths don't also keep up with military size gives me doubts that population growth alone explains a decline in fatality rates.

I find shifting norms around war the least compelling, mainly because the risk of war has remained fairly stable over time. Also, total war deaths have remained constant over time as well, suggesting that just as many people can die in wars today as in the past. If norms and ideology are really driving trends in a meaningful way, I'd expect to see movement in these metrics as well. That movement just doesn't exist.

This, of course, is just my opinion. You may draw different conclusions from the data.

4.4 Summary

In this chapter I introduced concepts and methods essential for testing whether war is becoming less deadly. On the theoretical side, there are lots of issues you need to contend with, and on the conceptual and measurement side, there are tough judgment calls that need to be made.

While the analysis I presented supports certain conclusions, I hope you're starting to appreciate that data analysis in the peace science tradition involves some subjectivity about what counts and what doesn't. I think this fact isn't something to bemoan but to embrace, because it means the peace scientist's work is never finished. Many unanswered questions remain, and more perspectives are necessary if further progress is to be made.

5 Dyadic Models of Conflict

Main ideas:

- **The dyadic approach dominates peace science research:** Most conflict research focuses on country-pairs (dyads) over time rather than macro-level trends, largely due to professional incentives, researcher degrees of freedom, and the massive number of data points dyadic datasets provide.
- **Building dyadic datasets with `{peacesciencer}`:** The R package `{peacesciencer}` streamlines constructing valid directed or non-directed dyad-year datasets and merging in commonly used variables (contiguity, major power status, distance, MIDs, etc.).
- **Logistic regression (logit) is the standard model for dyadic conflict:** Because conflict onset is a binary, rare event, the logit model is used to map predictors to probabilities of conflict while respecting the 0–1 bounds of probability.
- **Handling statistical dependence in dyadic data:** Since dyads are observed repeatedly over time, standard errors need adjustment either via cluster-robust standard errors or by modeling dyadic random effects directly in the specification.

5.1 Why do countries fight?

In the previous chapters, I spent a lot of time talking about how to measure war, and how to analyze macro-level trends in war propensity and severity over time. Despite how many chapters I dedicated to these subjects, the attention I paid to them isn't proportionate to the attention peace scientists pay to them.

This seems odd to me, because I think understanding macro trends in international conflict has a good deal of practical relevance to people, and is intrinsically interesting. But the field of peace science treats these issues as niche. The bulk of the field's focus, instead, lies in trying to understand conflict at the level of the *dyad*, or country-pair.

Because this is the main focus of peace research, best-practices and norms for what's acceptable for studying conflict at the dyad-level are well-established, so much so that if you deviate from these best practices you have to do a lot of extra work to justify your unconventional choices.

Because dyadic conflict research is so ubiquitous, and a narrow range of methodological choices are the norm, in this chapter I want to spend some time explaining the dyadic approach and

the statistical model most commonly applied in peace science research. I'll then walk you through a simple example and the R code that implements it.

Mind you, because I can't leave well enough alone, I'll offer some of my own thoughts about how I might tweak existing approaches. And, I'll test whether and how much this tweak alters conclusions from the replication analysis I'll show you.

But, before I get to that, let me talk a bit about the dyadic approach to conflict research.

5.2 Studying dyads

If I randomly sampled a peace science study on international conflict from the past few decades, odds are pretty good that the authors of that study conducted their analysis of conflict initiation between *pairs* of countries, or dyads, over time.

I believe this kind of analysis dominates the field for two reasons. The first is that researchers really want to know why countries go to war. The second is that there's a lot of untapped variation in dyadic conflict patterns left to explain. A rich set of theories are available to researchers to extend and modify to propose new predictions about which country dyads are at greatest risk of fighting. And new and improved measures get proposed all the time to time to test if old findings hold up with new data.

In short, dyadic analyses of conflict initiation have a lot of "researcher degrees of freedom," which creates space for lots of different theories and measures to get tested. It also leaves plenty of room for disagreement.

Conversely, there's much less space to make a novel contribution to research on global trends in conflict. Any budding peace scientist looking to publish their work has a much better chance of doing something novel with a dyadic analysis than one centered on global trends.

Think about it, the best conflict datasets available, such as the MIE dataset introduced previously, only cover about 200 years worth of data. That means an analysis of global conflict trends over time will at best have up to 200 data points (perhaps 12 times that value if doing monthly trends). Meanwhile, a directed-dyad-year dataset covering the same period has 1,836,630 data points. If your goal is to explain variation in a dataset with *millions* of data points, there's always a chance of finding something new.

This isn't necessarily a great, objective justification for focusing so much attention on dyadic studies. But peace scientists (and all scientists for that matter) *are* people who are subject to professional incentives and constraints. Often times, the research that gets done is the research that seems publishable. The more work that gets published in an area, the more publishable it seems, and then the more that kind of research gets published. On and on the cycle goes.

This is the idea of “path dependence,” and because of it, if you want to make a contribution to peace science, the best place to start is with a dyadic study. But how exactly do you do a dyadic analysis in peace science?

You need two things: (1) a dataset where each observation is a pair of countries in a given year and (2) a statistical model well-suited to taking a variety of different observed characteristics of country pairs and mapping them to probabilities of conflict initiation.

Because this kind of analysis is so common, tools are widely available for streamlining the process.

For a long time, many conflict scholars relied on tools such as [EUGene](#) for building their datasets. This tool provides a consistent way of defining the universe of cases you want to study, whether that’s individual countries or dyads over time. More recently, the political scientist Steve Miller made an R package called `{peacesciencer}` which makes the process of making and merging datasets much easier, and and possible to do directly in an R session on your computer (Miller 2022).

I mentioned this package in a previous chapter, and some of the code I’ve had you use already uses `{peacesciencer}` under the hood to perform various routines. In this chapter, I want to introduce the package more formally, and talk about its advantages. I personally use `{peacesciencer}` a lot in my own research because of these advantages.

While this package solves a lot of problems for doing peace science research, the most important problem it solves is making sure you have all the right units of observation in your dataset without duplicates or phantom cases, and without missing any relevant cases, too. If your unit of analysis in your data is fundamentally flawed, that’s bad news.

The second, somewhat less important problem the package solves (though it’s still quite important) is the drudgery of accessing the wide variety of commonly used datasets by peace scientists for doing their research. While `{peacesciencer}` doesn’t give you access to everything you could possibly want, it does provide access to many of the most important “go-tos.” It also provides access to an eclectic sample of other interesting and novel datasets that you might want to explore (for a person like me, looking through the package documentation feels very kid-in-a-candy-shop-esque).

I’ll talk more about different datasets that are available through the package in the coming chapters as I introduce different theoretical arguments for why countries go to war. For now, I just want to focus on the basic syntax the package provides for constructing a dataset with your desired universe of cases and then populating it with variables of interest.

Here’s some R code to get started. `{peacesciencer}` is meant to fit nicely into the tidyverse approach to R programming, so any time I use the package, I open the `{tidyverse}` as well (as I do in the below code). The next thing I do is construct a dataset with the universe of cases I want to study. I can do this with a wide variety of `create_*` functions. You can check out the package documentation to see all the available options. Below, because I’m

interested in doing a dyadic conflict analysis, I'm going to use `create_dyadyears()` which creates a dataset where each row is a unique country pair observed in a given year.

```
## open {tidyverse} and {peacesciencer}
library(tidyverse)
library(peacesciencer)

## make a directed-dyad-year dataset
create_dyadyears(
  subset_years = 1816:2014
) -> ddy
```

If you peek at the first few rows of the data, you can see what a dyad-year dataset looks like. `create_dyadyears()` returns a dataset with three columns: `ccode1`, `ccode2`, and `year`. The first is the Correlates of War (COW) country code for country 1 in the dyad, and the second is the COW code for country 2. The year column is the year a country pair is observed.

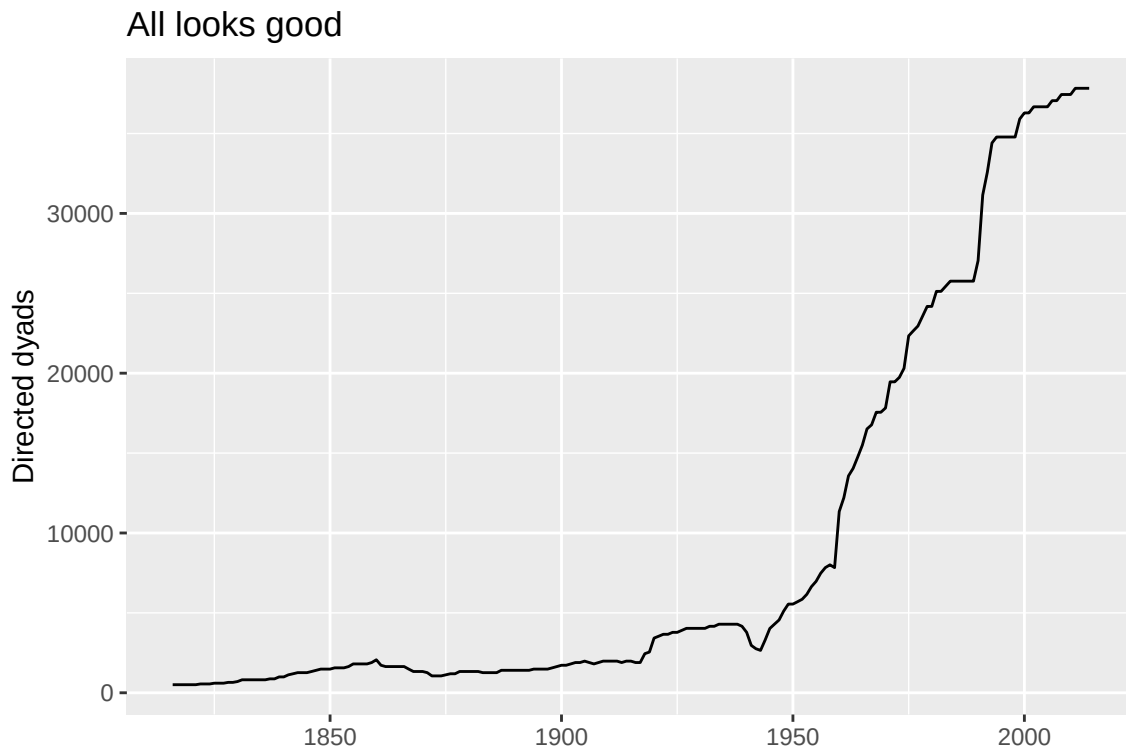
```
ddy |>
  slice_head(n = 5)
```

```
# A tibble: 5 x 3
  ccode1 ccode2  year
  <dbl>  <dbl> <int>
1      2      20  1920
2      2      20  1921
3      2      20  1922
4      2      20  1923
5      2      20  1924
```

Importantly, when `create_dyadyears()` constructs this dataset, it only returns *valid* country pairs. That means both countries must be recognized as members of the COW state system in the relevant year. To check that the dataset isn't just hallucinating invalid dyads, I can run some code to quickly check that the number of dyads in a given year changes over time. If it were constant, that'd be a sign that the dataset is falsely including country pairs that could not possibly have existed at one point in time.

```
ddy |>
  count(year) |>
  ggplot() +
  aes(year, n) +
  geom_line() +
```

```
labs(
  title = "All looks good",
  x = NULL,
  y = "Directed dyads"
)
```



Another important thing to note about the data `create_dyadyears()` produced is that it returned a *directed*-dyad-year dataset, which it does by default. Dyadic datasets usually take one of two forms: *directed* or *non-directed*. Which kind you use depends on what kind of question about conflict you want to answer.

A directed dataset is one where a country pair in a given year appears *twice*, but the order of countries 1 and 2 is flipped in the dyad's second appearance. For example, as the output from the below code shows, the U.S. (COW code of 2) and Canada (COW code of 20) dyad appears twice in 1920, but the U.S. and Canada switch sides depending on which directed dyad is observed.

```
ddy |>
  filter(ccode1 %in% c(2, 20), ccode2 %in% c(20, 2)) |>
  slice_min(year, n = 1)
```



```
# A tibble: 2 x 3
  ccode1 ccode2 year
  <dbl>  <dbl> <int>
1      2      20  1920
2     20       2  1920
```

This kind of dataset is called “directed” because it lets you analyze the *direction* of country behavior within a country pair. For example, say you were interested not just in understanding which country pairs are most prone to fighting, but which side in a country pair is most prone to initiating a conflict. In this case, you’d want to use a directed-dyad-year dataset.

To produce a non-directed-dyad-year dataset, all you need to do is indicate `directed = FALSE` when you run `create_dyadyears()`:

```
create_dyadyears(
  subset_years = 1816:2014,
  directed = FALSE
) -> udy
```

Joining with ``by = join_by(ccode1, ccode2, year)``

Now, when I run the same code I applied to the directed dataset to check whether a dyad appears twice on the non-directed dataset, I can clearly see that the U.S.-Canada dyad only appears once in 1920. In such a dataset the order of the countries (usually) doesn’t matter.

```
udy |>
  filter(ccode1 %in% c(2, 20), ccode2 %in% c(20, 2)) |>
  slice_min(year, n = 1)
```

```
# A tibble: 1 x 3
  ccode1 ccode2 year
  <dbl>  <dbl> <int>
1      2      20  1920
```

I hope this method of constructing a dataset sounds easy, because the process of getting the universe of valid cases right without a tool like this is incredibly painful, and much more error prone, which I know from personal experience.

Of course, creating a dataset of cases is just the beginning. The next step is to start adding in variables. `{peacesciencer}`, thankfully, makes the process of getting common variables easy as well by providing you with a variety of `add_*()` functions that will add data from different commonly used datasets to your dataset.

Let's construct a really simple one for analysis. Say I wanted to test how the factors used to construct the Braumoeller-Carson dyad measure of relevance introduced a couple chapters ago predict conflict onset between countries. Recall that this measure uses a statistical procedure convert information about dyadic contiguity, major power status of one of the countries in the pair, and bilateral distance into probabilities that countries can reach each other to fight. I can use `{peacesciencer}` to quickly construct a relevant dataset to confirm that these factors do, indeed, predict a higher chance of conflict. Because I'm only interested in the chance of conflict between countries and not the factors that lead one of the countries to initiate a conflict versus the other, I need to construct a non-directed-dyad-year dataset. Observe:

```
## start with non-directed-dyad-year base then pipe to add MIDs
create_dyadyears(
  subset_years = 1816:2014,
  directed = FALSE
) |>
  ## add COW MIDs data then pipe
  add_cow_mids() |>
  ## add peace spells then pipe
  add_spells() |>
  ## add contiguity data then pipe
  add_contiguity() |>
  ## add major power data then pipe
  add_cow_majors() |>
  ## add distance data then save as "dt"
  add_cap_dist() -> dt
```

At the moment, `{peacesciencer}` doesn't provide direct access to the MIE dataset, but it does offer access to the COW MIDs dataset by way of the function `add_cow_mids()`. As I discussed in a previous chapter, the COW MID dataset has some issues that were well-documented by the creators of the MIE dataset. But, for my purposes here, this dataset is good enough. `{peacesciencer}` also has functions that add contiguity data, major power status, and pair-wise distance between country capitals. Using a succession of relevant functions connected with the R pipe `|>` operator, I can string together a dataset that includes variables from all the relevant datasets I need to do my statistical analysis. If I use the `glimpse()` function, I can quickly take a look at what's in the final data:

```
glimpse(dt)
```

```
Rows: 918,315
```

```
Columns: 29
```

```
$ ccode1      <dbl> 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, ~
$ ccode2      <dbl> 20, 20, 20, 20, 20, 20, 20, 20, 20, 20, 20, 20, 20, 20, 20, ~
```

```

$ year      <dbl> 1920, 1921, 1922, 1923, 1924, 1925, 1926, 1927, 1928, 19~
$ dispnum   <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, ~
$ cowmidongoing <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, ~
$ cowmidonset <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, ~
$ sidea1     <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, ~
$ sidea2     <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, ~
$ fatality1  <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, ~
$ fatality2  <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, ~
$ fatalpre1  <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, ~
$ fatalpre2  <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, ~
$ hiact1     <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, ~
$ hiact2     <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, ~
$ hostlev1   <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, ~
$ hostlev2   <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, ~
$ orig1      <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, ~
$ orig2      <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, ~
$ fatality   <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, ~
$ hostlev    <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, ~
$ mindur     <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, ~
$ maxdur     <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, ~
$ recip      <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, ~
$ stmon      <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, ~
$ cowmidspell <dbl> 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16~
$ conttype   <dbl> 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, ~
$ cowmaj1    <dbl> 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, ~
$ cowmaj2    <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, ~
$ capdist    <dbl> 734.8694, 734.8694, 734.8694, 734.8694, 734.8694, 734.86~

```

The data has over 900,000 rows and 29 columns. It requires some extra cleaning to make the data more usable for a dyadic analysis, but let's just take a second to appreciate how little code I needed to run to generate this. Just five years ago, it would have taken me hours to download each of the individual datasets I would have needed to make this dataset, and then merge them together, and check for errors. Using `{peacesciencer}`, in less than a minute I was able to write and run some code to accomplish the same goal.

But, as I said, I need to do a bit more cleaning, and I do only mean a *bit*. I actually have most of what I need already, but I need to make a binary measure of contiguity and binary measure to indicate if either country in a dyad is a major power. At the moment the relevant column for contiguity, `conttype`, is just a series of numerical codes corresponding to different levels of contiguity. Based on what I know from the package documentation, I want to code all cases with a code of 1 or higher as contiguous (this is the most inclusive measure of continuity, encompassing countries that border each other and also those separated by less than 400 miles of water). For major powers, at the moment the relevant information is spread across two

different columns. One is an indicator for if country 1 is a major power, and the other is an indicator for if country 2 is a major power. I need a measure that equals 1 if either of these columns equals 1. Finally, for reasons I'll talk about in just a second, I want to create a unique dyad numerical code. Thankfully, it only takes me a bit of code to make these changes:

```
dt |>
  mutate(
    ## if conttype != 0, code as contiguous
    cont = ifelse(conttype > 0, 1, 0),
    ## return the maximum major power indicator for each country
    majpower = pmax(cowmaj1, cowmaj2),
    ## dyad ID
    dyad = 1000 * ccode1 + ccode2
  ) -> dt
```

Now, finally, to make my life easier I can use some additional `{tidyverse}` tools, like `select()`, to pair my data down just to the columns I need for my analysis:

```
dt |>
  select(
    dyad, ccode1, ccode2, year, cowmidonset, cowmidspell, cont, majpower, capdist
  ) -> dt
```

Now, the data looks a bit more manageable:

```
dt |>
  slice_head(n = 10)
```

A tibble: 10 x 9

	dyad	ccode1	ccode2	year	cowmidonset	cowmidspell	cont	majpower	capdist
	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
1	2020	2	20	1920	0	0	1	1	735.
2	2020	2	20	1921	0	1	1	1	735.
3	2020	2	20	1922	0	2	1	1	735.
4	2020	2	20	1923	0	3	1	1	735.
5	2020	2	20	1924	0	4	1	1	735.
6	2020	2	20	1925	0	5	1	1	735.
7	2020	2	20	1926	0	6	1	1	735.
8	2020	2	20	1927	0	7	1	1	735.
9	2020	2	20	1928	0	8	1	1	735.
10	2020	2	20	1929	0	9	1	1	735.

This finalized dataset has the following columns:

- **dyad**: a 4-6 digit numerical code unique to each dyad pair.
- **cocode1**: the COW code for country 1 in a dyad.
- **cocode2**: the COW code for country 2 in a dyad.
- **year**: the year a dyad exists.
- **cowmidonset**: a 0-1 indicator for whether a MID (militarized interstate dispute) began in a given year between a country pair.
- **cowmidspell**: a count of the number of years since the last ongoing MID between a pair of countries.
- **cont**: a 0-1 indicator for whether a dyad is contiguous.
- **majpower**: a 0-1 indicator for whether one of the countries in a dyad is a major power.
- **capdist**: the distance in kilometers between country capitals.

Alright, now that I have some data, now I need to model the chance of war, which is the subject of the next section.

5.3 Modeling dyadic data

If the dyadic dataset is the canvas for peace science research, the *logistic regression model*, also known as the *logit* model, is paint brush. If you've never heard of a regression model of any sort before, I highly recommend you check out [some resources I written on regression models](#) for my DPR 201 course.

While regression models, and statistical models in general, can seem intimidating to the uninitiated, they actually aren't too complicated. They are tools for measuring how one or several different factors explain variation in an outcome of interest, and they often express this relationship as a linear function, sometimes with a non-linear transformation applied.

For example, a linear regression model specifies that an outcome variable is just a linear additive function of separate explanatory variables and random noise:

$$\overset{\text{outcome}}{\tilde{y}_i} = \underbrace{\beta_0 + \beta_1 x_i + \beta_2 z_i}_{\text{linear function}} + \overset{\text{noise}}{\tilde{\epsilon}_i}$$

A logit model is just a variation on this idea. It has some of the same moving pieces, but also has some important differences that are tailored to solving a specific problem with data. Here's what a logit model looks like:

$$\Pr(y_i = 1) = \Lambda(\beta_0 + \beta_1 x_i + \beta_2 z_i)$$

The $\Lambda()$ represents the *logistic* function, which when written explicitly, looks like this:

$$\Lambda(x) = \frac{e^x}{e^x + 1}$$

The logistic function is where the logit model, or *logistic* regression model, gets its name. But what's the point of this transformation? Why do peace scientists, with rare exceptions, use the logit model to study war?

The answer is simple: (1) conflict onset between a pair of countries is a *binary* variable that can only take the values 0 or 1, and (2) conflict between countries is rare enough that using a linear model will lead to erroneous linear predictions of the probability of conflict. The latter problem is especially important. Probabilities are restricted to the unit interval (a fancy way of saying they can't be less than 0 or greater than 1). Linear regression models don't abide by this limitation. Logit models, to the contrary, do.

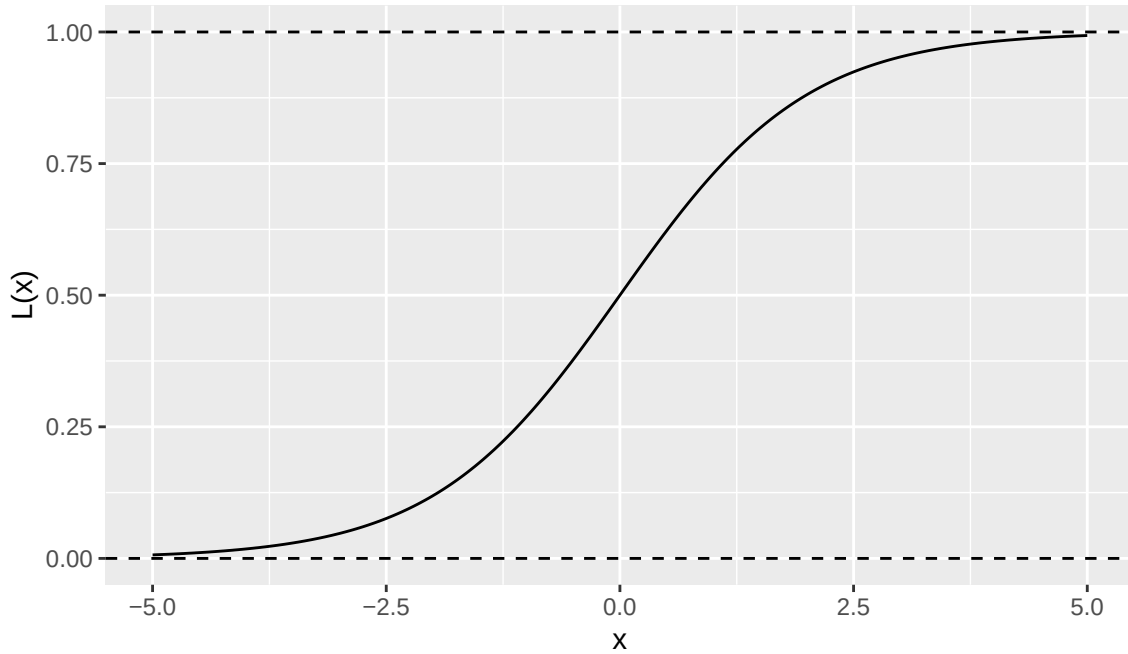
Here's why. The logistic transformation can take any continuous variable x and restrict it to the unit interval (between 0 and 1). Let me show you with some simple R code:

```
## create the logistic function and some x values
L <- function(x) exp(x) / (exp(x) + 1)
x <- seq(-5, 5, by = .01)

## plot how L(x) varies given x
ggplot() +
  aes(x, L(x)) +
  geom_line() +
  geom_hline(
    yintercept = 0:1,
    lty = 2
  ) +
  labs(
    title = "The logistic function in action",
    subtitle = "The classic 'S' or 'sigmoid' logistic transformation"
  )
```

The logistic function in action

The classic 'S' or 'sigmoid' logistic transformation



The above example shows what the logistic function does. It's a sigmoid or s-shaped function that takes numerical data with any real numbered value and forces it to fit between 0 and 1. You can see why it would be used to model probabilities.

Let's apply the logit model specifically to the problem of modeling dyadic conflict onset with the data I created above. I might write out the following formal model specification to represent what I plan to do with my statistical software:

$$\Pr(\text{MID}_{dt} = 1) = \Lambda[\beta_0 + \beta_1 \text{contiguity}_{dt} + \beta_2 \text{major}_{dt} + \beta_3 \ln(\text{distance}_{dt}) + \tau(\text{spell}_{dt})]$$

The above model, in English, says that the probability that a militarized interstate dispute (MID) breaks out between a dyad d in year t is predicted by dyadic contiguity, whether one of the countries is a major power, and the natural log of the distance between them (distance is a skewed variable, so researchers commonly take the log of distance to shrink in big outliers). The $\tau(\text{spell}_{dt})$ expression is shorthand for a cubic peace spell trend. This is a term that ends up in nearly all peace science logit models. It helps adjust for the the history of peace versus conflict between a pair of countries. A peace spell is just the number of years since the last ongoing conflict between a pair of countries.

Note that in the logit model, each of the predictors of interest is multiplied by a *coefficient*, denoted by β . The direction and magnitude of this coefficient controls how much and in which

direction the probability of conflict onset changes based on that predictor. The problem is that these coefficients are unknown. They need to be estimated. How?

All models require two things for estimation: (1) an objective function and (2) an estimator. An objective function provides a measure of how well different combinations of unknown parameters fit the data. An estimator is a rule for identifying the best set of parameters in light of the objective function.

For the logit model, the objective function is a likelihood function specified as:

$$\sum_i \Lambda(\beta_0 + \beta_1 x_i)^{y_i} \times [1 - \Lambda(\beta_0 + \beta_1 x_i)]^{1-y_i}$$

The best parameters are the ones that maximize this likelihood function. Thankfully, in practice, you don't have to solve this maximization problem. Your software will do it for you.

Executing this model in R is quite easy to do. R is a statistical programming language after all, and this is a statistical model. Here's simple code for estimating the regression model I specified earlier:

```
glm(  
  cowmidonset ~ cont + majpower + log(capdist) + poly(cowmidspell, 3),  
  data = dt,  
  family = binomial  
) -> fit
```

The function `glm()` stands for *generalized* linear model. It can be used to estimate linear models, but also models such as logit that apply a non-linear transformation to the right-hand side of the model equation to force the data to better fit certain characteristics of the outcome.

The first thing I gave `glm()` was details about my regression model using R's formula syntax. This lets me tell it what I want my outcome variable to be and which variables in my data I want to use to predict it.

I then tell `glm()` the dataset it needs to go looking in to find these variables by writing `data = dt`.

Finally, I specify `family = binomial` to indicate that I want to estimate a logit model. This syntax is actually shorthand for saying `family = binomial(link = "logit")`.

If I want to take a look at the model results, I can use the `summary()` function on the model object called `fit` I created in the code above:


```
summary(fit)
```

Call:

```
glm(formula = cowmidonset ~ cont + majpower + log(capdist) +  
     poly(cowmidspell, 3), family = binomial, data = dt)
```

Deviance Residuals:

Min	1Q	Median	3Q	Max
-1.0670	-0.0663	-0.0412	-0.0299	6.0427

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-4.22455	0.17067	-24.753	<2e-16 ***
cont	2.13119	0.05247	40.620	<2e-16 ***
majpower	1.67629	0.03652	45.897	<2e-16 ***
log(capdist)	-0.32679	0.01991	-16.410	<2e-16 ***
poly(cowmidspell, 3)1	-633.63571	31.81307	-19.917	<2e-16 ***
poly(cowmidspell, 3)2	99.72789	52.44293	1.902	0.0572 .
poly(cowmidspell, 3)3	-733.55661	39.22941	-18.699	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 45541 on 918314 degrees of freedom
Residual deviance: 33636 on 918308 degrees of freedom
AIC: 33650

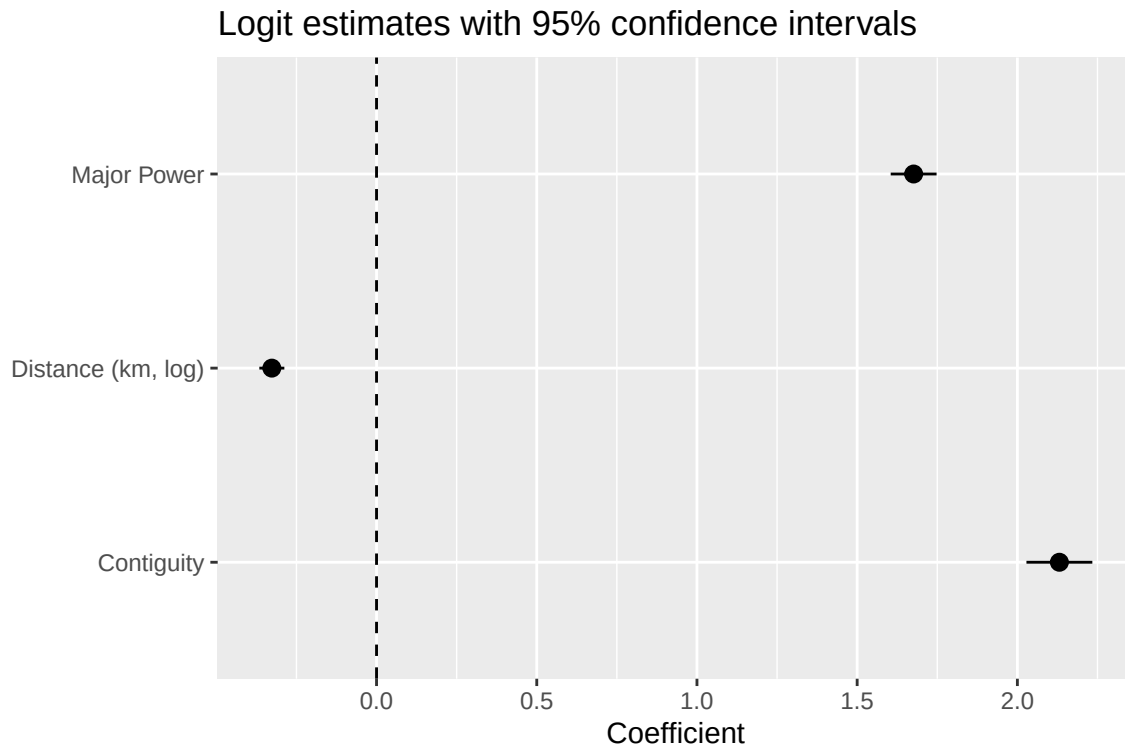
Number of Fisher Scoring iterations: 10

There's a lot of info that this function returns, but most of the action is under the **Coefficients:** heading. This is a table that, for each explanatory variable in the model, reports the coefficient, standard error, test statistic (known as a z-statistic), and p-value. The coefficient is a measure of how much the chance of conflict changes based on a change in the explanatory variable. The remaining values capture statistical uncertainty (the role of random noise) in generating these coefficients. If the p-value is less than 0.05, this means that the coefficient value is statistically significant. That is, the null hypothesis that the coefficient should be zero can be rejected. Looking at the results above, it looks like contiguity, major power status, and distance all have a statistically significant relationship with the probability of a MID between a pair of countries. Based on what you know about the Braumoeller-Carson

measure of opportunities for war, these values make sense. The coefficients for contiguity and major power status are positive, meaning contiguous dyads and those where at least one is a major power have a higher probability of fighting. The coefficient for the log of distance is negative, meaning dyads where countries are farther apart have a lower probability of fighting.

These results can be more accessibly summarized in a coefficient plot, as I do in the below code. Coefficient plots summarize model output by reporting the coefficient as a point and summarizing statistical significance using 95% confidence intervals, which have the appropriate coverage for determining if an estimate is statistically different from some value of interest (in this case, zero). The below code will make a coefficient plot specifically for the three factors of interest: contiguity, major power status, and distance.

```
fit |>
  broom::tidy() |>
  mutate(
    lower = estimate - 1.96 * std.error,
    upper = estimate + 1.96 * std.error
  ) |>
  slice(2:4) |>
  ggplot() +
  aes(estimate, term, xmin = lower, xmax = upper) +
  geom_pointrange() +
  geom_vline(
    xintercept = 0,
    lty = 2
  ) +
  scale_y_discrete(
    labels = c("Contiguity", "Distance (km, log)", "Major Power")
  ) +
  labs(
    title = "Logit estimates with 95% confidence intervals",
    x = "Coefficient",
    y = NULL
  )
```



Beyond noting statistical significance and the direction of relationships, the raw output from logit models isn't very intuitive to understand. Because of the nature of the logistic transformation, the coefficient for a given predictor represents the change in the logged odds of the outcome taking the value 1 per a unit change in the predictor. I don't know about you, but I don't really know what that means in practical terms.

For this reason, it's often recommended that you report simulated predicted values from logit models rather than only report the raw model coefficients. This task is easy to do in R as well, with the help of the `{ggeffects}` package. Check out the below code. It uses the `ggpredict()` function from the `{ggeffects}` package to get predicted values for MID onset from the logit model based on contiguity while keeping the other model factors fixed at their mean values (this is the default. You can change this if you want.). These values can then be plotted however you want using `ggplot()`.

The below code shows the conditional probabilities of conflict based on contiguity. Note that when I show two or more conditional probabilities from a logit model, I use 84% CIs instead of 95% CIs. As I discussed in a previous chapter, I can use the overlap or non-overlap of 84% CIs to judge statistical significance, just as I can use 95% CIs to judge statistical significance based on the overlap or non-overlap of these intervals with zero.

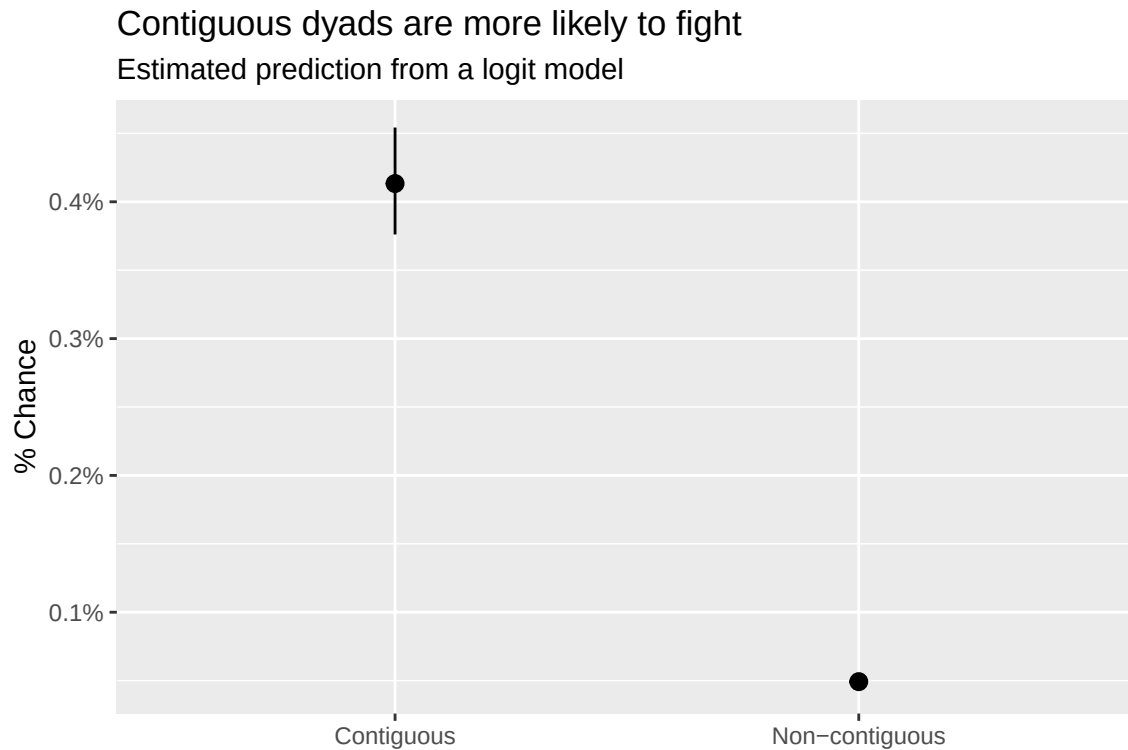
```

## open {ggeffects}
library(ggeffects)

## get predictions
ggpredict(fit, terms = "cont", ci_level = .84) |>
  as_tibble() -> preds

## plot results
preds |>
  mutate(
    x = ifelse(x == 1, "Contiguous", "Non-contiguous")
  ) |>
  ggplot() +
  aes(x, predicted, ymin = conf.low, ymax = conf.high) +
  geom_pointrange() +
  scale_y_continuous(
    labels = scales::percent
  ) +
  labs(
    title = "Contiguous dyads are more likely to fight",
    subtitle = "Estimated prediction from a logit model",
    x = NULL,
    y = "% Chance"
  )

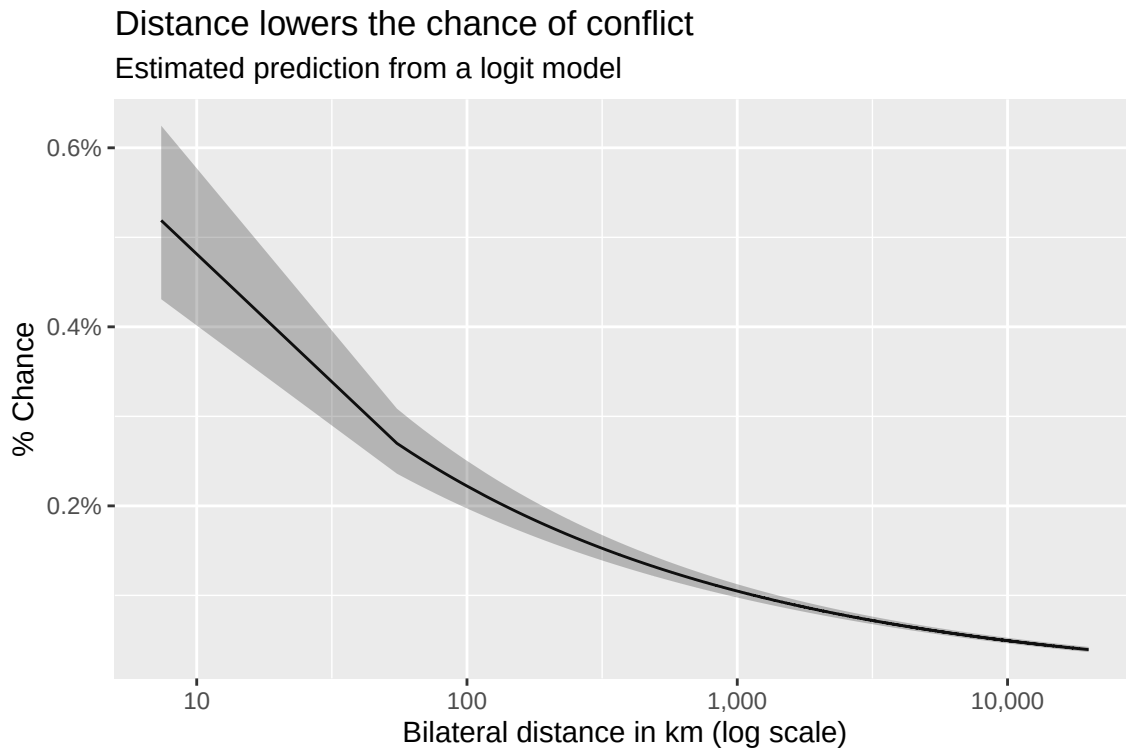
```



You can do a similar process for each of the other covariates, but depending on whether the predictor is binary or continuous, the execution will be a bit different. For example, here's how this process would look for distance, which is continuous:

```
ggpredict(fit, "capdist", 0.84) |>
  as_tibble() |>
  ggplot() +
  aes(x, predicted, ymin = conf.low, ymax = conf.high) +
  geom_line() +
  geom_ribbon(
    alpha = 0.3
  ) +
  scale_x_log10(
    labels = scales::comma
  ) +
  scale_y_continuous(
    labels = scales::percent
  ) +
  labs(
    title = "Distance lowers the chance of conflict",
    subtitle = "Estimated prediction from a logit model",
```

```
x = "Bilateral distance in km (log scale)",  
y = "% Chance"  
)
```



All in all, the data offers some good confirmation of the logic beyond the Braumoeller-Carson measure of conflict opportunities. The countries most prone to “rolling the iron dice,” to resurrect a metaphor I introduced in previous chapters, are those that are neighbors, major powers, or those that are relatively close geographically.

5.4 Some additional recommendations

The model results shown above nearly conform to the best practices in peace science research, but one big factor remains: robust inference. Logit models don’t on their own account for the fact that dyads interact over time, which means one dyad in a given year is also correlated with the same dyad in other years. Controlling for peace spells, as I did in the model, helps a bit, but not perfectly, because the problem has to do with the uncertainty created by repeated interactions over time.

There are two ways around this problem. The first is to simply estimate standard errors that adjust for this, known as cluster-robust standard errors. The second is to model this feature of the data directly in your logit specification. My preference is the second, but the first is more the norm, mainly because it's easier to do.

Before I tell you how to implement each approach, let me quickly explain why I like the second one better.

As I teach my 201 students, an assumption like independence of observations (the idea that no two or more data points influence each other) only matters when you're doing statistical inference for linear regression models. Independence has no bearing on the linear model estimates themselves. This isn't the case with many *generalized* linear models which make stronger assumptions of the data. In the case of logit models, a violation of independence (e.g., the fact that one dyad in one year is correlated with this same dyad in other years) means that your model estimates (coefficients) will be biased in addition to your standard errors. Therefore, taking your model coefficients as is and then applying robust standard errors after the fact only fixes half of the problem.

Modeling dependence directly fixes both problems at once. Importantly, using this approach doesn't guarantee that everything is fine. You still can get biased results if you have a poorly specified model. But in practice I think it offers better precision in model estimates and reduces bias in your model coefficients if done well.

Because this is just my preference, let me show you how to implement both approaches. As you'll see, in both cases I need to use the dyad identifier column I added to the data to indicate which observations should be treated as clustered (dependent).

First, robust standard errors:

```
## open {lmtest} and {sandwich}  
library(lmtest)
```

Loading required package: zoo

Attaching package: 'zoo'

The following objects are masked from 'package:base':

```
as.Date, as.Date.numeric
```

```
library(sandwich)

## rerun coefficient test with robust standard errors clustered by dyad
fit |>
  coeftest(
    vcov. = vcovHC(fit, clusters = dt$dyad)
  ) |>
  broom::tidy() |>
  mutate(
    lower = estimate - 1.96 * std.error,
    upper = estimate + 1.96 * std.error,
    Method = "Robust SE"
  ) |>
  slice(2:4) -> app1
```

Second, here's how to model dyadic dependence directly. To implement this approach, I need to use the `gam()` function from the `{mgcv}` package, and then include `s(dyad, bs = "re")` in the formula. This tells `gam()` that I want to include *dyadic random effects* in the model. These let me account for between dyad differences, and within dyad dependence simultaneously, while also allowing me to make partial comparisons between dyads for my other model predictors. When I use this approach, I can use the standard errors directly produced from the model:

```
library(mgcv)
```

Warning: package 'mgcv' was built under R version 4.2.3

Loading required package: nlme

Attaching package: 'nlme'

The following object is masked from 'package:dplyr':

`collapse`

This is mgcv 1.9-0. For overview type 'help("mgcv-package")'.


```

gam(
  cowmidonset ~ cont + majpower + log(capdist) + poly(cowmidspell, 3) +
    s(dyad, bs = "re"),
  data = dt,
  family = binomial
) -> new_fit

new_fit |>
  coeftest() |>
  broom::tidy() |>
  mutate(
    lower = estimate - 1.96 * std.error,
    upper = estimate + 1.96 * std.error,
    Method = "Random Effects"
  ) |>
  slice(2:4) -> app2

```

Once I have the results I can compare them:

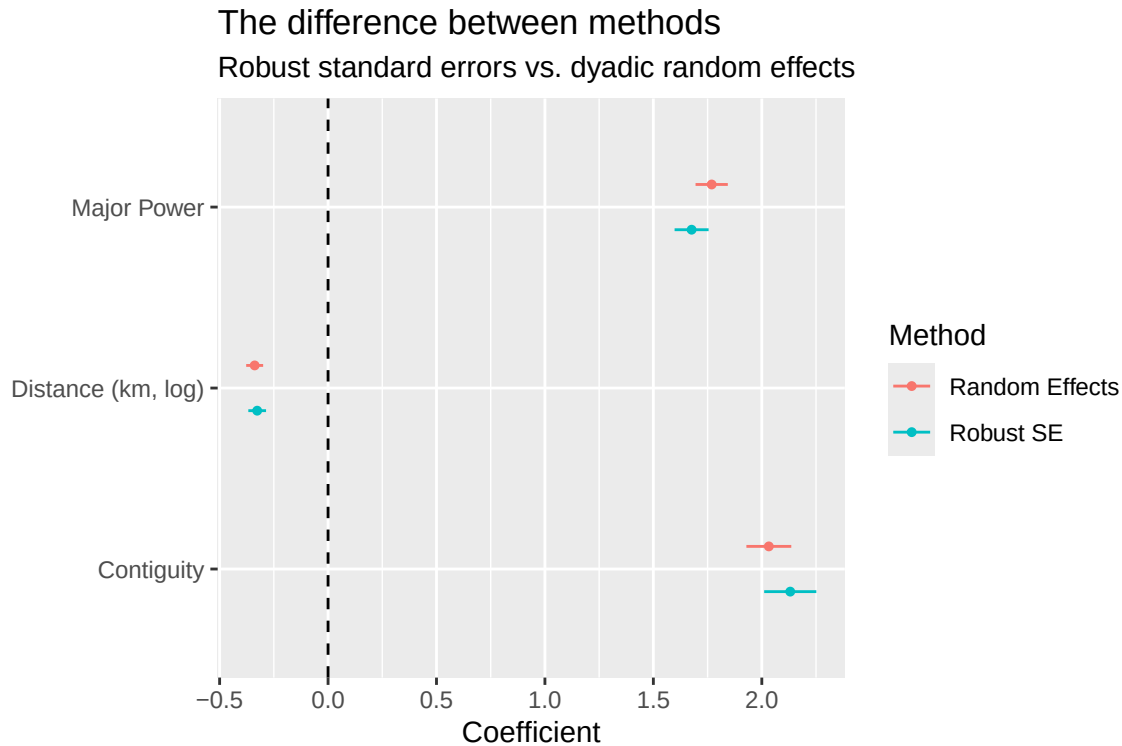
```

bind_rows(app1, app2) |>
  ggplot() +
  aes(estimate, term, xmin = lower, xmax = upper) +
  geom_pointrange(
    aes(color = Method),
    position = position_dodge(-.5),
    size = .1
  ) +
  geom_vline(
    xintercept = 0, lty = 2
  ) +
  scale_y_discrete(
    labels = c(
      "Contiguity",
      "Distance (km, log)",
      "Major Power"
    )
  ) +
  labs(
    title = "The difference between methods",
    subtitle = "Robust standard errors vs. dyadic random effects",
    x = "Coefficient",

```

```
y = NULL  
)
```

Warning: `position\_dodge()` requires non-overlapping x intervals.



The differences aren't substantial, but they do exist, and depending on what all you include in your model, differences can be potentially more substantial. I've personally seen variables shift from being statistically significant to not (and vice versa), and some variable estimates shift direction. Going forward, I'll primarily use the robust standard error approach since this is more the norm, but I wanted to make sure you knew about a different option that I think offers some important advantages (useful to keep in mind for a final project).

5.5 Summary

I covered a lot of ground in this chapter. For that I'm sorry. To do peace science requires understanding some more -than-basic statistical tools. But the good news is that few peace science studies deviate very far from the methods I used here. You can take what you learn from here and apply it general to nearly any international conflict study you can conceive.

However, while the methods change very little from one study to the next, as you'll see in the coming chapters when I introduce different theoretical arguments for why countries fight, the real art of peace science is measuring sometimes difficult to measure concepts so that you can plug them into a statistical model like logit. This requires making difficult judgment calls and an ability to *write* a compelling argument that convinces other researchers that your theoretical and empirical contribution to the field is sound and worth publishing.

6 The Balance of Power and War

Main ideas:

- **Balance of Power vs. Power Transition:** Power as a cause of war has been long-debated by conflict scholars, and two different theoretical approaches, BoP and PT, hypothesize very different relationships between power and the chance of war.
- **What is power?** Power is usually measured in material terms. The most commonly used measure is called the Composite Indicator of National Capability or CINC.
- **CINC and War:** Using CINC to measure the distribution of power between countries, a dyadic conflict analysis offers contradictory results depending on whether you use a directed or non-directed dataset.

6.1 What is power?

Long ago, the famed political scientist Robert Dahl (1957) defined power as the ability of A to get B to do what B otherwise would not. While Dahl's understanding of power remains debated to this day, one thing is clear from his definition: power is relational. It doesn't exist in a vacuum.

Another thing is also clear: power's influence is hard to measure. By Dahl's definition, to identify the effects of power you need to observe what B would have done absent the intervention of A. In international relations, where A and B are countries, this is almost impossible.

Hans Morgenthau (1978), an important international relations scholar and theorist, took a more pragmatic approach to conceptualizing power. He identified nine components of national power: geography, resources, industrial capacity, military preparedness, population, national character, national morale, quality of diplomacy, and quality of governance.

A strength of this way of conceptualizing power over Dahl's is that it provides researchers with many tangible metrics. Today, most conflict scholars measure power in terms more akin to Morgenthau than Dahl. The most commonly used measure is the Correlates of War (COW) Composite Indicator of National Capability (CINC) (D. J. Singer 1987; J. D. Singer, Bremer, and Stuckey 1972). A quick glance at Google Scholar for the search term "CINC score" returns 1,200 search hits between 2000 and 2025. It's hard to find a conflict study that doesn't use this measure.

CINC scores are composed of six different metrics, each intended to capture a distinct dimension of national power:

- Military expenditures
- Military personnel
- Iron and steel production (in tons)
- Primary energy consumption (in coal-ton equivalents)
- Total population
- Urban population

CINC is calculated as a country's world share of the six component measures above. It can take values anywhere from 0 to 1. A 1 would mean that a country is 100% responsible for all six of the above metrics; a 0 would mean that a country is responsible for 0%.

Other measures of power exist as well, all of which focus on observable metrics like CINC, but which are used much less widely. In the early years of peace science research, prior to the development of CINC, some scholars used gross domestic product (GDP) to measure power because it captures the capacity to build up military forces. More recently, Anders, Fariss, and Markowitz (2020) proposed a concept they call *surplus* domestic product, which is a country's GDP minus minimum expenditures required to keep a population alive at the bare minimum level of subsistence. The idea behind this measure is to quantify resources available to a country after meeting basic needs, or as they put it, spending on "bread before guns or butter."

Carroll and Kenkel (2019) offer yet another approach to quantifying power using machine learning and predictive modeling. Using an "ensemble" learner, they use each of the individual components used to construct CINC as separate predictors of military victory in dyadic conflicts. Their measure of power is equivalent to the probability that, if country A and B fight, A wins.

Other new measures are coming out on a regular basis, and I suspect even more will emerge over the coming years (especially as more researchers leverage AI to come up with fixes for older approaches).

Having looked at and compared many these measures myself, I can say one thing definitively: These measures aren't identical, but they are all strongly correlated with each other. Practically speaking, this means using one measure over the others is unlikely to make a massive difference in your research. But, you should still use these measures wisely and transparently. If your statistical findings change dramatically based on the measure you use, it's unwise (and dishonest) to hide this fact from your readers.

In the analysis that follows, I'll use CINC since this is the most commonly used. But don't assume that it's the only correct measure of power.

Ultimately, when it comes to defining and measuring power, conflict scholars have adopted a purely empirical approach, focusing on observable metrics of economic size or realized military might. They tend to ignore harder to quantify metrics like national morale or successful diplomacy and soft power (the ability to persuade). For this reason, it's helpful to think of all of the measures of power I just identified as *proxies* for power. They aren't power itself, but they are correlated with it (at least, most conflict scholars think so).

6.2 Why measure power?

What's the point of trying to measure power? The simple answer is that different theoretical arguments exist about power and war. I can't possibly cover all the competing arguments here, so I'll focus on two of the most prominent.

The first argument is rooted in the international relations school of thought known as *Realism*. For realists, the defining characteristic of international relations is a state of *anarchy*, or the absence of a world government to enforce laws. In such a world, countries must adopt self-help strategies to survive, which includes amassing military forces for self-defense or, even, offense. Many realist theories argue that creating a *balance of power* (BoP) is required to ensure peace. If one country has a preponderance of power, no others can serve as a check on its ability to impose its will on other countries, which inevitably leads to conflict and war. Power parity, therefore, is predicted to lead to a greater chance of peace.

BoP's chief competitor is *power transition* (PT) theory, which views international relations as hierarchical rather than anarchic. This view holds that the balance of power isn't as important for explaining patterns in conflict as *changes* in the balance of power. PT holds that rising powers are likely to be more aggressive in their foreign policies and, thus, more prone to be a source of conflict and war in the international system. Part of what contributes to their aggression is the presence of a *hegemon*, a country that possesses a preponderance of power that acts as the hierarch of the international system. Rising challengers, in the PT view, challenge the hegemonic power which has largely succeeded in shaping the international system in favor of its own interests. The rising power, dissatisfied with the status quo, seeks to change the international system in ways that better align with its own preferences, and this could lead to war.

BoP and PT have many sub-variants, but at their core, they consistently adopt opposing understandings of the role of power in international conflict. BoP predicts that parity promotes peace. PT predicts that parity promotes war. BoP further predicts that stronger states are more conflictual, while PT predicts weaker, rising powers are more conflictual.

More than an academic issue, identifying which of these frameworks is the best has important policy implications for world leaders. The obvious policy prescription with BoP is for countries to pursue parity in their military capabilities. For any one country, this might look like maintaining parity (or near parity) with your peers and taking steps to prevent any other

country from amassing too much military might. PT prescribes different policies, such as maintaining a preponderance of power and offering proportionate concessions to rising powers to minimize their dissatisfaction as they grow in strength.

The stakes are quite high, because the policies one approach prescribes, the other proscribes. Operating as if one approach is true when the other actually has better explanatory power could be disastrous.

6.3 Testing BoP and PT

Let's go to the empirical record and see what the data says. Do countries with more power than others start fights more often, as BoP predicts, or do countries with less power tend to start fights more often, as PT predicts?

I want to walk you through a quick example (with R code) of a simple analysis that tests the competing predictions of BoP and PT head-to-head. While there are a wide variety of approaches to testing these theories, and arguable counter points to the approach I'll take here, I think you'll see that even my simple approach helps to explain an ongoing tension in the BoP-PT debate: a simple change to my research design yields contradictory findings.

I'll start with gathering some data. I'll use `{peacesciencer}` to get started, which is the package I introduced in the previous chapter. I'll also use tools from the `{tidyverse}` and some extra helper functions I've created to make analyzing data made with `{peacesciencer}` less of a headache, and for bringing in the MIE-based measures of conflict initiation rather than using the COW MID dataset. Here's the code you need to run to get started:

```
library(tidyverse)
library(peacesciencer)
source(
  "https://raw.githubusercontent.com/milesdwilliams15/death-destruction-data/refs/heads/main,
)
```

The first thing I need to do is make a dataset. When it comes to doing peace science research, remember that most studies use dyadic data, but I have to make a choice. Should the data be *directed* or *non-directed*?

The below code will make a *directed*-dyad-year dataset. Remember that a directed dataset helps answer questions about *which* of two countries is most likely to initiate a fight with the other. This is a natural choice for testing competing claims from BoP and PT, because these theories make claims about which country is more likely to be aggressive based on its power relative to other countries. BoP says more powerful countries will start more fights, while PT says weaker countries pose a more serious threat. To test this, I need to model the *direction* of conflictual behavior. Hence, I need a directed-dyad-year dataset.

The below code will produce one that has all the variables I need. I start by using `create_dyadyears()` to make a directed-dyad-year dataset from 1816 to 2014. I then pipe in some conflict data using `add_icd_mics()`, which provides information at the dyad-level about militarized interstate confrontations (MICs) as summarized in the MIE dataset (Gibler and Miller 2024b), which I used in previous chapters. This function provides information not just about whether a MIC began or is ongoing between a pair of countries, but also information about which country initiated the conflict and a count of peace years (or peace “spells”) for each country pair, which I need to control for in my analysis. I then bring in data from the COW National Material Capabilities dataset, which provides CINC scores for all countries in the COW state system over time (D. J. Singer 1987; J. D. Singer, Bremer, and Stuckey 1972). As noted above, these are a 0-1 scale of a country’s global share of material capabilities in a given year. Finally, I bring in data to control for opportunities countries have to fight: contiguity, major power status, and bilateral distance.

```
## create a directed-dyad-year dataset
create_dyadyears(
  subset_years = 1816:2014
) |>
  ## add dyadic MIC dataset
  add_icd_mics() |>
  ## add CINC scores
  add_nmc() |>
  ## add contiguity, major power, and distance controls
  add_contiguity() |>
  add_cow_majors() |>
  add_cap_dist() -> dt
```

Now that I have the data, I just need to implement a few recodes. First, I need to calculate the dyadic power ratio by dividing country 1’s CINC score by country 2’s. This is a common measure in conflict research used to account for the balance of power between a pair of countries. It’s somewhat ambiguous, however, whether this should be the raw ratio, or if it’s better to take the log. I prefer the latter, so this is what I’ve done in my code below. I also included recodes to clean up my control variables for opportunities for war and made my dyadic identifier, which I need for calculating clustered standard errors. (Note that the code to make the dyadic identifier is a bit different. Because dyads appear twice per year, I need to account for this.)

```
## some recodes
dt |>
  mutate(
    ratio = log(cinc1 / cinc2),
    cont = ifelse(conttype > 0, 1, 0),
    major = pmax(cowmaj1, cowmaj2, na.rm = T),
```



```
ldist = log(capdist),
dyad = 1000 * pmin(ccode1, ccode2) + pmax(ccode1, ccode2)
) -> dt
```

Now that I have everything finalized, I can analyze the data. The below code will estimate a logit model of MIC onset initiation by country 1 in each directed-dyad-year pair as a function of the log of the power ratio, contiguity, whether one of the countries is a major power, and the log of bilateral distance. It also includes a cubic peace years trend. Note that the function I'm using is `glm_robust()` rather than `glm()` which I used in the last chapter. This is a function that I've added to my helper functions. It's a wrapper for `glm()` that automatically specifies that it should estimate a logit model, and it reports robust standard errors, which can be clustered based on whatever variable you specify with the `clusters` option. The output is saved as an object called `fit1`, which is a tidy data frame of model estimates and summary statistics (an ideal format for plotting).

```
## estimate logit model with cluster-robust 95% cis
glm_robust(
  miconset_init1 ~ ratio + cont + major + ldist +
    micspell + I(micspell^2) + I(micspell^3),
  data = dt,
  clusters = "dyad"
) -> fit1

## report the output
fit1
```

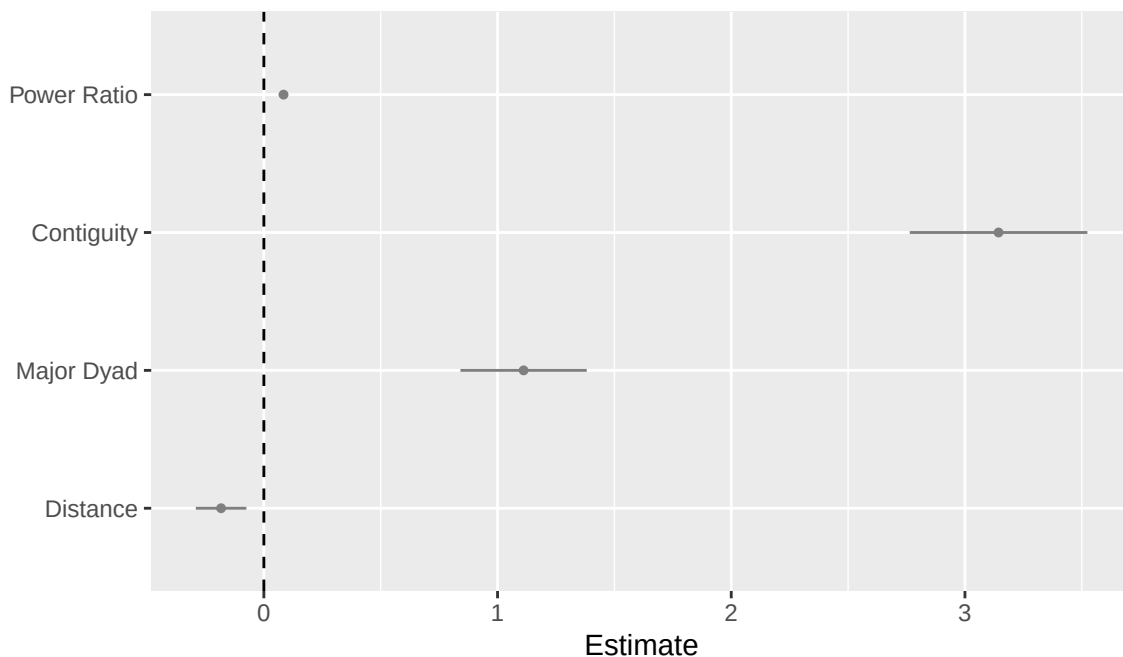
```
# A tibble: 8 x 7
  term          estimate std.error statistic  p.value    lower    upper
* <chr>          <dbl>     <dbl>     <dbl>   <dbl>    <dbl>    <dbl>
1 (Intercept)   -5.22      0.493     -10.6  3.32e-26 -6.19     -4.26
2 ratio         0.0843   0.00832     10.1  3.67e-24  0.0680     0.101
3 cont          3.14     0.194      16.2  4.42e-59  2.76       3.52
4 major         1.11     0.138      8.06  7.39e-16  0.841      1.38
5 ldist        -0.183   0.0552     -3.31  9.26e- 4 -0.291     -0.0747
6 micspell     -0.151   0.0109    -13.8  1.48e-43 -0.172     -0.129
7 I(micspell^2) 0.00232   0.000320     7.25  4.23e-13  0.00169    0.00294
8 I(micspell^3) -0.0000101 0.00000215  -4.67  2.99e- 6 -0.0000143 -0.00000584
```

The results are easier to see in a coefficient plot. I added a function called `coef_plot()` to my helper functions file that takes the output from `glm_robust()` and runs ggplot code under the hood to produce a decent looking coefficient plot so I don't have to write a bunch of extra code to get the same result.

```
## coef plot of results
fit1 |>
  slice(2:5) |>
  coef_plot(
    coef_map = c(
      "Power Ratio",
      "Contiguity",
      "Major Dyad",
      "Distance"
    )
  ) +
  labs(
    title = "The results support BoP",
    subtitle = "Logit estimates with cluster-robust 95% CIs"
  )
```

The results support BoP

Logit estimates with cluster-robust 95% CIs



As you can see, the results support BoP rather than PT. According to the model estimates, after controlling for opportunities to fight and peace years, the greater country 1's power relative to country 2's, the more likely 1 is to initiate a conflict with 2.

Before jumping to any hasty conclusions, let me show you how a subtle change to the data

yields a very different finding. One factor that probably too few peace scientists pay attention to is the conceptual importance of *directed* versus *non-directed* datasets. Say I didn't care about modeling the direction of aggression between countries and instead did a non-directed dyad analysis. The below code will make the necessary changes to the data, re-estimate the regression model, and report the results, this time comparing them to those shown above in the same coefficient plot. Do you see what's changed?

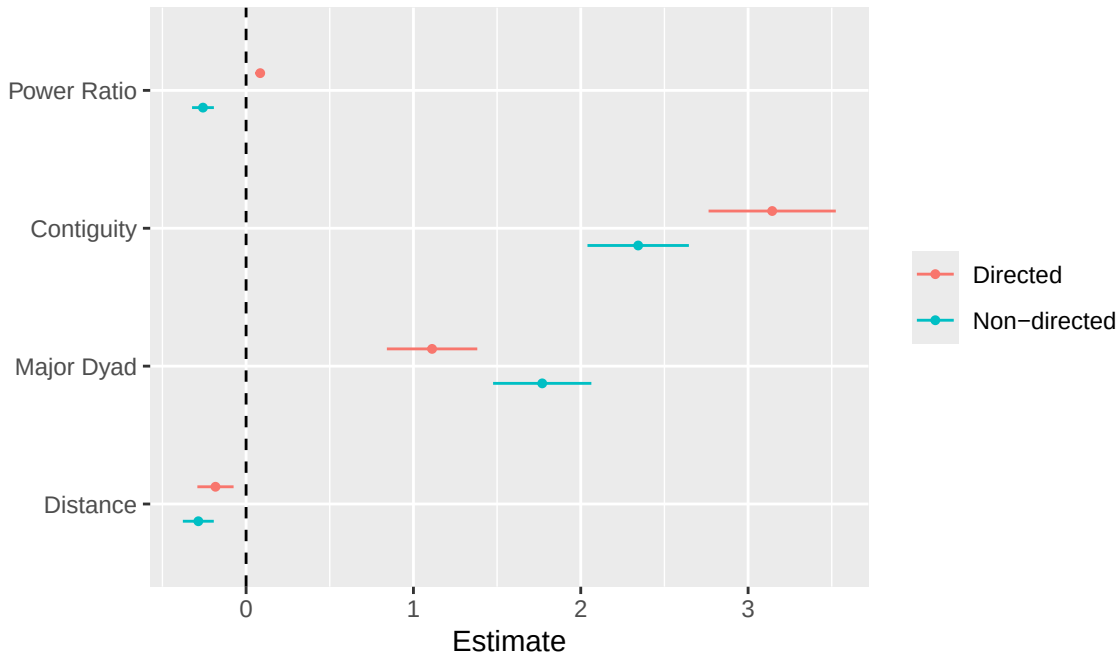
```
## make the data non-directed
dt |>
  filter(ccode1 < ccode2) |>
  mutate(
    ratio = log(pmax(cinc2 / cinc1, cinc1 / cinc2, na.rm = T))
  ) -> ndt

## estimate logit model
glm_robust(
  miconset ~ ratio + cont + major + ldist +
    micspell + I(micspell^2) + I(micspell^3),
  data = ndt,
  clusters = "dyad"
) -> fit2

## summarize results and compare
bind_rows(
  fit1 |> slice(2:5) |> mutate(model = "Directed"),
  fit2 |> slice(2:5) |> mutate(model = "Non-directed")
) |>
  coef_plot(
    coef_map = c(
      "Power Ratio",
      "Contiguity",
      "Major Dyad",
      "Distance"
    )
  ) +
  labs(
    title = "The results support PT in non-directed dyads",
    subtitle = "Logit estimates with cluster-robust 95% CIs"
  )
```

The results support PT in non-directed dyads

Logit estimates with cluster-robust 95% CIs



Amazingly, in a non-directed dyad analysis, the data supports PT rather than BoP. Country pairs where there's a bigger imbalance of power are less likely to fight. What's going on here?

I have to confess that I don't really know, but this slight, seemingly innocuous tweak that gave me statistically significant results that contradict the first set of findings, should raise alarm bells. And this is the point I wanted to make: it's possible to analyze the data in such a way that supports either theory.

This is the most important challenge associated with peace science (and social science generally), and it demonstrates why you shouldn't just start cobbling together a grab-bag of variables and running analyses without first thinking very carefully about the choices you're making and why. I think this point is even more important now in an age of pattern-engine-driven workflows, where AI agents can easily make subtle choices, against your wishes, or on their own without your input, that dramatically affect results. For this reason, even if down the road a tool like Claude ends up writing and running 99% of your code, you still need to understand what it's doing, and *you* need to be in the driver's seat. Control the tool; don't let it control you.

If all you remember from this exercise is that subtle choices make big differences, that will be enough for me.

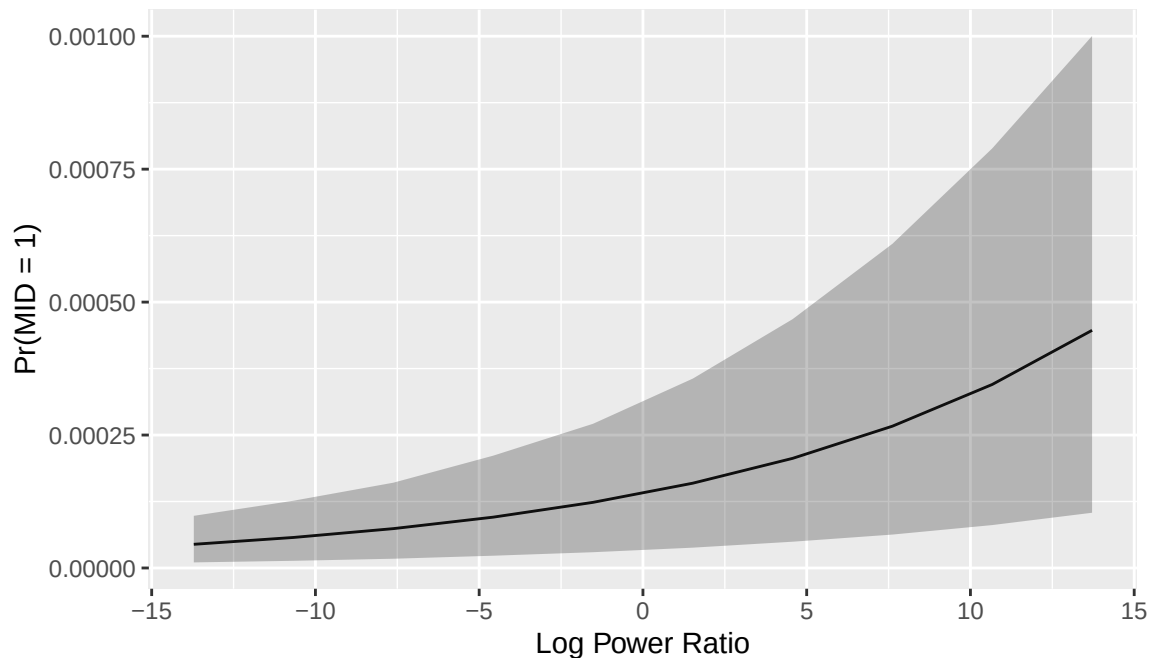
And one more thing before I wrap up. In addition to the results appearing contradictory, they also have little practical significance. The below code will produce a prediction plot

based on the model fit with the directed dyad dataset, and the results appear below. The predicted change in the probability of side 1 in a country pair initiating a conflict with side 2 is negligible.

```
tibble(
  ratio = seq(min(dt$ratio), max(dt$ratio), len = 10),
  cont = mean(dt$cont),
  major = mean(dt$major),
  ldist = mean(dt$ldist),
  micspell = mean(dt$micspell)
) -> ndata
sim_pred(fit1, ndata) |>
  group_by(ratio) |>
  summarize(
    pred = mean(p),
    lower = quantile(p, .08),
    upper = quantile(p, .92)
  ) |>
  ggplot() +
  aes(ratio, pred, ymin = lower, ymax = upper) +
  geom_line() +
  geom_ribbon(alpha = .3) +
  labs(
    title = "Power ratio has little practical significance for conflict",
    subtitle = "Directed dyad model",
    x = "Log Power Ratio",
    y = "Pr(MID = 1)"
  )
)
```

Power ratio has little practical significance for conflict

Directed dyad model



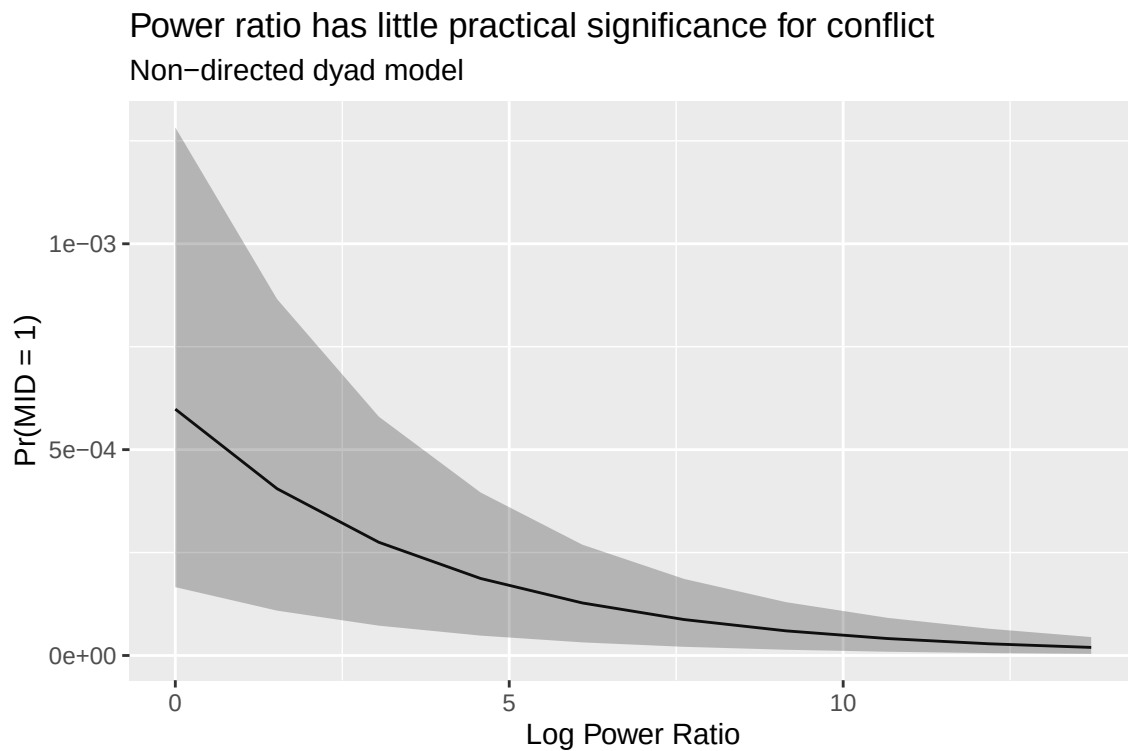
The same is true for the non-directed model. The below code produces another prediction plot like the first, but this time for the model fit with the non-directed dyad data. Again, you can see that the balance of power predicts a negligible difference in the chance of conflict between a pair of countries. What this tells me is that BoP and PT, if they are useful theories, have limited utility in explaining most of the variation in international conflict.

```
tibble(  
  ratio = seq(min(ndt$ratio), max(ndt$ratio), len = 10),  
  cont = mean(ndt$cont),  
  major = mean(ndt$major),  
  ldist = mean(ndt$ldist),  
  micspell = mean(ndt$micspell)  
) -> ndata  
sim_pred(fit2, ndata) |>  
  group_by(ratio) |>  
  summarize(  
    pred = mean(p),  
    lower = quantile(p, .08),  
    upper = quantile(p, .92)  
  ) |>  
  ggplot() +
```

```

aes(ratio, pred, ymin = lower, ymax = upper) +
geom_line() +
geom_ribbon(alpha = .3) +
labs(
  title = "Power ratio has little practical significance for conflict",
  subtitle = "Non-directed dyad model",
  x = "Log Power Ratio",
  y = "Pr(MID = 1)"
)

```



6.4 Summary

Social science research involves tough judgment calls and subjective choices where there isn't one correct option. This becomes an issue when your findings depend on your choices. Ideally, you want to see findings that aren't so brittle. When it comes to BoP and PT, unfortunately, the results are brittle indeed.

This doesn't mean all is lost. As a social scientist, seeing contradictory findings is an invitation to solve a puzzle. The question you should ask about BoP and PT is: *why* are the results

contradictory? A good follow up would be: is something else going on?

These are the kinds of questions that can become the basis for new research. Maybe you're the one to start looking for answers.

7 The Democratic Peace

Main ideas:

- **Democratic Peace:** Democracies rarely fight each other, though they fight non-democracies at normal rates. This empirical pattern is one of the most widely accepted findings in conflict research.
- **Debate Over Causes:** Scholars disagree about why the democratic peace exists. Some argue democracy itself promotes peace through accountability, audience costs, or shared norms, while others argue peace is driven by correlated factors like economic interdependence (the “capitalist peace”).
- **Measuring Democracy:** Researchers use different datasets to measure democracy—such as the Polity 2 index, the V-Dem polyarchy index, and the extended Unified Democracy Scores (eUDS)—which capture different aspects of democratic governance but generally correlate with one another.
- **Empirical Testing:** A directed-dyad-year dataset can be used to test whether the likelihood of conflict depends on the democracy levels of both countries, but to test the complex and condition relationship implied by the democratic peace, this requires estimating an *interaction model*.
- **Key Finding and Puzzle:** Evidence supports the democratic peace (democracies avoid fighting each other), but results can also suggest a “non-democratic peace,” raising questions about measurement, model specification, and alternative explanations for peace.

7.1 What is the Democratic Peace?

I don’t recall when I first learned about the “democratic peace,” or even when I first heard the term. For most peace scientists narrowly, and political scientists generally, the democratic peace is common knowledge and one of the few regularities in the social sciences that occurs with *law-like* consistency. It belongs in the pantheon of laws in the natural and hard sciences, placed squarely between the law of gravity and the second law of thermodynamics.

I’m being hyperbolic, but only a little. The democratic peace is a persistent pattern that shows up in the data, and few question its empirical accuracy.

So what is the *democratic peace*? Dean Babst (1964) coined the term decades ago, and it describes the fact that democracies rarely fight each other, even while they fight other countries just as much as everyone else. This finding persists across a wide range of datasets and definitions of conflict and measures of democracy.

The irony with the democratic peace is that, while widely accepted as an empirical fact, little agreement exists on exactly *why*. An entire cottage industry of research has coalesced around trying to explain the democratic peace. It's a subject written about so much that once a graduate school professor of mine told me not to do a project on it because it'd been beaten like a dead horse. As of March 2026, a Google Scholar search for "Democratic Peace" returns over 6,400 articles and books published since 2022. If I extend my search to articles published any time, the number of pieces found totals almost 45,000. A dead horse indeed.

So much work exists on the subject because the implications for global peace are profound. Could spreading democracy reduce the chance of war? Or is democracy correlated with something else that seems to profoundly lower the chances of conflict between countries?

Each of these questions gestures toward two main schools of thought that have developed around the democratic peace. The first argues that democracy itself is the cause. The second argues that democracy and peace are both correlated with other factors that are actually the cause of peace.

Work in the first school tends to theorize that the leaders of democratic regimes pay attention to the regime of other countries when making a determination about whether to go to war. This idea helps answer why the democratic peace is localized specifically to democratic dyads. Other accounts emphasize the idea that democracies are less apt to go to war because citizens would bear most of the costs of fighting. Elected leaders have to pay attention to the interests of their constituents, and this moderates their aggression toward other countries. However, this argument fails to account for the fact that democracies fight non-democracies at normal rates. Clearly, if democracy is the cause of peace, part of the mechanism must be that democracies pay attention to the regimes of potential targets.

Even among those who accept that leaders base decisions on the regimes of other countries, further disagreement exists on exactly why. Some argue that leaders of democracies have to be more reliant on popular support for their policies, which makes them more formidable adversaries in a conflict when they actually go to war. Others argue that leaders in democracies face *audience costs* if they back down from a threat or conflict, which increases their resolve. The idea is that citizens will punish their leaders at the ballot box for appearing weak on the global stage. Since democracies know this about each other, they are less apt to provoke fellow democracies who will face strong domestic pressure to respond to international threats.

Others argue that democracies behave differently toward each other because of shared norms. Negotiation and compromise (theoretically) are central to the democratic process domestically. Therefore, the argument goes, democracies extend these same domestic norms to their interactions with other democracies. Conversely, democracies don't apply these norms to countries they view as non-democratic.

Unfortunately, while these arguments make a lot of sense, definitive evidence that any of these mechanisms is at work remains elusive. This creates space for alternative arguments which posit that democracy and peace are simply correlated with other confounding factors. One prominent alternative explanation, which I'll introduce in greater detail in the next chapter, centers on trade. Many democracies also tend to be important trading partners, and this could explain why democracies rarely fight each other. This school of thought is more closely aligned with the "capitalist peace," which argues financial and economic interdependence reduces the chance of war. This idea isn't inherently in conflict with the democratic peace (both could be true at the same time), but some invoke the capitalist peace as competing explanation rather than a complementary one.

Nonetheless, plenty of researchers remain committed to the idea that democracy itself promotes peace (at least among democratic countries), and in the applied example below I'll introduce you to a particular approach to analyzing the data that reveals the conditional nature of the democratic peace: namely, that democracies appear to behave differently toward fellow democracies compared to everyone else.

But first, I think it's important to spend some time thinking about democracy as a concept and how to measure it.

7.2 Measuring Democracy

Just as the democratic peace is a hotly debated and widely written about subject, so is democracy itself. An R package called `{democracyData}` provides access to 39 distinct datasets that measure democracy, and on Google Scholar a search for the term "Democracy" returns almost 5 *million* results.

With so much literature to contend with, knowing where to start is hard to do. To keep things simple, I'll introduce you to just three distinct measures of democracy that are directly accessible using the `{peacesciencer}` package. Compared to many of the alternatives, these three measures are among the most commonly used.

First, there's the Polity 2 Index from the Polity IV Project (Marshall, Gurr, and Jaggers 2017). Only recently has Polity advanced to the Polity 5 Project, but as of now `{peacesciencer}` only provides access to version IV. This isn't really a big deal, though, because little has actually changed, and version IV provides sufficient data coverage for the time-frame covered by the Militarized Interstate Events (MIE) dataset I've been using to measure international conflict in previous chapters (Gibler and Miller 2024b).

Polity's approach to measuring democracy emphasizes constraints on country executives (presidents or prime ministers in democracies, and "supreme leaders" of one sort or another in non-democracies). Countries receive numerical scores across a range of dimensions by expert coders for things such as how executives are selected, whether and to what extent they can act without impunity, and whether and to what extent other governing bodies can check their

power. Once all the relevant scores are noted, they're added up. The result is a score from -10 to 10 which quantifies how democratic or autocratic a country is. This score is known as the Polity 2 Index. A 10 on the scale is as democratic as a country can possibly be, while a -10 is as autocratic as one can be.

One of the chief competitors with Polity's measure of democracy is the Variety of Democracy (V-Dem) Project's polyarchy index (Coppedge et al. 2020). Much like Polity 2 scores, the polyarchy index quantifies quality of democracy on a single numerical scale. In this case, scores range from 0 to 1, where 1 is as democratic as a country can be, and 0 is as autocratic as a country can be. However, the difference between V-Dem and Polity's measures extends well beyond a difference in scale. V-Dem has an entirely different approach to measuring democracy.

The first main difference is reliance on expert surveys rather than a team of dedicated in-house coders. V-Dem regularly reaches out to a sample of democracy or country/area specialists and asks them questions about the regimes of specific countries.

The second main difference is that V-Dem takes into consideration a broader set of issues than Polity when quantifying democracy's quality, such as the extent to which laws constrain who has the right to vote in elections, freedom of expression, and other factors many people would associate with "liberal" societies (and "liberal" $\neq$ "progressive" in this context, but in the broader sense, rooted in liberal ideals such as individual rights, rule of law, freedom of speech and assembly, and so on).

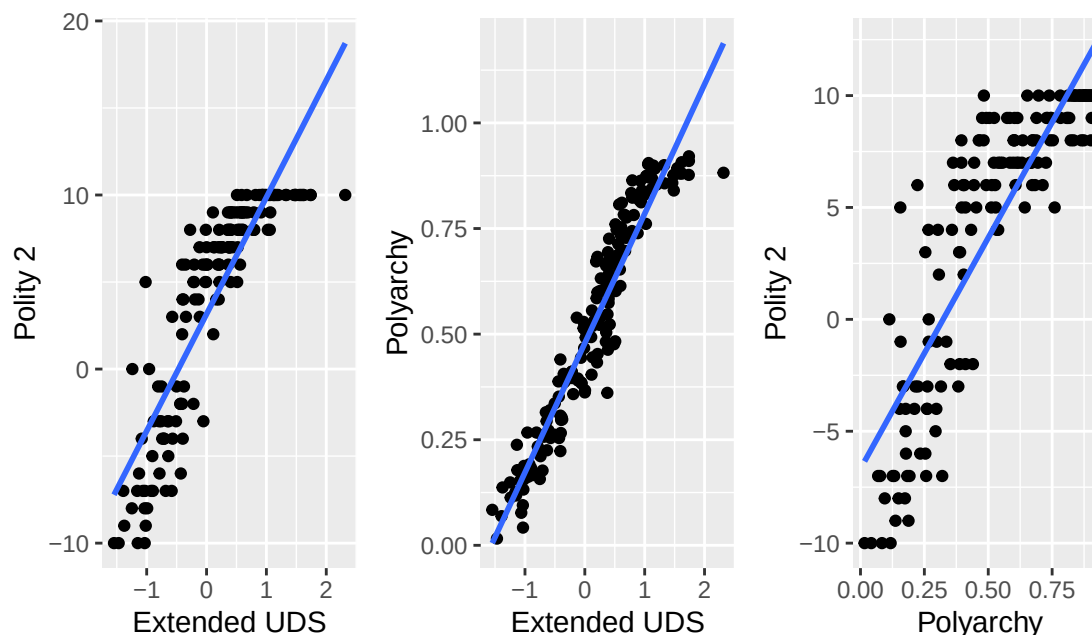
The final measure accessible with `{peacesciencer}` is the Unified Democracy Scores index originally created by Pemstein, Meserve, and Melton (2010) and later extended by Marquez (2016). The extended UDS (eUDS) index takes yet another philosophical approach to quantifying the quality of democracy. It is actually based on a combination of different unique democracy indexes, and it's constructed via a Bayesian model that creates a novel *latent* democracy score based on the range of unique democracy scores countries have across a range of different datasets. You can think of eUDS as a compromise or pragmatic approach that treats different prominent democracy measures as potentially equally informative about a country's quality of democracy at a given point in time while not giving any single measure pride of place.

These three measures (Polity 2, polyarchy, and eUDS) have their pros and cons. Which you select might depend on the kind of research question you want to answer. For example, if you're most concerned about the impact of leader constraints, you might have a preference for using Polity 2. Alternatively, if you care about other dimensions of democracy, you might prefer to use V-Dem's polyarchy index.

Whichever measure you choose, you should be aware of two important features of the data. The first is that these measures of democracy are closely correlated with each other, which should tell you that, whatever their differences in philosophy, they're all picking up similar trends. You can see this in the plot below which shows the correlation between each of these three measures of democracy in 2018.

All three measures of democracy are correlated

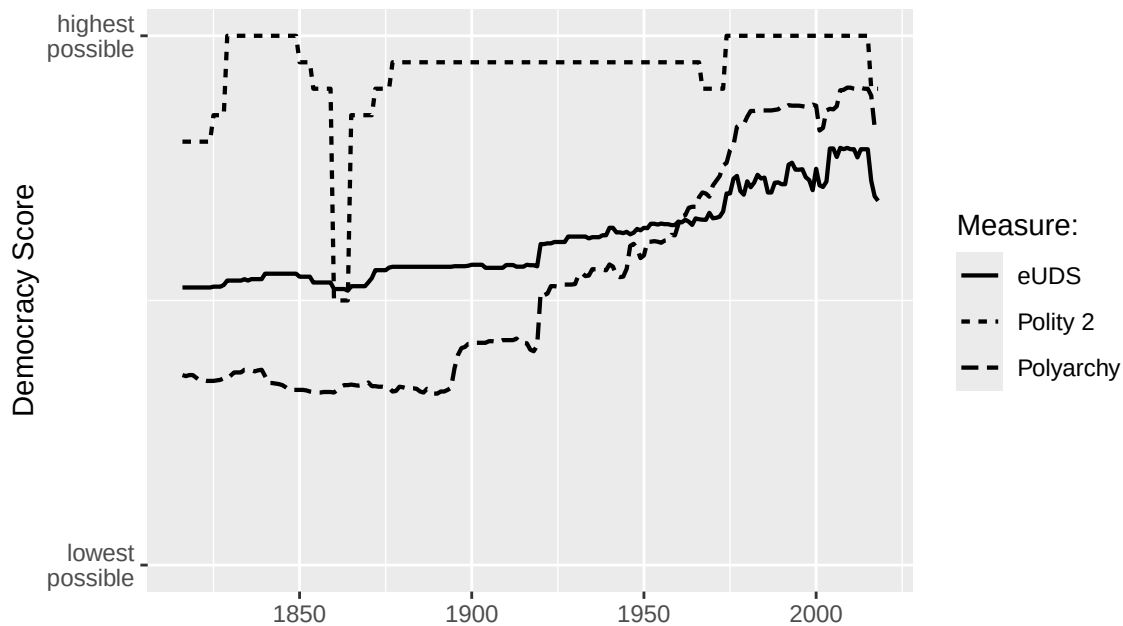
Measure by measure scatter plots for 2018



The other thing you should be aware of is that these measures can diverge in potentially substantial and important ways, especially the farther back you go in time. This can happen, not just for marginal cases, but very important ones such as the United States. The below plot shows the trend in America's quality of democracy based on the three measures I've been discussing. I put them all on a common scale so that values could be easily compared. The farther back in time you go, the more discrepancy you can see between these measures in their evaluation of America's quality of democracy. They really only start to converge after 1950, and in the 1800s the gaps are substantial. According to Polity's measure, the US has been as strong as a democracy can possibly be for most of its history, but according to V-Dem's measure, the US was closer to being an autocracy than a democracy for all of the 19th century and the early decades of the 20th century. This shows just how much the different emphases these measure place on dimensions of democratic rule can influence the results.

Major discrepancies in quality of democracy for the US

Trends from 1816 to 2018. All measures rescaled from lowest to highest possible values.



Given the similarities and differences across measures of democracy, you should be careful about how you proceed in your analysis. Think carefully about what you consider to be essential elements of democratic rule, or more narrowly about the elements you consider relevant to your research. Behind each measure of democracy is a theory of democracy and a set of value judgments about what matters for judging democracy's quality.

7.3 Testing the Democratic Peace

The `{peacesciencer}` package doesn't provide you access to every measure of democracy that exists, but it does give you access to three measures that capture a range of approaches to quantifying democracy. At one extreme you have Polity 2, which places substantial emphasis on chief executives. At the other extreme you have V-Dem's polyarchy index, which considers a range of issues many consider essential to "liberal" democracy broadly defined. In the middle there's the extended UDS index which relies on a wide variety of democracy measures (including Polity 2 and polyarchy, and many others) to generate a single measure of democracy.

Let's take one of these measures for a test drive to see if it supports the democratic peace. Since I'm asking you to consider it the "compromise" measure, I think it makes sense to use

extended UDS. I'll leave it to you to think about the alternatives and to figure out on your own if they offer complementary or conflicting findings.

I'll start with some code to create the data. Because the theories that explain the democratic peace argue that democracies behave differently toward fellow democracies versus non-democracies, a *directed*-dyad-year dataset makes sense. The goal is to test how countries with certain characteristics modify their behavior toward others based on their characteristics. The below code does this and saves the output as an object called `dt`. Note the use of the `add_democracy()` function. This is the `{peacesciencer}` function that populates a dataset with information about country democracy scores based on the three approaches discussed in the previous section.

```
## packages and tools
library(tidyverse)
library(peacesciencer)
source(
  "https://raw.githubusercontent.com/milesdwilliams15/death-destruction-data/refs/heads/main,
)

## make the data
create_dyadyears(
  subset_years = 1816:2014
) |>
  add_icd_mics() |>
  add_democracy() |>
  add_contiguity() |>
  add_cow_majors() |>
  add_cap_dist() -> dt

## clean up the controls
dt |>
  ## controls
  mutate(
    cont = ifelse(conttype > 0, 1, 0),
    major = pmax(cowmaj1, cowmaj2),
    ldist = log(capdist),
    dyad = 1000 * pmin(ccode1, ccode2) + pmax(ccode1, ccode2)
  ) -> dt
```

If you look at the democracy measures in the dataset, you'll see 8 in total. Four for country 1 in the dyad, and four for country 2. There are four for each because there are two versions of the extended UDS measure. One has the raw scores, which are on a continuous scaled normalized to standard deviation units (`euds1` and `euds2`). The other is adjusted so that zero

is the mean (`aeuds1` and `aeuds2`). It doesn't really matter which you use, but as a matter of personal taste I prefer standardized measures that are centered around zero, so this is the measure I'll go with in my analysis.

As in the previous chapter, to analyze the data I'll use a logit model that controls for contiguity, major power status, and the distance between countries, along with a cubic peace years trend. However, to test the democratic peace, I need to do something more complicated than just throw measures of democracy in the model.

The democratic peace holds that the relationship between democracy and war is somewhat complicated. It proposes that the more democratic country 1 is, the less likely it is to initiate a conflict with country 2, *conditional* on how democratic country 2 is. This kind of hypothesized relationship calls for what's called an *interaction* model. In such models, the relationship between one predictor and the outcome is allowed to vary systematically as a function of another predictor in the model. The below formal logit specification shows what this implies:

$$\Pr(\text{MIC}_{dt} = 1) = \Lambda(\beta_0 + \beta_1 \text{eUDS1}_{dt} + \beta_2 \text{eUDS2}_{dt} + \beta_3 \text{eUDS1}_{dt} \times \text{eUDS2}_{dt} + \dots)$$

The “...” in the above just is a stand in for the standard set of controls I've used in all subsequent analyses (contiguity, major powers, distance logged, and cubic peace years). The thing I want you to pay attention to is the fact that democracy scores enter the equation in three different places. First, country 1's democracy score enters on its own, then country 2's enters on its own, and finally the product of both is included. This kind of model makes it possible to detect a more complex relationship between democracy and the chance that country 1 initiates a fight with country 2. It ensures that the predicted change in the probability of conflict initiation based on country 1's democracy score isn't just determined by a single slope parameter β_1 ; it's also determined by β_3 .

To spell it out more clearly, in this model, the change in the probability that country 1 initiates a conflict with country 2 is determined by:

$$\beta_1 + \beta_3 \text{eUDS2}_{dt}$$

The coefficient β_1 tells you a baseline relationship between country 1's eUDS index and the chance that it starts a fight with country 2. The coefficient β_3 tells you how much this relationship changes up or down depending on country 2's eUDS index.

Here's how to estimate this model in R. The below code uses the `glm_robust()` function to estimate the chance that country 1 initiates a militarized interstate confrontation (MIC) with country 2 based on each country's democracy score, controlling for their opportunities to fight and peace years. All should look familiar, except note the use of the `*` symbol between `aeuds1` and `aeuds2`. These are country 1 and country 2's democracy scores according to the extended UDS measure, respectively. This syntax is shorthand for specifying an interaction between a

set of variables. It specifies the model I formally laid out above that includes each measure of democracy as a separate term, and their product, as predictors in the model.

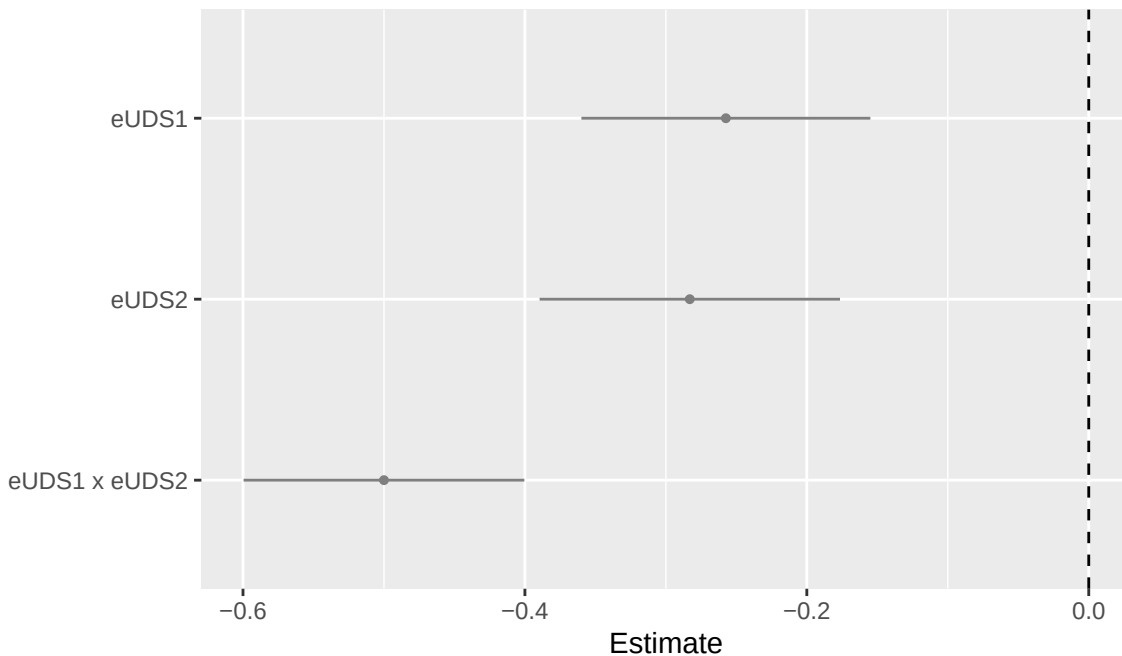
```
## using "*" introduces an interaction term
glm_robust(
  miconset_init1 ~ aeuds1 * aeuds2 +
    cont + major + ldist +
    micspell + I(micspell^2) + I(micspell^3),
  data = dt,
  clusters = "dyad"
) -> fit
```

You can see the results in the below coefficient plot. I just pulled out the estimates for democracy from the model. You can see the point estimate and its 95% confidence interval for each of the individual democracy measures, and for their product.

```
fit |>
  filter(
    str_detect(term, "aeuds")
  ) |>
  coef_plot(
    coef_map = c("eUDS1", "eUDS2", "eUDS1 x eUDS2")
  ) +
  labs(
    title = "Evidence of the democratic peace",
    subtitle = "Logit estimates for main and interaction terms of country democracy"
  )
```

Evidence of the democratic peace

Logit estimates for main and interaction terms of country democracy



When dealing with an interaction model, there are some things you need to keep in mind when interpreting the results. First, you need to be careful about making too much of the estimates for the individual measures that you interacted. In an interaction model these estimates have a very narrow interpretation. They tell you what the relationship is between one predictor and the outcome when the other predictor is zero. Many people mistakenly treat these as an overall average effect. Second, the estimate for the interaction term tells you how much each of the individual estimates will change as the other increases.

In the case of the results shown above, the estimates tell you that when country 2's democracy score is equal to zero (which is also equivalent to the mean democracy score, since the measure is centered around zero), the chance that country 1 initiates a MIC with country 2 goes down the more democratic country 1 is. Further, when country 1's score is zero (or average), as country 2 becomes more democratic the odds that country 1 initiates a MIC with 2 goes down as well. Further, the estimate for the interaction term (the product of country democracy scores) is negative, meaning the more democratic one or both of the countries is, the more this pulls the relationship between one country's democracy score and the chance country 1 initiates a MID with 2 in a negative direction.

All of that is quite complicated and unintuitive, which is why it's much better to simulate some predictions from interaction models to better understand how relationships differ under certain scenarios. The below code will generate a conditional prediction plot from the logit model I estimated above. It does so for two scenarios. First, it shows the probability that

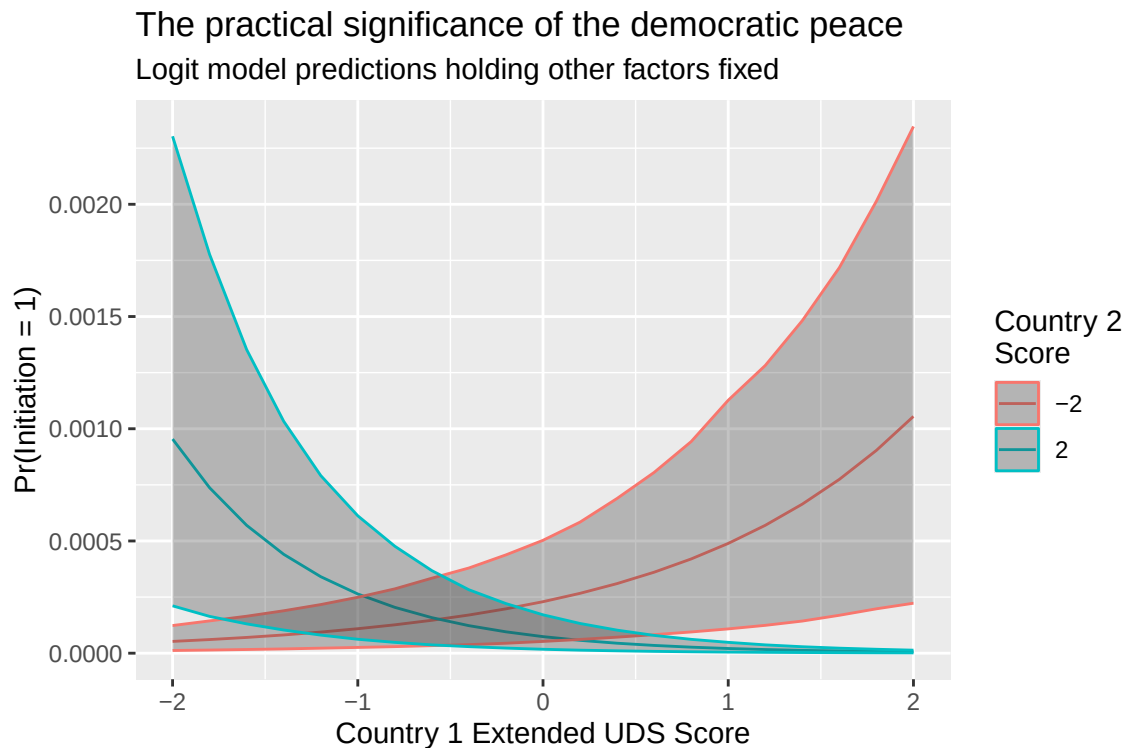
country 1 initiates a MIC with country 2 based on 1's democracy score holding country 2's fixed at -2 (because of the way the eUDS measure is constructed, this is equivalent to 2 standard deviations below the average eUDS score in the data). Second, it shows the probability that country 1 initiates a MIC with country 2 based on 1's democracy score holding country 2's fixed at +2 (or two standard deviations above the mean). The results clearly show that when country 2 is a non-democracy, the probability that it's attacked by country 1 *increases* the more democratic country 1 is. Conversely, if country 2 is a democracy, the probability that it's attacked by country 1 *decreases* the more democratic country 1 is.

```
sim_pred(
  fit,
  newdata = expand_grid(
    aeuds1 = seq(-2, 2, by = .2),
    aeuds2 = c(-2, 2),
    cont = mean(dt$cont),
    major = mean(dt$major),
    ldlist = mean(dt$ldlist),
    micspell = mean(dt$micspell)
  )
) -> preds

preds |>
  group_by(aeuds1, aeuds2) |>
  summarize(
    mean = mean(p),
    lower = quantile(p, 0.08),
    upper = quantile(p, 0.92)
  ) |>
  ggplot() +
  aes(aeuds1, mean, ymin = lower, ymax = upper,
      color = as.factor(aeuds2)) +
  geom_line() +
  geom_ribbon(alpha = .3) +
  labs(
    title = "The practical significance of the democratic peace",
    subtitle = "Logit model predictions holding other factors fixed",
    x = "Country 1 Extended UDS Score",
    y = "Pr(Initiation = 1)",
    color = "Country 2\nScore"
  )
)
```

`summarise()` has grouped output by 'aeuds1'. You can override using the

`` .groups`` argument.



In short, the model shows that democracies are less likely to attack fellow democracies compared to other countries, consistent with the democratic peace. But it also supports another conclusion that doesn't perfectly fit the conventional democratic peace story. The model shows that non-democracies are less likely to attack non-democracies than they are to attack democracies. That is, it also supports a *non*-democratic peace. Essentially, there are two pockets of peace, one among democracies and another among non-democracies.

Here's yet another puzzle to consider, which you either will find exciting or frustrating. What could be driving this? It could be that the eUDS measure is biasing the result. Maybe if you used one of the alternative democracy measures you'd see a pattern in the data that better aligns with the democratic peace.

It could also be that the model is mis-specified. Sometimes interaction terms can introduce unforeseen biases into an analysis. That could be going on here. Relatedly, it could also be the case that the model doesn't adequately control for confounding factors. While the logit model I estimated controls for sensible things (opportunities to fight and peace spells), it's fairly bare-bones by today's standards (which is on purpose since my goal is just to teach you something rather than engage in an exhaustive and rigorous analysis).

A third explanation is that democracy is picking up shared norms or common interests between democracies, on the one hand, and non-democracies, on the other hand. If this is true, it suggests there's room for modifying democratic peace theory.

Whatever the explanation, some follow-up analysis is needed. The work of the peace scientist is never done.

7.4 Summary

The democratic peace is probably the most ubiquitous and enduring finding in conflict research (the somewhat surprising results in my own example above notwithstanding). The central puzzle with the democratic peace centers on why it exists. The fact that it's real is generally accepted. The central tension in the literature is about whether democracy itself is the cause of peace, or if something else that both democracy and peace are correlated with is the actual cause. While many have tried to resolve this debate, definitive answers remain elusive.

Many other factors beyond democracy might offer more compelling, or at the very least, equally interesting explanations for why countries go to war as well. There is, of course, *power* which I talked about in the previous chapter, but another is economic interdependence (e.g., *trade*), which is the subject of the next chapter.

8 The Trading Peace

Main ideas:

- The **trading peace** describes the empirical reality that countries that have more trade interdependence are less likely to go to war
- **Trade interdependence theory** offers an explanation for why this pattern exists: trade dependence increases the opportunity cost of conflict
- **Three key criticisms** challenge the theory: trade rebounds quickly after wars end; war costs from trade loss correlate with confounding factors; and peace may cause trade rather than trade causing peace
- **Measurement:** trade interdependence is calculated as the minimum trade dependence (bilateral trade / GDP) between a country pair
- Empirical analysis produces **mixed results**, and significance only appeared after trying multiple model specifications, raising concerns about p-hacking
- The trading peace finding remains **empirically robust** in the broader literature, even if the exact causal mechanism is debated

8.1 Introduction

The **trading peace** is a close cousin of the democratic peace, which I discussed in the last chapter. This is mainly because both perspectives are situated in **liberal** international relations (IR) theories rather than **realist** IR theories. The latter are concerned mainly (in some cases only) with the distribution of power as a cause of war. The identities, regimes, history, and interconnections between countries are treated as irrelevant.

Both the democratic and trading peace perspectives take the opposing view. They assert that domestic institutions (democratic peace) and trade/commerce (trading peace) differentiate certain countries from others and affect how they behave toward each other.

This, however, is as far as their similarities go. Unlike some explanations for the democratic peace which emphasize shared norms of negotiation and compromise, the most prominent explanations for the trading peace center on material interests. While many schools of thought exist, the one I want to address in this chapter, which I'll call **trade interdependence theory**,

posits that greater mutual dependence on trade between a pair of countries raises the costs of war.

This theory adheres quite closely to the bargaining model of war. If you remember back to several chapters ago, this is the perspective Braumoeller (2013a) argues is most appropriate for defining “real war” as spelled out by Carl von Clausewitz (2003). The idea is that countries have a disagreement over a certain issue, and “rolling the iron dice” is one political choice leaders may take to successfully resolve this disagreement.

However, as a classic piece of IR literature argues, a negotiated settlement costs less than war (Fearon 1995). Why do countries fight then? As Fearon (1995) argues, they fight because they miscalculate, because of biases, or because of commitment problems. As Wagner (2000) would later argue, starting a conflict might be the only way countries can get a clear sense of who has the upper hand in a negotiation. That is, countries fight to learn the cost of war.

What does this have to do with trade? Simply put, the bargaining model of war assumes war is costly, and trade, in trade interdependence theory, is argued to be an important factor in calculating war’s potential cost. While there are exceptions, many analyses show that trade declines (often substantially) between countries when they go to war with each other (Anderton and Carter 2001). As the logic goes, the more countries trade with each other, the greater the cost of fighting, and thus the greater the incentive to resolve disputes peacefully.

Like the democratic peace, evidence of the trading peace is a fairly robust empirical finding. However, much like the democratic peace as well, this isn’t enough to save it from fierce debate over exactly why this pattern exists. Mousseau (2021) highlights three challenges to this perspective that remain unresolved.

The first is that research shows trade quickly returns to pre-war levels once a conflict ends, which helps to mitigate the cost of fighting. In addition, research shows that trade never actually ceases during war. Add to these facts that accurate numbers of trade flows during war are hard to validate, and clear evidence that trade makes war too costly to pursue starts to evaporate.

The second is that for opportunity costs to truly explain the trading peace, it must be that elasticity and substitutability of traded goods is random with respect to the level of trade between countries and conflict. This probably isn’t true. Some point out that more industrialized and developed countries can offset many of the costs to war created by a loss in trade with a potential adversary. Other factors unique to specific countries may offset costs as well. As a result, the cost of war due to loss of trade is at best uneven, and at worst deeply correlated with possible confounding factors.

The third problem is that of reverse causality. How do we know that countries don’t fight because of trade, or that countries trade to avoid fighting? While this concern doesn’t outright contradict trade interdependence theory, it does highlight a conceptual problem with the idea that trade *causes* peace. It can just as easily be argued that peace causes trade.

These debates aside, the trading peace remains an important finding in peace research, and the opportunity costs of trade interdependence a well-known explanation for why the trading peace exists.

Next, I want to show you a typical way to measure trade interdependence, and then I'll do an analysis to see if it supports the trading peace.

8.2 Quantifying Trade Interdependence

Oneal and Russett (1997) set the tone for future research on the trading peace with their measure of trade interdependence. Their analysis centered on non-directed dyads post World War II, and they defined trade dependence as the level of trade between two countries, divided by each country's gross domestic product (GDP), respectively. Basically, holding trade constant, the country with the *smaller* GDP is considered *more* dependent on trade. Trade *interdependence* is then measured as the minimum trade dependence between a pair of countries. Holding trade constant, trade dependence for the country with the *larger* GDP captures the level of trade interdependence for the dyad.

This measure can be constructed using the `{peacesciencer}` R package, but it does require some additional steps along the way. First, here's code to generate the raw data. It will produce a non-directed-dyad-year dataset from 1950 to 2014 that includes a measure of conflict onset using the MIE dataset (Gibler and Miller 2024b), all the usual control variables, bilateral trade data, and economic data that includes a measure of GDP. Trade data can be added to a dataset using the `add_cow_trade()` function, and economic data using the `add_sim_gdp_pop()` function. The original data comes from the Correlates of War Project (Barbieri, Keshk, and Pollins 2009), and for each country pair in the data, the function to add trade information returns trade flows into country 1 (imports) from country 2, and vice versa, in millions of current USD. (Note that to get this to work for dyadic data, you may need to run `download_extdata()` in the R console first).

```
## packages and tools
library(tidyverse)
library(peacesciencer)
source(
  "https://raw.githubusercontent.com/milesdwilliams15/death-destruction-data/refs/heads/main,
)

## make the data
create_dyadyears(
  subset_years = 1950:2014,
  directed = F
) |>
```



```

add_icd_mics() |>
add_cow_trade() |>
add_sim_gdp_pop() |>
add_contiguity() |>
add_cow_majors() |>
add_cap_dist() -> dt

## clean up the controls
dt |>
  ## controls
  mutate(
    cont = ifelse(conttype > 0, 1, 0),
    major = pmax(cowmaj1, cowmaj2),
    ldist = log(capdist),
    dyad = 1000 * pmin(ccode1, ccode2) + pmax(ccode1, ccode2)
  ) -> dt

```

To construct the measure proposed by Oneal and Russett (1997), I next need to measure imports plus exports for each dyad to measure total trade. Then, to measure how much country 1 is dependent on trade with country 2, I need to divide by country 1's GDP, and do the same for country 2 with its GDP. Finally, I'll take the minimum trade dependence between countries as my measure of trade interdependence.

I have all the raw data to do this, but there's just one problem. The measure of GDP I have in my data is in millions of 2017 *constant* USD, while import and export values are in millions of *current* USD. That is, researchers adjusted GDP data for inflation, but not trade. That's no good. So how do I fix it?

Thankfully, lots of people use the R programming language to study financial data, and adjusting for inflation is a fairly common task for these folks. That means, of course, there's an R package that solves this problem. It's called `{priceR}`. You can install it by running `install.packages("priceR")` in the R console.

Once you have the package installed, you can run the below code which will do all the work necessary to help adjust for inflation. What it will do, specifically, is add a new column to the dyadic dataset called `k` which is a constant that I can multiply trade values by that will convert them into 2017 constant values.

```

## get constant to adjust for inflation
library(priceR)
inflation_dataframe <- retrieve_inflation_data("US")

```

Validating iso2Code for US

Generating URL to request all 296 results
Retrieving inflation data for US
Generating URL to request all 66 results

```
countries_dataframe <- show_countries()
```

Generating URL to request all 296 results

```
tibble(  
  year = 1950:2014,  
  k = afi(  
    rep(1, len = length(year)),  
    year,  
    "US",  
    2017,  
    inflation_dataframe,  
    countries_dataframe  
  )  
) -> inf_dt  
  
## merge with the dyadic data  
dt |>  
  left_join(inf_dt) -> dt  
  
## calculate total dyadic trade and adjust for inflation  
dt |>  
  mutate(  
    trade = flow1 + flow2,  
    trade2017 = k * trade  
  ) -> dt
```

Now that I have a measure of trade that matches the 2017 constant USD values for GDP, I can construct a measure of trade interdependence, as I do in the below code:

```
dt |>  
  mutate(  
    interdependence = pmin(trade2017/pwtrgdp1, trade2017/pwtrgdp2, na.rm = T)  
  ) -> dt
```

Now, let's analyze the data to see if we find support for the trading peace.

8.3 Analysis

As I've done in all previous analyses, I'll use the `glm_robust()` function to estimate a logit model and return robust standard errors clustered by dyad.

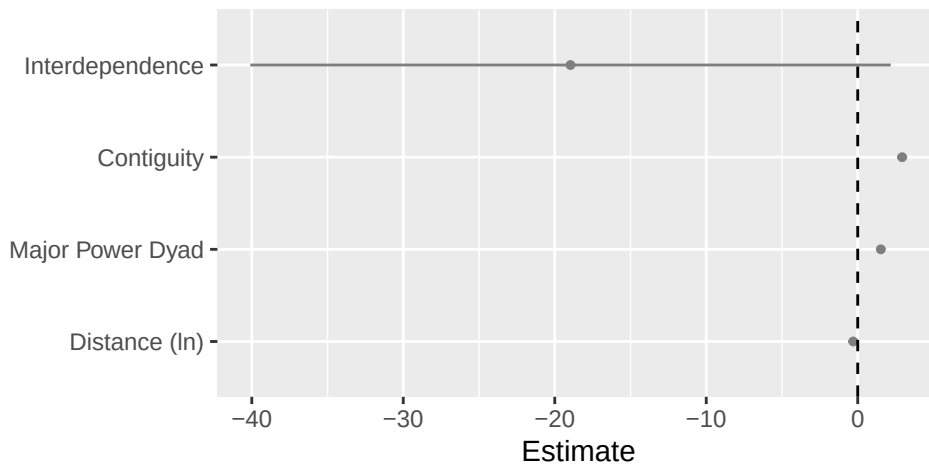
```
glm_robust(  
  miconset ~ interdependence + cont + major + ldist +  
    micspell + I(micspell^2) + I(micspell^3),  
  data = dt,  
  clusters = "dyad"  
) -> fit
```

Next, I'll make a coefficient plot and look at the results. Take a look. Does the analysis support the trading peace?

```
coef_plot(  
  fit |> slice(2:5),  
  coef_map = c(  
    "Interdependence",  
    "Contiguity",  
    "Major Power Dyad",  
    "Distance (ln)"  
  )  
) +  
  labs(  
    title = "Logit estimates for the trading peace",  
    subtitle = "Point estimates with cluster-robust 95% CIs shown"  
  )
```

Logit estimates for the trading peace

Point estimates with cluster-robust 95% CIs shown

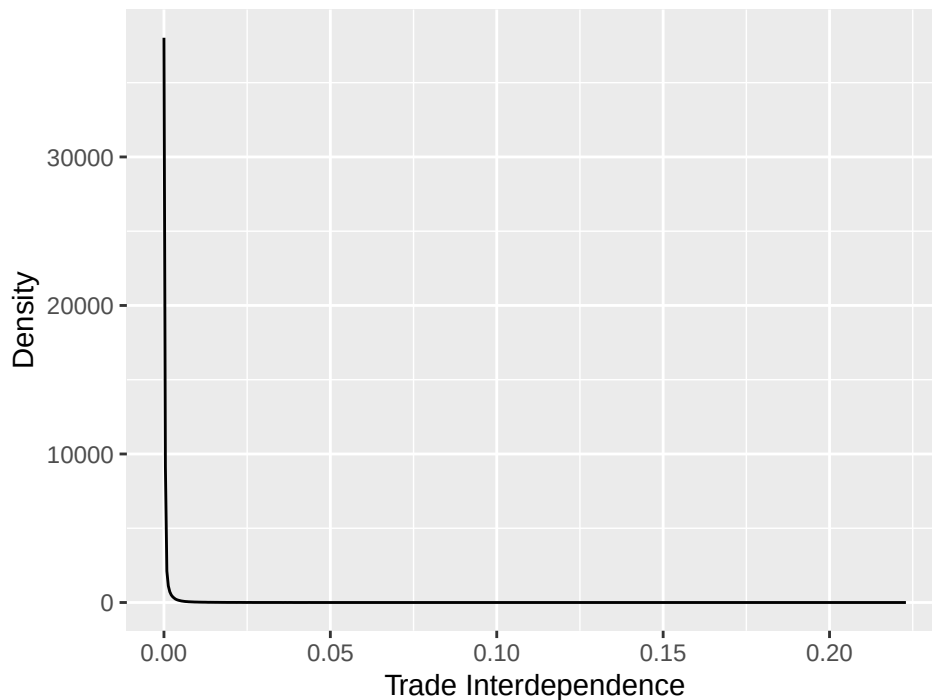


Much to my surprise, it does not. The estimate for trade interdependence is negative, as expected. It also is quite substantial. But it isn't statistically significant (the 95% confidence interval overlaps with zero). What's going on here?

I decided to do a bit of digging into the data. The first thing I noticed is that trade interdependence is a heavily skewed variable, which you can see in the density plot I made below. This could cause some problems.

```
ggplot(dt) +  
  aes(interdependence) +  
  geom_density() +  
  labs(  
    title = "Trade interdependence is heavily skewed",  
    x = "Trade Interdependence",  
    y = "Density"  
  )
```

Trade interdependence is heavily skewed

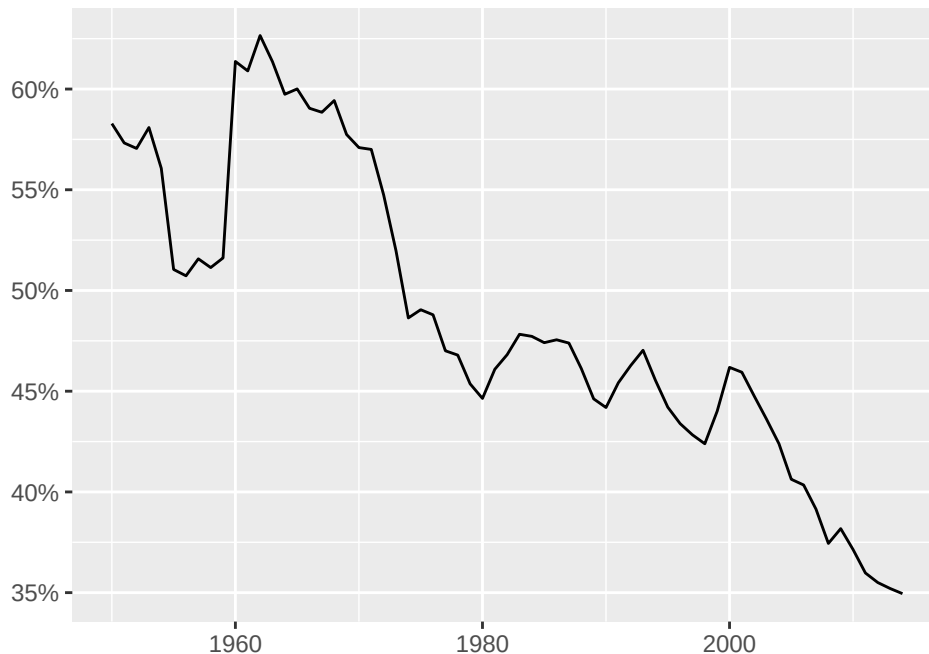


I also noticed that lots of dyads have *zero* trade interdependence, which the next plot shows. From 1950 to 2014, anywhere between 35-65% of dyads had no trade interdependence whatsoever. So not only is trade interdependence heavily skewed, it has lots of zeros.

```
dt |>
  group_by(year) |>
  summarize(
    mean = mean(interdependence == 0, na.rm = T)
  ) |>
  ggplot() +
  aes(year, mean) +
  geom_line() +
  labs(
    title = "Lots of dyads have zero trade interdependence",
    subtitle = "% of cases where measure = 0",
    x = NULL,
    y = NULL
  ) +
  scale_y_continuous(
    labels = scales::percent
  )
```

Lots of dyads have zero trade interdependence

% of cases where measure = 0



I decided to play around with the data a bit to see if some changes made any difference to the results. The first thing I tried was making trade interdependence binary. Maybe the level of trade interdependence isn't as important as the existence of interdependence. To test this idea, I estimated another logit model, but this time instead of using `interdependence` as a predictor, I wrote `I(interdependence > 0)`. This will turn this measure into a binary variable based on whether a dyad has any trade interdependence.

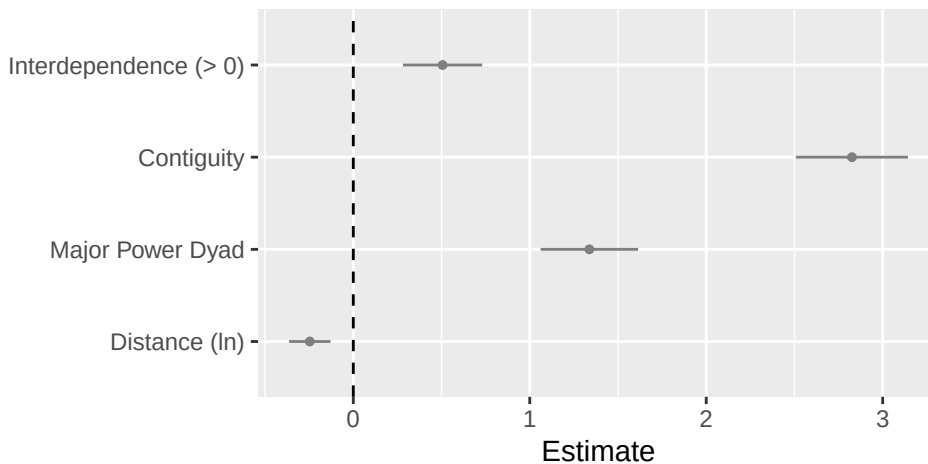
```
glm_robust(  
  miconset ~ I(interdependence > 0) + cont + major + ldist +  
    micspell + I(micspell^2) + I(micspell^3),  
  data = dt,  
  clusters = "dyad"  
) -> fit2
```

You can see the new results in the coefficient plot below. With this one simple change, the estimate for trade interdependence is now statistically significant! It also now runs in the *wrong* direction. It's positive; not negative. Countries that have any level of trade interdependence as opposed to none have a higher probability of fighting. That can't be right, can it?

```
coef_plot(
  fit2 |> slice(2:5),
  coef_map = c(
    "Interdependence (> 0)",
    "Contiguity",
    "Major Power Dyad",
    "Distance (ln)"
  )
) +
labs(
  title = "Logit estimates for the trading peace",
  subtitle = "Point estimates with cluster-robust 95% CIs shown"
)
```

Logit estimates for the trading peace

Point estimates with cluster-robust 95% CIs shown



Next I decided to adjust for the heavy skew in the data. I could try something like the natural log or (or better, the inverse hyperbolic sine, since trade dependence has zero values, and the log won't work on zero). However, even these transformations don't adequately normalize the data. So instead I decided to implement a rank-based transformation. This is a common transformation that researchers will use with heavily skewed data. What it does is assign a numerical score to observations based on their order. For example, if I had two data points, and one was 15 and the other 200, the first would get a rank of 1 and the second a rank of 2. They would receive these same scores even if the data points had different values. All that matters is the order; nothing else.

The below code implements this transformation and estimates another logit model.

```

myrank <- function(x) x |>
  as.factor() |>
  as.numeric()

dt |>
  mutate(
    intrank = myrank(interdependence)
  ) -> dt

glm_robust(
  miconset ~ intrank + cont + major + ldist +
    micspell + I(micspell^2) + I(micspell^3),
  data = dt |> filter(interdependence > 0),
  clusters = "dyad"
) -> fit3

```

Here are the results. Alas, the results aren't encouraging. It's hard to see, because the new estimate for trade interdependence is so small. It's still positive, but now it's no longer statistically significant.

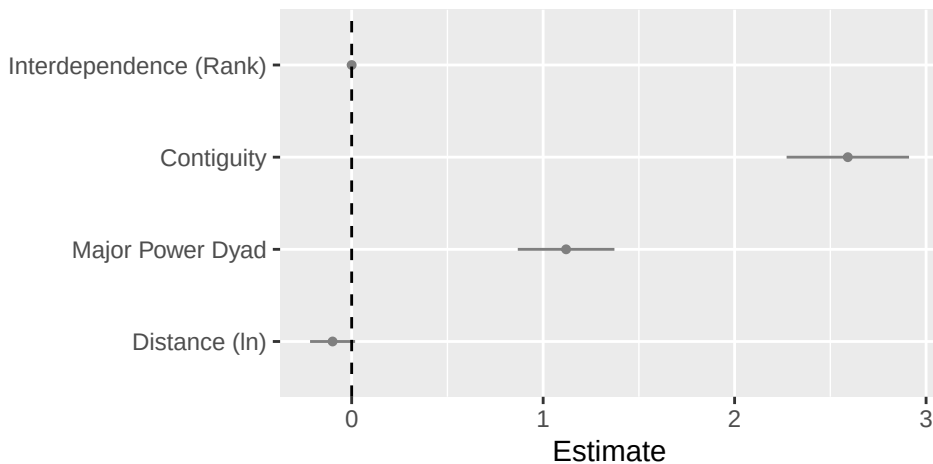
```

coef_plot(
  fit3 |> slice(2:5),
  coef_map = c(
    "Interdependence (Rank)",
    "Contiguity",
    "Major Power Dyad",
    "Distance (ln)"
  )
) +
  labs(
    title = "Logit estimates for the trading peace",
    subtitle = "Point estimates with cluster-robust 95% CIs shown"
  )

```


Logit estimates for the trading peace

Point estimates with cluster-robust 95% CIs shown



In my desperation, I decided to try one more thing. I mentioned a few chapters ago that my preferred approach for modeling dyadic data with logit models is to model dependence within dyads directly using a multilevel model with dyadic random effects rather than an *ad hoc* adjustment to model standard errors. So I tried that below, using the original model specification for trade interdependence.

```
library(mgcv)
library(lmtest)
gam(
  miconset ~ interdependence + cont + major + ldist +
    micspell + I(micspell^2) + I(micspell^3) +
    s(dyad, bs = "re"),
  data = dt,
  family = binomial
) -> fit4
```

Here are the results, which I decided to compare side-by-side with the original model. At last, some success! The estimate for trade dependence is somewhat attenuated compared to the original model with cluster-robust standard errors, but it is still negative *and* now statistically significant. Maybe the trading peace has some legs to stand on after all.

```
fit4 |>
  coeftest() |>
  broom::tidy() |>
  mutate(
    lower = estimate - 1.96 * std.error,
```

```

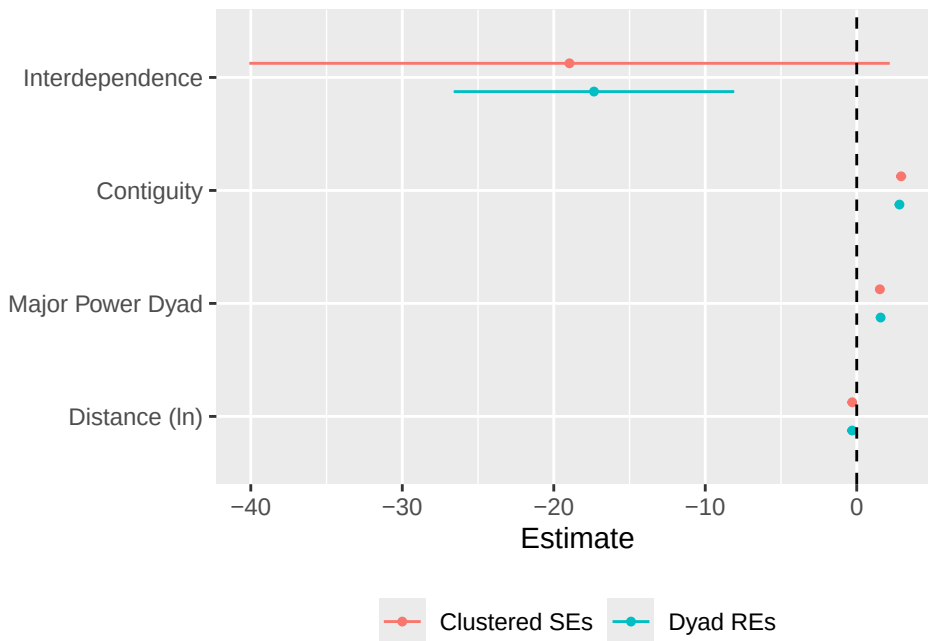
    upper = estimate + 1.96 * std.error,
    model = "Dyad REs"
  ) -> fit4_tidy
bind_rows(
  fit |>
    mutate(model = "Clustered SEs") |>
    slice(2:5),
  fit4_tidy |>
    slice(2:5)
) |>
coef_plot(
  coef_map = c(
    "Interdependence",
    "Contiguity",
    "Major Power Dyad",
    "Distance (ln)"
  )
) +
labs(
  title = "Comparison of logit estimates",
  subtitle = "Point estimates with 95% CIs",
  color = NULL
) +
theme(
  legend.position = "bottom"
)

```

Warning: `position\_dodge()` requires non-overlapping x intervals.

Comparison of logit estimates

Point estimates with 95% CIs



Or maybe it doesn't. While all the above might look like careful detective work, you could also call it "p-hacking," which is the process of finagling a dataset until it gives you the significant results you were looking for. You could also call it an "inquisitional data analysis." That is, you torture the data until it confesses. Either way, this isn't the most principled way to go about doing research.

I should admit that I live in a glass house. Whether consciously or unconsciously, I've felt pressure to find the "right" results in data. This is quite human. You believe so strongly in an idea that you refuse to give up when things don't first go your way. There's something commendable about that, but also dangerous.

There's no perfect way to solve the human tendency to seek out a desired result, but certain practices can help. One is pre-registration. That is, think really hard about your data before you see it, specify ahead of time how you plan to analyze it, and stick to your plan.

Another is transparency about your process. If you tried multiple specifications but only showed the significant results, that's not transparent. But if you tried several and showed them all, that's better. In academic research, many reviewers and editors expect you to try multiple approaches as "robustness checks" anyway, so the fact that you already tried a few different modeling and measurement approaches puts you ahead.

8.4 Summary

The trading peace, much like the democratic peace, describes a pattern in international conflict that many agree exists but disagree on *why*. One of the more popular arguments in favor of the idea that trade causes peace is trade interdependence theory, which holds that trade increases the opportunity cost of war. As a result, countries that trade a lot with each other have incentives to resolve their disputes peacefully rather than resort to military force.

This argument isn't immune to criticism. Three problems that remain unresolved are: (1) the fact that declines in trade during war often quickly rebound once the fighting is over, while trade during war never truly disappears; (2) the fact that the cost of war due to trade is correlated with other factors that might also influence the cost of war; and (3) the fact that peace can just as easily be argued to be the cause of trade as trade can be argued to be the cause of peace. If you find trade interdependence theory compelling, research that confirms the opportunity cost of war among trading countries, accounts for hard to address confounders, or addresses reverse causality would be a good place to start.

My own analysis confirms the empirical reality of the trading peace, but only after a good deal of p-hacking. Of course, because my analysis is only intended to be an example, I advise against taking the results too seriously. Another weakness with my analysis is that I failed to control for certain factors that also have become commonplace in peace science research. Many of these are factors that I already discussed in previous chapters, like the balance of power and democracy. Many studies that examine the trading peace in particular also control for economic growth, as Oneal and Russett (1997) do in their canonical study. Plenty of studies also account for alliances, which is the subject of the next chapter.

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